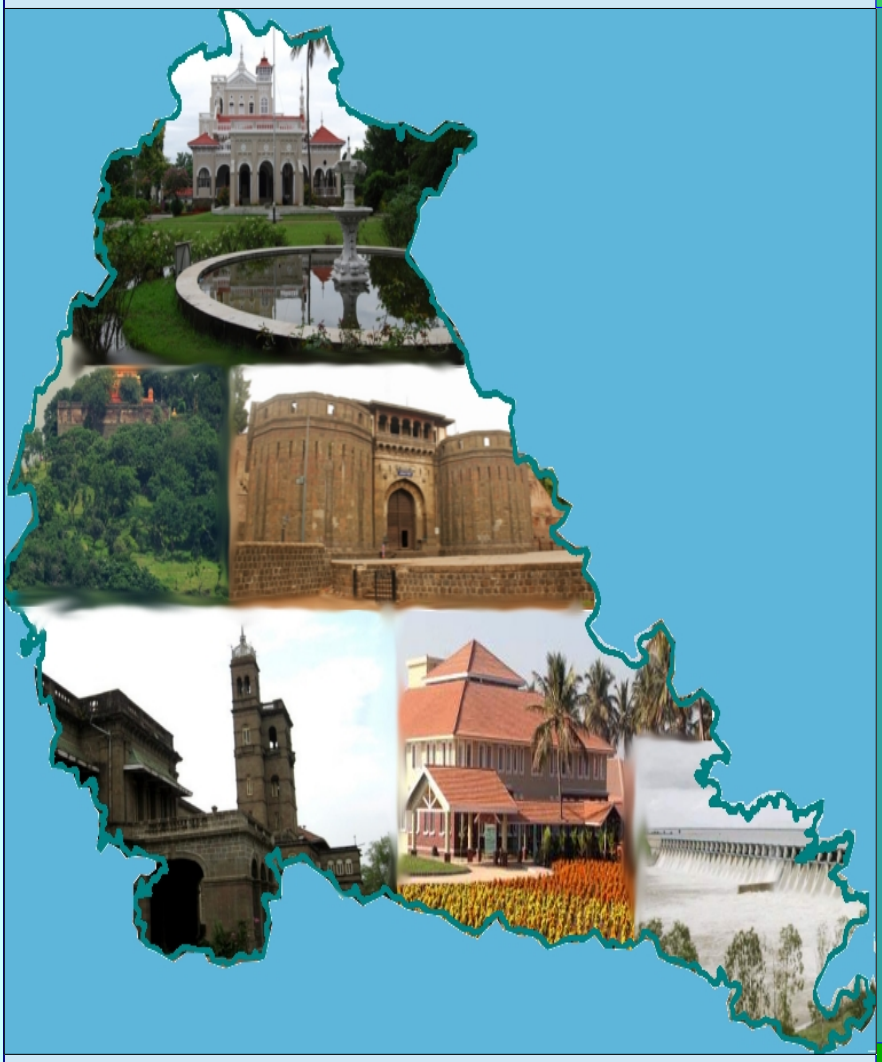
Pradhan Mantri Krishi Sinchayee Yojana

PMKSY

District Irrigation Plan
Pune



District Superintending Agriculture Officer

Pune

Index

Sr. No.	Table of Content	Page no.
1	**Chapter- 1 General Information of the District**	
	1.1 District Profile	1
	1.2 Demography	4
	1.3 Biomass and Livestock	6
	1.4 Agro-Ecology,Climate, Hydrology and Topography	10
	1.5 Soil Profile	13
	1.7 Land Use Pattern	20
2	**Chapter- 2 District Water Profile.**	
	2.1 Area Wise, Crop W ise irrigation Status	25
	2.2 Production and Productivity of Major Crops	29
	2.3 Irrigation based classification	33
3	**Chapter- 3 Water Availability of District**	
	3.1 Status of Water Availability:	37
	3.2 Status of Ground Water Availability	41
	3.3 Status of Command Area	44
	3.4 Existing type of irrigation	47
4	**Chapter- 4 Water Requirement /Demand**	
	4.1 Domestic Water Demand	51
	4.2 Crop Water Demand	55
	4.3 Livestock Water Demand	58
	4.4 Industrial Water Demand	61
	4.6 Total Water Demand of the District for Various sectors Water Budget	63
5	**Strategic Action Plan for Irrigation in Pune District under PMKSY**	68
6	**Action Plan**	71
7	**Annexure – I Strategic Action Plan of Irrigation for Blocks of Pune**	77
8	**Annexure – II Production and Productivity of Major Crop- Block-wise**	91
9	**Annexure – III Crop Water requirement - Block-wise**	105
10	**Annexure – IV Strategic Action Plan – Year wise (5 years)**	120
11	**Annexure – V Areawise Cropwise Irrigation Status**	127

List of Tables

Sr. No.	Name of Table	Page No.
1.1	District Profile	2
1.2	Demography	5
1.3	Livestock of Pune	7
1.4	Agro-Ecology,Climate, Hydrology and Topology	11
1.5	Soil Profile	15
1.7	Land use Pattern	21
2.1	Area wise crop wise irrigation status	27
2.2	Production and Productivity of Major crops	30
2.3	Irrigation based Classification	33
3.1	Status of Water Availability	38
3.2	Status of Ground water Availability	41
3.3	Status of Command area	45
3.4	Existing type of Irrigation	47
4.1	Domestic Water Demand	51
4.2	Cropwise Water Requirement	56
4.3	Livestock Water Demand	59
4.4	Industrial Water Demand	62
4.6(a)	Total water Demand of District for various section	63
4.7	Water Budget	66
5	Strategic Action for Irrigation in District under PMKSY	75-76

Document Compiled and Prepared by

AFARM - PUNE

Chapter-I: General Information of the District

1.1 District Profile

Geography

Pune is the second largest district of Maharashtra State in respect of area. The district has a geographical area of 15642 sq.km., which is 5.08% of the total area of State. It is situated in the western part of the State and lies between north latitude 17°54’ and 19°24’ and east longitudes 73°29’ and 75°10’ and falls in parts of Survey of India degree sheets 47-E, 47-F, 47-I, 47-J, 47-K, 47-N and 47-O. It is bounded by Ahmadnagar district in the north and east. Satara and Solapur districts in south and south east respectively and Thane and Raigarh districts in north west and west respectively. For administrative convenience it is divided in 14 talukas namely Pune City, Haveli, Khed, Ambegaon, Junnar, Shirur, Daund, Indapur, Baramati, Purandhar, Bhor, Velhe, Mulsi and Maval.

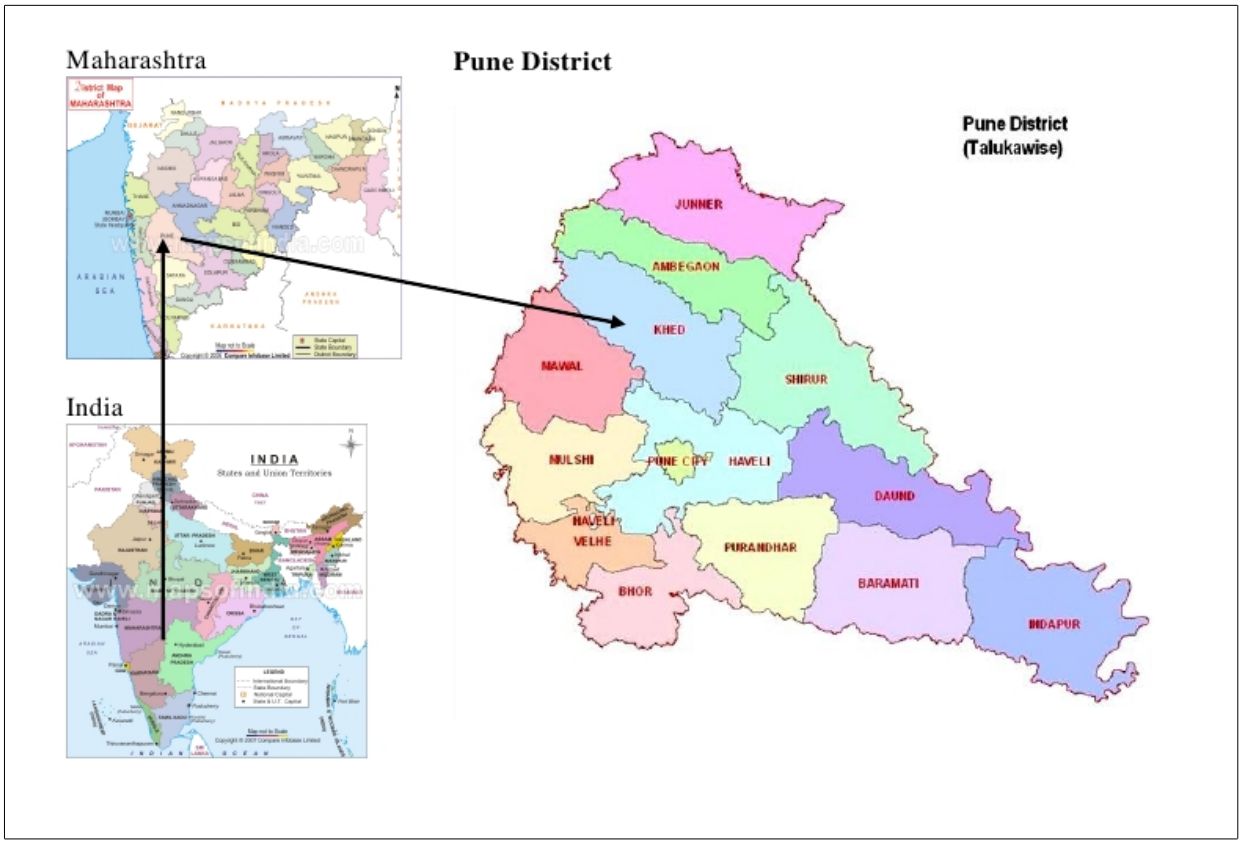


Figure 1.1 Location of Pune District

Pune district lies in the Bhima and Nira basins. It has a shape of triangle with its base in Sahyadri mountains on the West and its apex in the extreme South-East corner near Nira River. Taking into consideration the height from sea level, rainfall, soil pattern etc. It is divided into three zones viz.

Western Zone, Central Zone and Eastern Zone. The Sahayadri ranges are spread from North to South in the district.

District Profile – Pune	
State – Maharashtra	
District Code	521
Geographical Area	15642 sq.km
Latitude and Longitude	N 17°54' and 19°24' E 73°29' and 75°10'
Area under Forest	171809 Ha
Area under Agriculture use	1172951 Ha
Net Sown Area	1042282 Ha
Gross Cropped Area	1172951 Ha
Net Irrigated(Rabi/kharif)	375053 Ha
Net Rainfed	521686 Ha
Total Number of block	13
Total Number of Panchayat	1407
Total Number of Villages	1877
Total Population	9429408
Population Growth Rate(Decadal)	30.34%
Total Male Population	4924105
Total Female Population	4505303
Total Population ST	348876
Total Population SC	1180703
Total Population GEN	7899829
Total livestock	1948183
Poultry	18537999
Stored Surface water	0.402975 BCM

Table 1.1 District Profile

History

In the beginning of the Rashtrakuta rule we get reference to a 'Vishaya' or district with headquarters at Poona or Punya and Punak as it was then called. Hence the word 'Pune' might have been originated from the word 'Punya' or sacred. This Sacredness might have been attached to the town because of the confluence of two rivers viz. Mula and Mutha in its vicinity. Punak with the passage of time degenerated into 'Pune'. The district has the shape of a triangle with its base in the Sayadri Mountains on the west and its apex in the extreme southeast corner near the point of confluence of the Bhima and Nira rivers. It appears to have been a centre of highways for trade and commerce for over 2000 years. For

administrative convenience and industrial and agricultural development, Pune district was divided into 4 sub divisions namely Pune, Haveli, Baramati and Junnar. The story revealed by the numerous Buddhist caves in the area dated the 2nd Century A.D. indicates the Buddhist influence. The inscriptions in Buddhist caves make mention of a number of important trading places within and without the country of various social and occupational sections of the people and of recognized industrial associations. Junnar was the capital of a king of the Deccan. It will not be an exaggeration to say that, Junnar which for a long time during the historical period has been the chief city of the Poona district is in fact one of the oldest towns in Western India.

The chalukyas Rashtrakutas and Yadavas ruled Poona during the Hindu period (300-1294 A.D.). With the conquest of Devgiri by Alauddin Khilji in 1294 A.D. and the acceptance by Ramdeorao Yadav of the suzerainty of the Delhi rulers. Poona passed indirectly under the Muslim rule. The BahamaniNazimshah's and Adilshah's ruled Poona during the medieval Muslim period (1295-1720). The Marathas reigned Poona during 1720-1818 till the capture of Poona by the British troops. The year 1876 witnessed one of the worst famines in India History and parts of Poona especially the eastern ones, were lead sufferers. Then onwards Britisher's ruled Poona from 1819-1947, until it was finally freed from the British Clutches by the Great Indian Freedom Fighters. There were 1213 villages in the district in 1820. These villages were grouped under various subdivisions. In all 9 subdivisions were formed. In 1878, headquarters of Bhimthadi taluka were transferred from Petas to Supa, Baramati, Peta of Bhimthadi was abolished and the headquarters of Bhimthaditaluka were shifted to Baramati and a new peta was carved out in 1892 with headquarters at Dhond. After 1949 Bhimthadi taluka was designated as Baramati taluka. In 1948-49 Petas of Ambegaon, Dhond. Sirur and Mulshi were elevated to full-fledged taluka status. As a consequence of the merger of former states a part of Bhor state was annexed with the district and one village from Poona was transferred to the North Satara district. Taluka of Bhor and VelheMahal were newly carved out in 1949. Re-organization of States resulted in the inclusion of the district in Bombay state in 1956 and since 1960 it forms a part of Maharashtra state. The district comprised of 1498 inhabited villages in 1961, which rose to 1753 in 1981. There were 22 towns in the district in 1961 and 1971 and the number rose to 26 in 1981. In 1991 the district comprises 1844 inhabited villages and 34 towns spread over 14 tahsils. In 2001 the district covered same 1844 inhabited villages and 25 towns over 14 tahsils. In 2011 census, the district has 14 tahsils, 35 towns and 1877 villages (including 25 uninhabited villages).

Administrative Setup

There were Major changes in the administrative set-up of Maharashtra after the 1991 Census, resulting in increase of five districts. Greater Bombay district of 1991 census divided into two forming Mumbai and Mumbai suburban districts. Similarly new districts Nandurbar, Washim,Hingoli and Gondiya district were formed bifurcating, Dhule, Akola, Parbhani and Bhandara district respectively. At the time of 2001 Census Konkan division included 6 districts, Nashik division had 5 districts, Pune division had 5 districts, and Aurangabad division had 8 districts, Amravati division had 5 districts and Nagpur division had 6 districts. Pune district is included in Pune division. Thus for 2011 census the State has 35 districts spread over 6 divisions in Maharashtra. In 2001 Pune district comprised 1866 villages and 25 towns. 2 new villages have been created after 2001Census. The district has now 35 towns, 1877 villages spread over 14tahsils viz., Junnar (183), Ambegaon (143), Shirur (115), Khed (186), Mawal (187), Mulshi (145), Haveli (123), Pune City (1), Daund (102), Purandhar (108), Velhe (130), Bhor (195), Baramati (116) and Indapur (143) tahsils. For administrative purposes the district is divided into 5 sub-divisionsof Pune, Mawal, Baramati, Junnar, and Bhor. Pune sub-division includes Pune City and Haveli tahsils, Mawal sub-division includes Mawal, Shirur and Mulshi tahsils, Baramati sub-division includes Baramati, Indapur and Daund tahsils, Junnar sub-division includes Junnar, Ambegaon and Khed tahsils and Bhor sub-division includes Bhor, Purandhar and Velhe tahsils.The district collector along with the district Judge, Superintendent of police, chief executive officer of ZillaParishad and other senior officers of the State Government look after the development and regulatory functions in the district. At tahsil level, the Tahsildar, Block Development officer, Judicial Magistrate, Deputy Engineers and other officers look after their respective departments for development and regulatory functions.

1.2 Demography:

According to census of 2011, the population of Pune district has gone up by 30.34 per cent to reach 94.26 lakh in the past decade. With growth hovering around 30 per cent in the past two decades, which registered 35.9 percent growth during last decade. The average growth rate for the state is 15.99 percent and Pune city at 8.4 percent.

The population of Pune district grew from 72,24,224 in 2001 to 9429408 in 2011. While the number of males was 49,24,105 and 45,05,303 were females. Literacy rate for Pune was 87.19 per cent with male literacy at 92.72 per cent and female literacy at a lower 81.13 per cent.

The scheduled caste population was 1180703 in 2011, whereas scheduled caste population was

348876. The population of general categories of the district was 7899829. The total number of household in the district was 2151503 in 2011.

1.2 Demography									Source : Census 2011
Name of the state : Maharashtra									
Name of the disrict :Pune									
Name of Block/G.P	No of the villages covered	Population			SC	ST	General	Total	
		M	F	Total	No. of Members	No. of Members	No. of Members	No. of household	No. of Members
Pune city	1	1700867	1604021	3304888	446602	36708	2821578	781507	3304888
Haveli	123	1316346	1119235	2435581	375246	50677	2009658	586551	2435581
Maval	187	198487	179072	377559	36325	29208	312026	78233	377559
Mulshi	145	90053	80953	171006	18092	7418	145496	38014	171006
Bhor	195	94158	91958	186116	11090	5414	169612	39007	186116
Velha	130	27504	27012	54516	2635	2032	49849	11428	54516
Juunar	183	202360	196942	399302	17922	80922	300458	85864	399302
Khed	186	237868	212248	450116	30409	49291	370416	99477	450116
Ambegaon	143	119226	116746	235972	9757	50704	175511	50357	235972
Shirur	115	201152	184262	385414	32885	12030	340499	80917	385414
Baramati	116	221094	208506	429600	64334	3957	361309	90912	429600
Indapur	143	198801	184382	383183	61080	4766	317337	79683	383183
Daund	102	196283	184213	380496	57273	9673	313550	78294	380496
Purandar	108	119906	115753	235659	17053	6076	212530	51259	235659
Total	**1877**	**4924105**	**4505303**	**9429408**	**1180703**	**348876**	**7899829**	**2151503**	**9429408**

Table 1.2 Demography

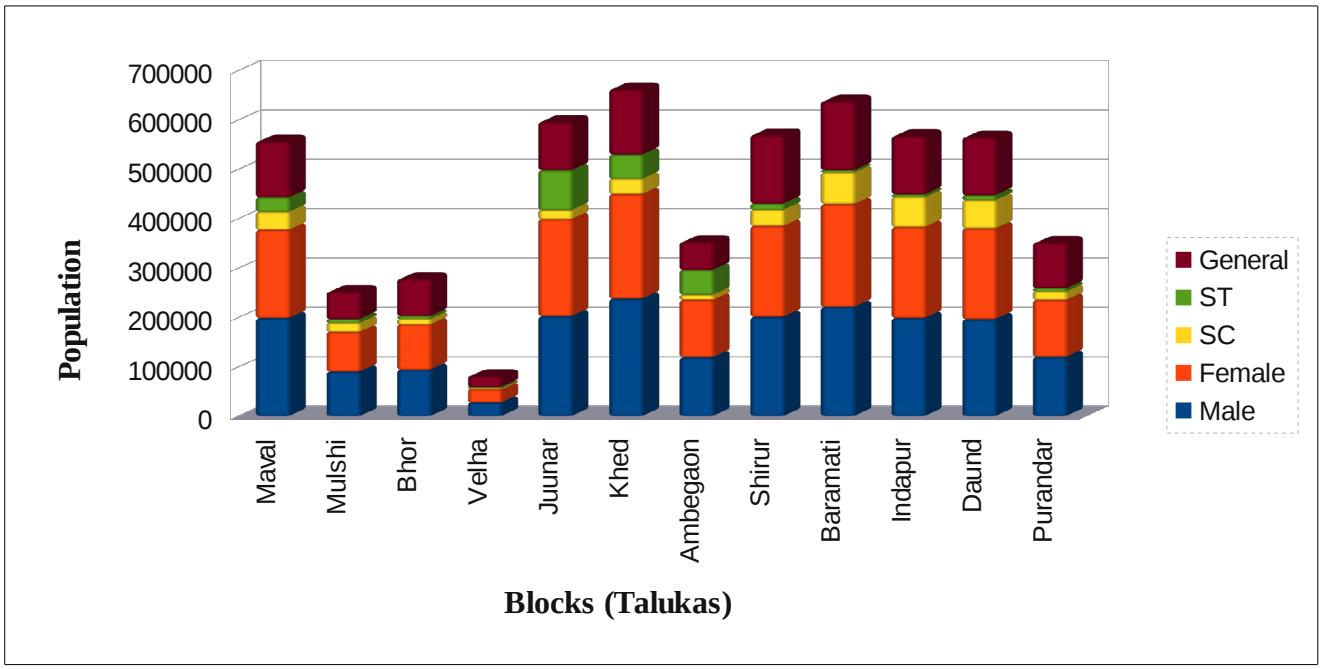


Figure 1.2.1 Urban Blockwise Rural Population

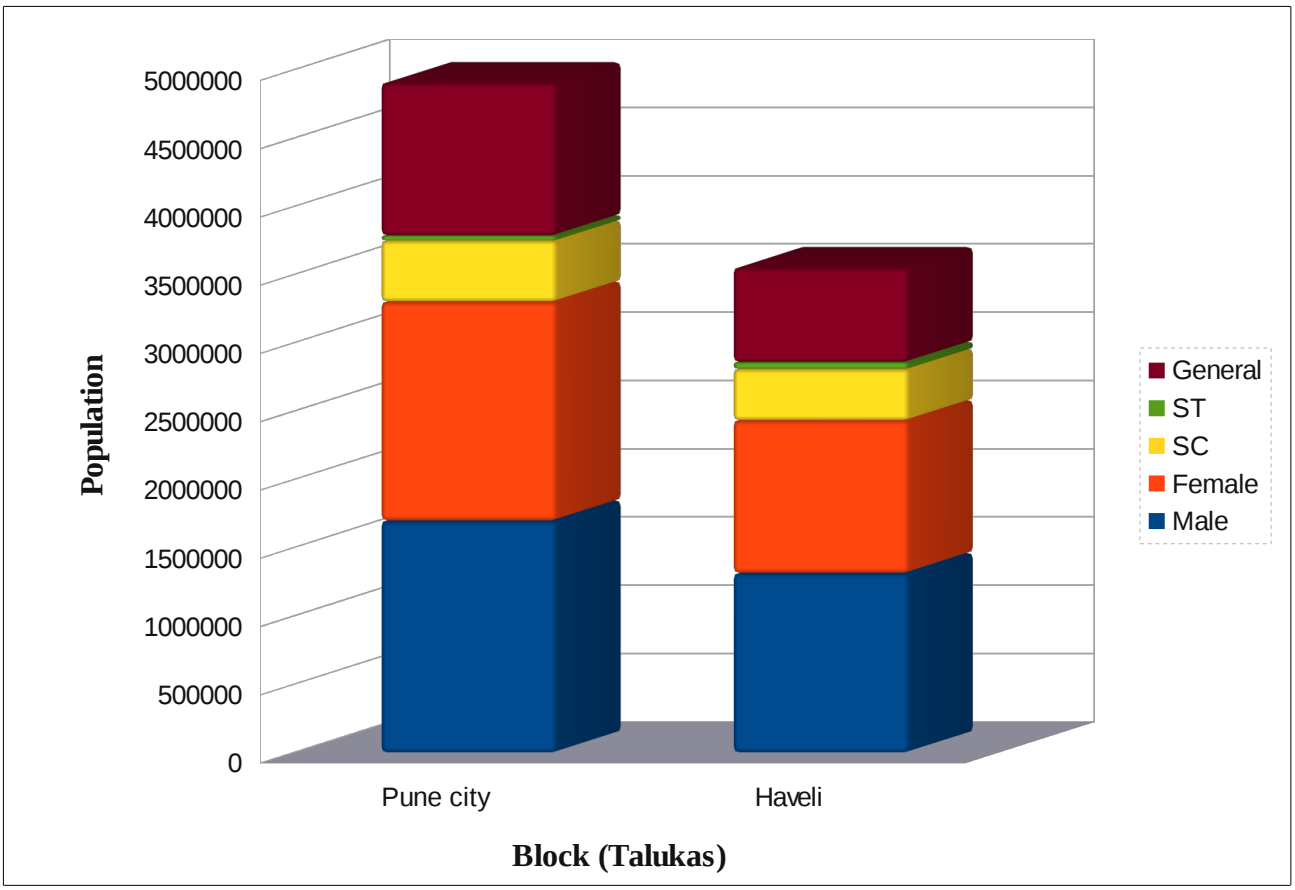


Figure 1.2.2 Urban Population (Pune City) and Haveli Taluka

The average family size of the district is 4 person. For every 1000 males, there were 919 females in 2011. The district has a population density of 603 inhabitants per square kilometre.

Out of total 1180703 SC population of the district , the blocks like Baramati, Indapur and Daund witness more population, whereas the Scheduled Tribe population is occurring more in blocks like Ambegaon, Khed, Junnar, Maval etc.

1.3 Biomass and Livestock:

Livestock :

The livestock husbandry has been a major component of agricultural production systems since time immemorial. Animal Husbandry, Dairy and Fisheries sectors play a crucial part in district economy. Livestock provides vital additional income for sustenance of poor in rural areas through gainful employment. Livestock render insurance cover for the poor during drought and famine. Livestock provide cheap nutritional food and indispensable proteins to the people.

In the district the milching animals like cow, buffalo's, goats etc are reared on large scale since it

contributes economical gain of the family, this is also supported with the poultry businesses in the different part of the district. There are number of commercial poultry units as well as house hold level poultry units existed in the district.

1.3 Biomass and Livestock													Source : Livestok Census of India 2013-14
Name of the state : Maharashtra													
Name of the disrict : Pune													
Block	Small Animals					Large Animals			Draft Animal (Buffalo/ yak/bulls/any other (Nos.)				
	Poultry (Nos.)	Ducks (Nos.)	Pigs (Nos.)	Goats (Nos.)	Sheeps (Nos.)	Indigenous Cow (Nos.)	Hybrid Cow (Nos.)	In descriptive Buffalo (Nos.)	bull	he buff	Horse	Donkey	Camel
Pune city	19303	71	2589	4582	771	1200	1860	6471	157	434	0	254	0
Haveli	2015065	118	808	17123	33771	16733	19894	33836	9626	2426	0	68	24
Maval	1471307	117	898	9252	646	27917	4166	22704	18401	1555	0	8	0
Mulshi	337878	44	194	4808	494	16453	4093	11848	9681	1155	0	0	0
Bhor	1024813	0	23	5638	1079	17268	14298	17867	10547	1152	0	0	0
Velha	89074	0	0	4565	345	12072	1949	10287	6190	651	0	0	0
Juunar	2610506	28	635	47802	4336	34881	56055	21157	22447	1669	230	175	0
Khed	2001263	5	447	25754	3388	42136	29865	28982	30233	2801	0	17	0
Ambegaon	2577945	21	120	33388	7022	26567	53058	9882	18410	808	0	112	0
Shirur	2720328	200	1223	34212	46722	34056	63749	24584	17653	1867	0	58	1
Baramati	201323	64	520	63840	89858	16566	57789	13358	9011	540	45	34	1
Indapur	1033090	130	949	83359	22854	25559	68112	40675	8694	3150	0	314	0
Daund	1924964	81	454	43777	73956	23447	54806	25550	11054	1626	2	40	4
Purandar	511140	0	645	16623	18667	12227	26485	6563	8165	573	41	38	0
Total	**18537999**	**879**	**9505**	**394723**	**303909**	**307082**	**456179**	**273764**	**180269**	**20407**	**318**	**1118**	**30**

Table 1.3 Livestock of Pune

Livestock of the Pune district is divided largely in Four types namely Poultry, Small Animals, Large Animals and Draft Animals. The Poultry of the district is flourishing industries with total of 18537999, which is considerable size. In small animals categories Ducks, Pigs, Sheeps and Goat is counted with their total count of 709016. The large animal count is 1037025 which is substantial. The draft animal total of the district comes to 202142.

The figure 1.3.1 shows the graph of livestock of Poultry for Pune District. The Shirur taluka has the largest number of Poultry birds 2720328 which is highest in the district. The blocks Amebgaon, Junnar also shows significant number of Poultry birds.

Pune city has least number of poultry units due to the dense population and lack of space availability.

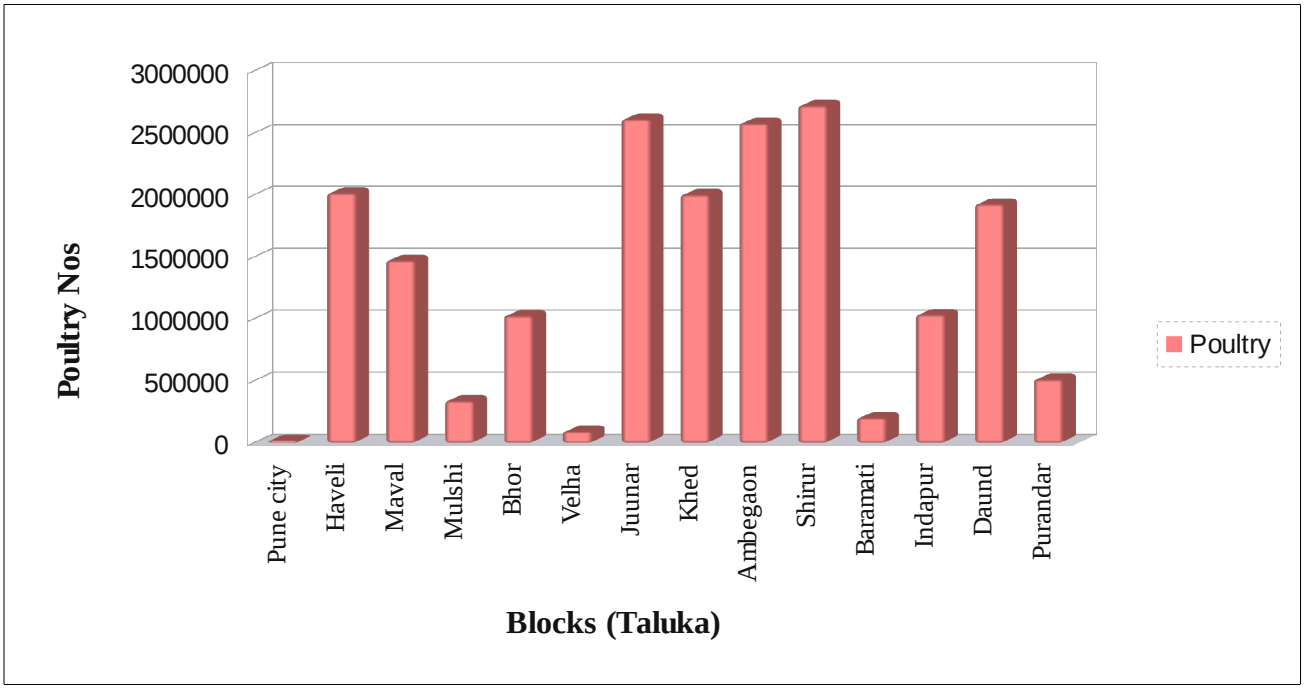


Figure 1.3.1 Livestock - Poultry

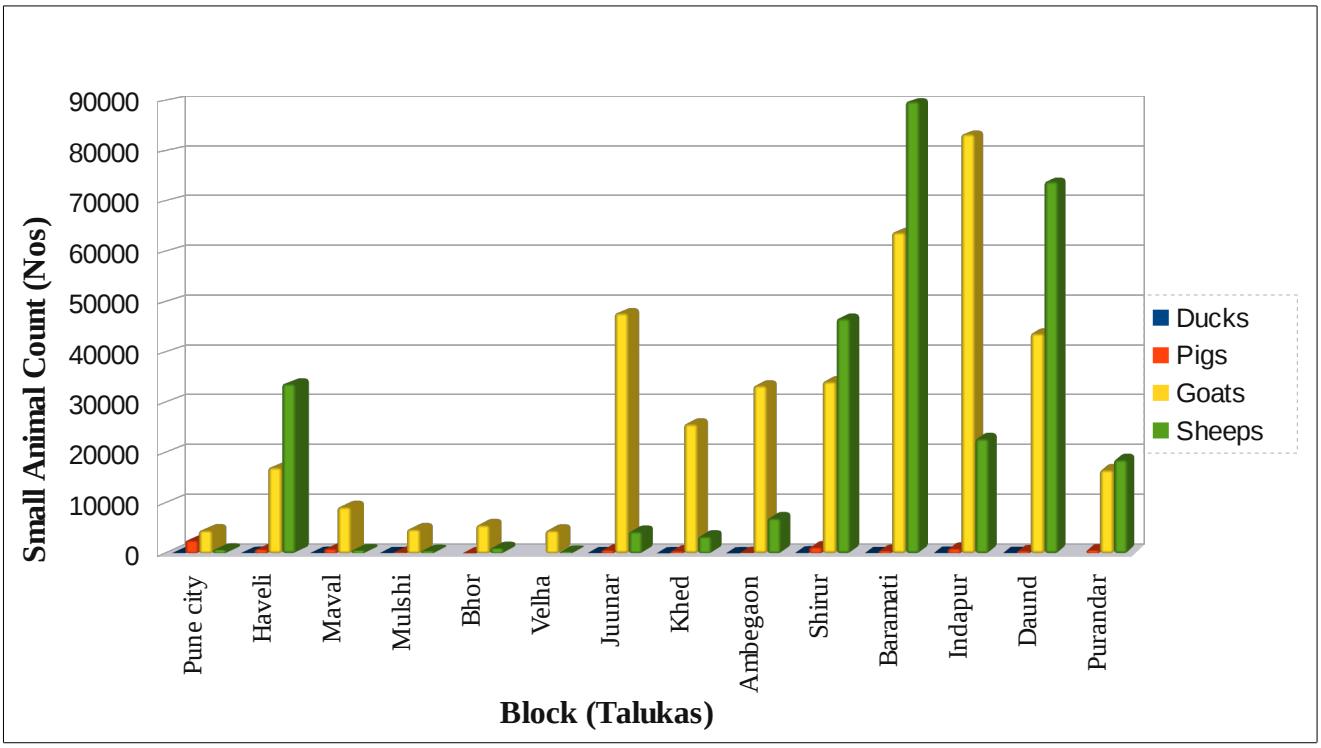


Figure 1.3.2 Livestock - Small Animals

Livestock of Small Animals like Goats and sheeps are large in Baramati, Indapur, Shirur, Junnar and Daund. There is considerable scope for goat and sheep rearing in other parts of the district as well, since this is a supplementary source of income generation specially landless and marginal farmers.

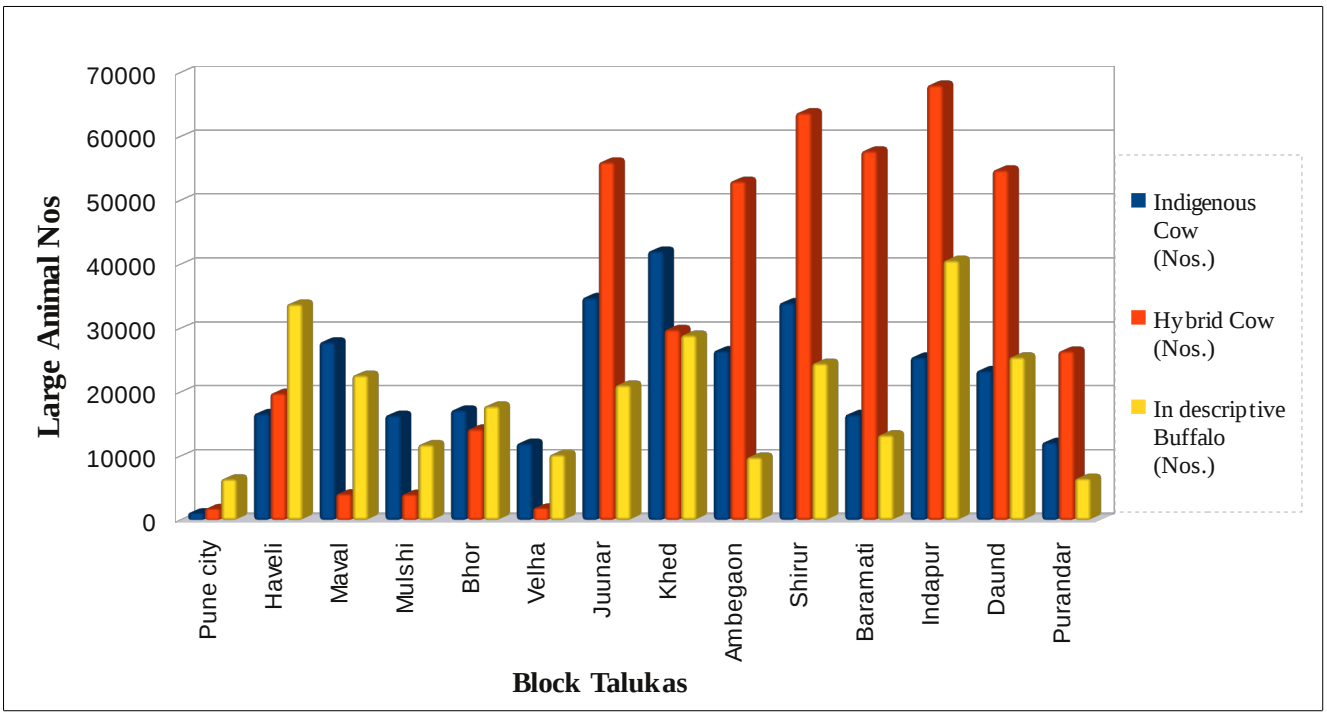


Figure 1.3.2 Livestock – Large Animals

There is sizable number of cattle in the district. Total number of cow count both indigenous and hybrid variety is 763261 and buffalo count is 273764. In the district there occurs huge scope for dairy business provided assurance of optimum water and green fodder during peak period of the year. Presently the scarcity of water and availability of green fodder across the year inhibits animal husbandry and dairy business of the district.

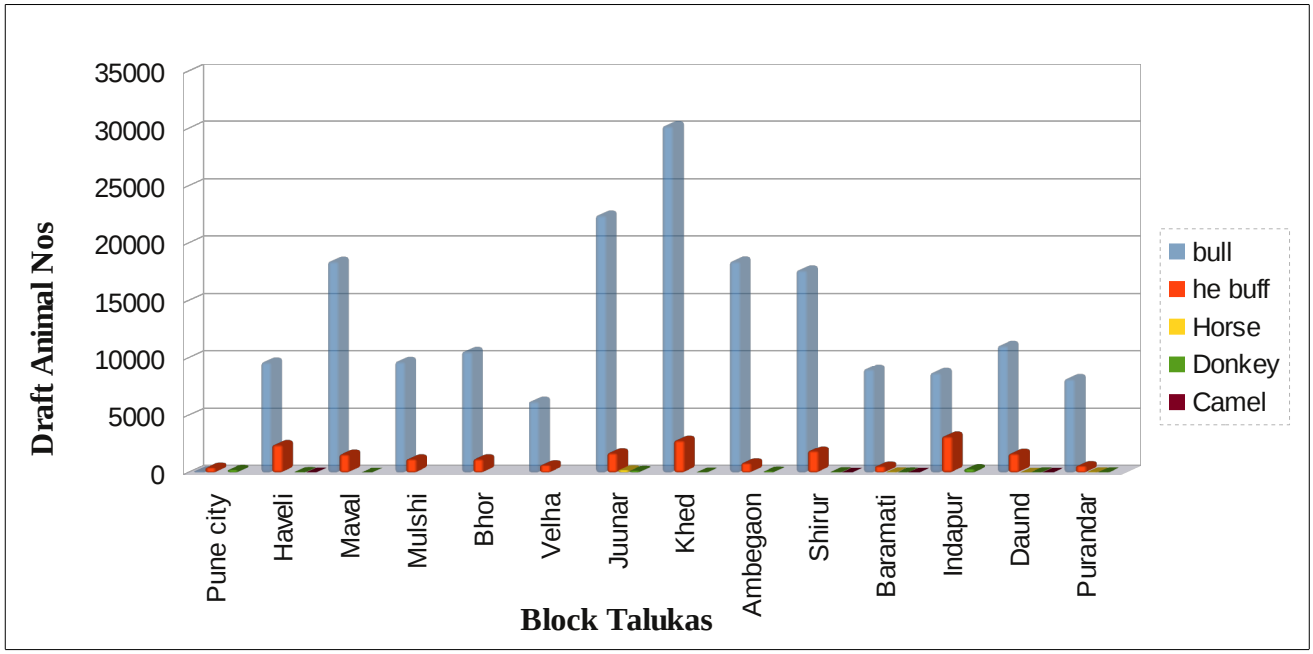


Figure 1.3.4 Draft Animals

Draft animals with the total count of 202142 in the district of which number of 180269 are bullocks followed by 20407 he buffalo's, 1118 donkeys, 318 horses and 30 camels. The bullock and he buffalo power available in the district is predominantly used for agricultural operations whereas horses donkey and camel are used for enhancing income from other supplementary activities.

Unavailability of dry and green fodder in summer discourages farmers from increasing their livestock. With increasing pressure on land for growing remunerative crops, production of fodder takes a back seat. The nutritive value of feed and fodder has a significant bearing on productivity of livestock. Hence, producing high yielding varieties of fodder to make fodder production more remunerative is critical to bridge the gap between demand and supply.

1.4 Agro-Ecology, Climate, Hydrology and Topography:

The Pune district is divided in six agro-ecological zones namely AES-I to VI. The first AES-I comprises of Ghat section, high rainfal, shallow soils which covers western part of Velha, Bhor Mulshi, Khed, Junnar, Ambegaon and Maval. This part of the district receives annual rainfall ranging from 3000-5000. The average monthly rainfall of this area is 750 to 1250mm.

The second AES-II covers sub mountaineous region with medium rainfall having lateric soils. This mainly consists of western part of Velha, Bhor, Mulshi, Khed, Junnar, Ambegaon and Maval blocks of the district. This zone receives annual rainfall of 1750 to 2500mm.

The zone AES-III is sub-mountaineous with assured rainfall and medium soil. This covers western part of Purander, Haveli Bhor, Eastern part of Maval, Mulshi & Central parts of Khed, Ambegaon, Junnar. This area receives average monthly rainfall 250 to 425mm.

The next zone AES-IV is a plain area with a assured rainfall and medium soil. This contributes Central part of Haveli ,Khed, Purandar, Ambegaon & Junnar with annual rainfall of 950-1250mm and average monthly rainfall from 238 to 313mm.

The AES-V is a scarcity zone with low rainfall and light to medium soils covering parts of Purandar, part of Baramati,Daund, Haveli , Indapur, Shirur, Ambegaon and Khed. This area receives average monthly rainfall of 188 mm.

The last zone AES-VI is again a scarcity zone with light to medium soils and rainfall. This covers Parts of Shirur, Baramati, Indapur, Daund blocks. These blocks receive less than 750 mm ranfall annually.

1.4 Agro Ecology, Climate,Hydrology and Topography																							Source: IMD,Regional ICAR centre(s)SAUs,KVKs,etc.
Name of the state : Maharashtra																							
Name of the disrict : Pune (Jurisdiction- Junnar, Ambegaon, Khed, Maval, Mulshi, Shirur)																							
Sr. No.	Agro Ecological Zone Type	Type of terrain	Name of block	Block Area (ha)	Normal Annual Rainfall (mm)	Average Monthly Rainfall (mm)	No. of Rainy Days (Nos)	Average Weekly Temperature										Potential Evapo-Transpiration(PET)				Elevation		
								Period										Period			Cumula-tive Total			
								Summer (April-May)			Winter (Oct mar)			Rainy (June Sept)				Summer (Avg. April-May)	Winter (Avg. Oct.-March)	Rainy (Avg. June-Sept)		min (M)	max (M)	mean (M)
								min	max	mean	min	max	mean	min	max	mean								
1	AES-I Ghat Area, High Rainfall, Shallow Soils	High Rainfall, Shallow Soils	,western part of velha, bhor mulshi khed, junnar ambegaon Maval & Mulshi	-	3000-5000	750-1250	NA																	
2	AES-II Submountain, medium rainfall, lateratic soils	Submountain , medium rainfall, lateratic soils	,western part of velha, bhor mulshi khed, junnar ambegaon Maval & Mulshi	-	1750-2500	425-625	NA																	
3	AES-III Sub mountain, Assured rainfall, Medium Soil	Sub mountain, Assured rainfall, Medium Soil	western part of purander, haveli bhor, eastern part of of Maval, Mulshi & Central parts of Khed, Ambegaon, Junnar	-	1000-1700	250-425	NA																	
4	AES-IV Plain area, Assured rainfall, medium soil	Plain area, Assured rainfall, medium soil	Central part of haveli Khed, purandar, Ambegaon & Junnar	-	950-1250	238-313	26	20.60	36.90	28.70	30.88	13.76	22.32	30.40	20.20	25.30	9.6	6.4	4.2	20.2	-	656	656
5	AES-V Scarcity, Low rainfall, Light to medium soil, Non command area	Scarcity, Low rainfall, Light to medium soil, Non command area	purandar, part of baramati,daund, haveli indapur, Shirur, Ambegaon & Khed	-	Less than 750 mm	188	NA																	
6	AES- VI Scarcity, Low rainfall, Light to medium soil, command area	Scarcity, Low rainfall, Light to medium soil, command area	Parts of Shirur, Ambegaon,Khed & Junnar,baramati, indapur, daund,haveli		Less than 750 mm	188		21.7	38.50	30.10	15.40	31.60	23.50	22.10	32.20	27.15	17.3	24.9	18.8	20.33	549	1363	956

Climate and Rainfall

The winter season in the district starts from December to middle of February followed by summer season which last up to May. June to September is the south-west monsoon season, whereas October and November constitute the post-monsoon season. The mean minimum temperature is about 12°C and mean maximum temperature is about 39°C.

The normal annual rainfall over the district varies from about 468 mm to 4659 mm. It is minimum in the eastern part of the district around Daund (468 mm), Baramati (486 mm) and Jujuri (494 mm). This increases towards west and reaches a maximum around Khandala (4659 mm) in the western ghat.

The Chances of receiving normal rainfall are maximum (50 to 55%) in eastern part around Indapur and Daund, in the central part around Pune city and small area around Junnar in northern part of the district. Rainfall analysis also indicates occurrence of drought prone area in eastern, southern, south-eastern,

central and north western part around Indapur, Baramati, Jujuri, Daund, Talegaon-Damdhare ,Alandi, Shirur and Bhor covering about 50% area of the district.

Topography and Agro-Climatic Conditions:

The climate of the district is characterized by dry atmosphere except monsoon. The summer is moderately high and temperature varies from 36 c to 42 c. The average annual rainfall is 905 mm. The rainfall pattern varies between 5080 mm on the western hilly region to 457 mm in eastern plateau progressively declining eastwards. Pune district forms a part of the tropical monsoon land and therefore shows a significant seasonal variation in temperature as well as rainfall conditions. Climate of the western region of Pune is cool whereas the eastern part is hot and dry. Pune district is divided into four (3 to 6 zones) agro-climatic zones of Maharashtra which are described below

1) Zone - 3 : Western Ghat Zone:

The area included is this zone comprises of the hilly, high lying terrains round about the Ghats of Amboli, Fonda and Amba in Kolhapur District, Koyana and Mahabaleshwar in Satara district, Lonawala and Khandala in Pune district, Kalsoobai hills in Akola taluka in Ahmednagar district and Igatpuri and Trimbak in Nasik district. This zone covers 1.16 lakh hectares area of Pune district.

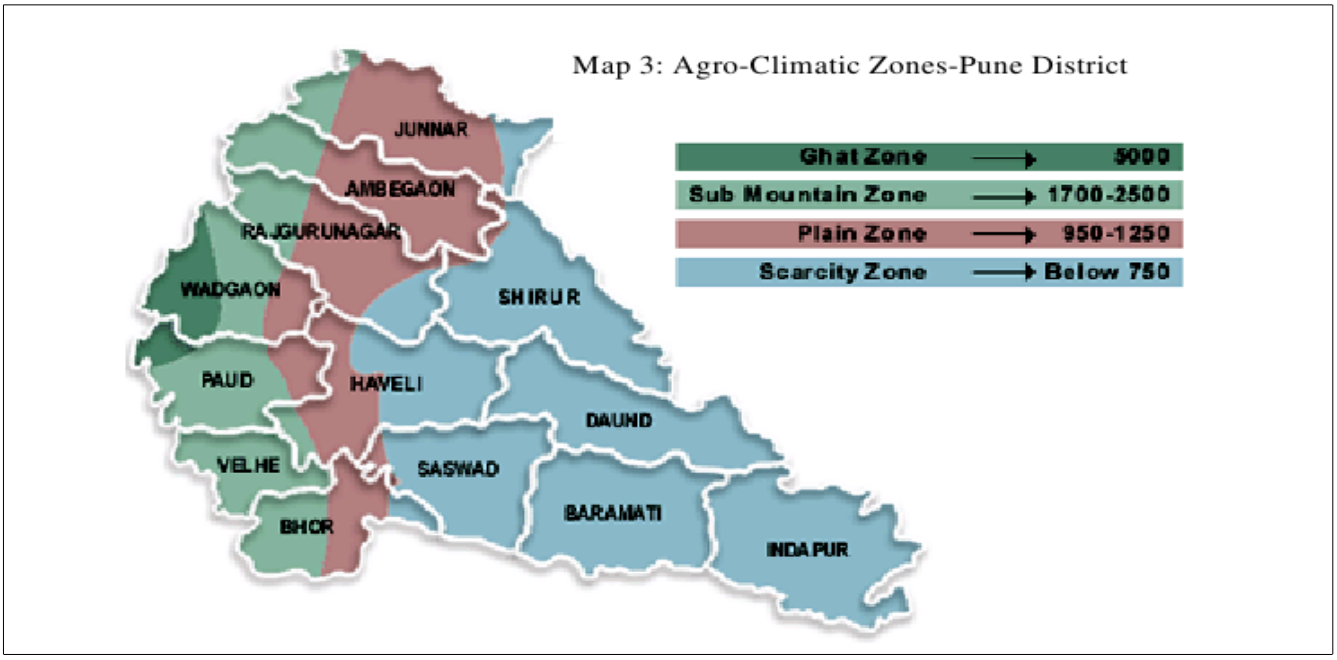


Figure Agro-Climatic Zones-Pune District

2) Zone - 4 : Sub-Mountain Zone – Transition Zone-1 with red to reddish brown soils:

It comprises a narrow strip running through the western hilly flank of Kolhapur district; western part of Bhor, Velhe and Mulshi, central part of Mawal, western strip of Khed, Ambegaon and Junnar talukas

of Pune district. The area covered by this zone is 3.08 Lakh Ha area of district.

3) Zone – 5 : Western Maharashtra Plain Zone – Transition Zone-II with Grayish Black Soils

This is a wider strip running parallel and to eastern side of Transition Zone-I and extends towards east up to the line where the Scarcity Zone starts. It extends to central part of Bhor, western part of Haveli and eastern parts of Maval, Khed, Ambegaon and Junnar talukas in Pune district covering 3.14 lakh hectares.

4) Zone- 6: Scarcity Zone- With kharif - cum Rabi Cropping

The vast area in mid-western Maharashtra is traditionally known as the famine area of the State. It comprises of Pune, Ahmedngar, Nasik and Dhule districts (excluding the portions covered under plain Zone). This zone is the largest zone of Pune district constituting 8.21lakh hectares and accounting for 53 per cent of the area of the district. It covers areas of Purandar, Baramati, Indapur, Daund, Shirur, Junnar, Ambegaon, Khed, Haveli and Bhor taluka of district.

1.5 Soil Profile

The soil profile plays significant role in landuse planning and productivity of the crops grown in the area, it mainly covers from the top of the soil to weathered zone of the land. The upper part of the soil profile contributes black soil with both decomposed and non-decomposed biomass. This part of soil profile has better water holding capacity. This highly fertile part of soil profile. Down to this the weathered yellowish colored part of land which normally contribute soft murum of the profile. This part has lower water holding capacity and enhances either gravitational or horizontal flow of subsurface runoff.

Soil Type

Pune district possesses mainly three types of soils, viz. black-fertile, brown and mixed type. In western region of the district the soils are brown in color and less fertile. On the other hand the eastern part has comparatively deep soils with black color and are more fertile. The richest alluvial soil track found in the Valley of Bheema River. The rivers Velu, Ghod are left side of Bheema and Indrayani, Bhama, Mula-Mutha etc. are at right side. Each tahsil of the district have minimum one river . Therefore, the agro-climatic condition of district is favourable. Soils of Pune district are categorized in to 5 types viz., Black (45 percent), Red (5 percent), Alluvial (8 percent), Sandy (12 percent) and Sandy loams (30 percent).

The district forms part of Western Ghat and Deccan Plateau. Physiographically the district can be divided in to three distinct belts i.e. (1) The western belt stretching from 16 to 31 km east of Sahayadri- an extremely rugged country cut by deep valleys, divided and crossed by hill ranges. (2) The central belt extending for about 30 km east of western belt across the tract whose eastern belt is roughly marked by a line drawn from Pabal in the north to south up to Purandhar through Pune. In this belt a

Geomorphology

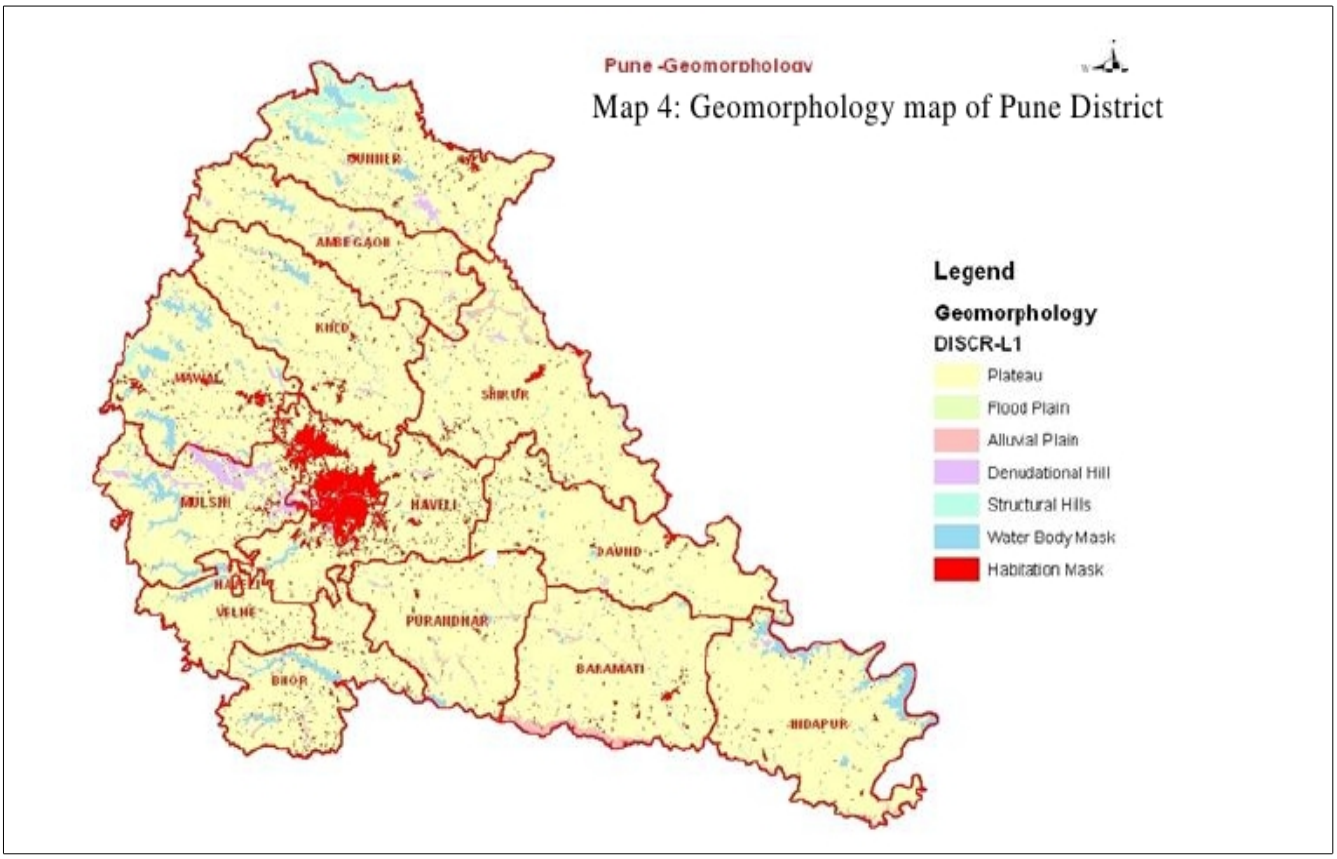


Figure 4 : Geomorphology map of Pune District

series of small hills stretch in to valleys and large spurs from Plateaux and (3) The eastern belt with a rolling topography and low hills sinking slowly in to the plains with relatively broader valleys. Therefore, the physiography of the district has given rise to four major characteristic land forms namely; (1) The hills and Ghats (2) The foot hills (3) the plateau and (4) the plains.

Drainage Network

The district has three major drainage systems namely (i) The Bhima-Ghod River System in northern, north-eastern and eastern part of which Bhima River has a total length of about 355 km and Ghod river

has a drainage of about 196 km. (ii) Mula-Mutha River System covering the central part and having total length of 242 km in the district. (iii) Nira River system covering south, south-east and eastern part and has total length of about 231 km in the district. The other important rivers that are flowing through the district are Bhima, Andhra, Karna, Shivganga, Pushpavati, Pauna and Indrayani. All the rivers have most semi-dendritic drainage pattern and the drainage density is quite high. Based on geo-morphological setting and drainage pattern the district is divided into 71 watersheds.

Soil Slope

Slope and Topography describes the shape and relief of the land. Slope is the percentage change of elevation over the certain distance. The angle of slope will affect the type, depth and moisture content of soil. It will also affect the rate of soil erosion. The degree to which a farm can be mechanized is influenced by slope.

Table 1.5.2 Soil Profile					**Source : Dept of Agriculture**
Name of the state : Maharashtra					
Name of the disrict :Pune					
		Land Slope			
S.N.	**Block**	**0-3 % (ha)**	**3-8% (ha)**	**8-25 % (ha)**	**> 25% (ha)**
1	Haveli	9568	25341	19518	1286
2	Maval	1155	13389	4254	618
3	Mulshi	6988	21491	38893	212
4	Bhor	365	1292	6133	85
5	Velha	626	4176	5700	1897
6	Juunar	11687	46667	29308	531
7	Khed	10267	44530	30219	1502
8	Ambegaon	4941	24694	28866	1061
9	Shirur	17350	86200	13260	0
10	Baramati	10212	62239	13851	0
11	Indapur	15574	79978	11022	0
12	Daund	17075	59843	12183	0
13	Purandar	13991	56731	18614	0
	Total	**119799**	**526571**	**231821**	**7192**

1.5.2 Soil Slope Map

The district has under taken soil survey on 885383ha of the districts of which 119799 ha area has 0-3% slope which is highly suitable for agriculture purpose. As indicated in above table 526571 ha area has 3-8% slope and is moderately suitable for agriculture whereas 231821 ha area with slope of 8-

25% is least suitable for agriculture. The remaining land of 7192 ha with slope greater than 25% is totally unfavorable for agriculture.

Blocks like Shirur, Baramati, Indapur, Daund have plain fields with the least variation of sloping conditions having slope lesser than 25% is merely used for agriculture.

Velhe block mainly comprised of hilly terrain and undulated area. Total 1897ha of this block has greater than 25% slope. This area shows maximum erosion due to higher degree of slope.

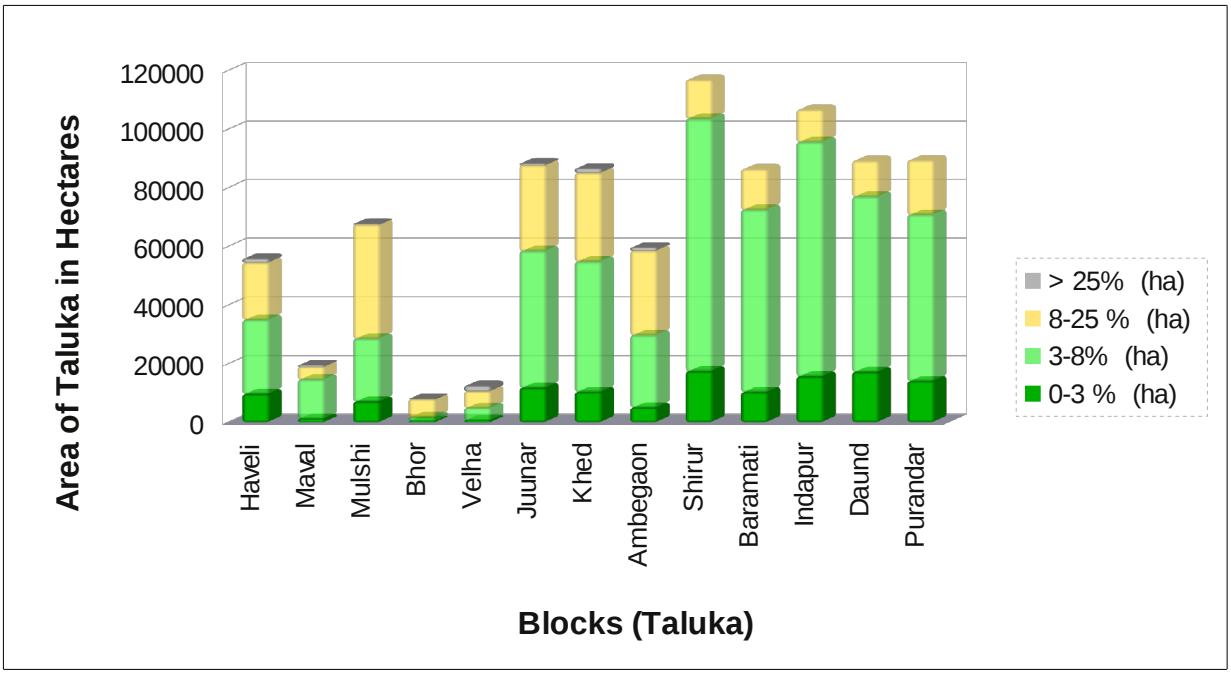
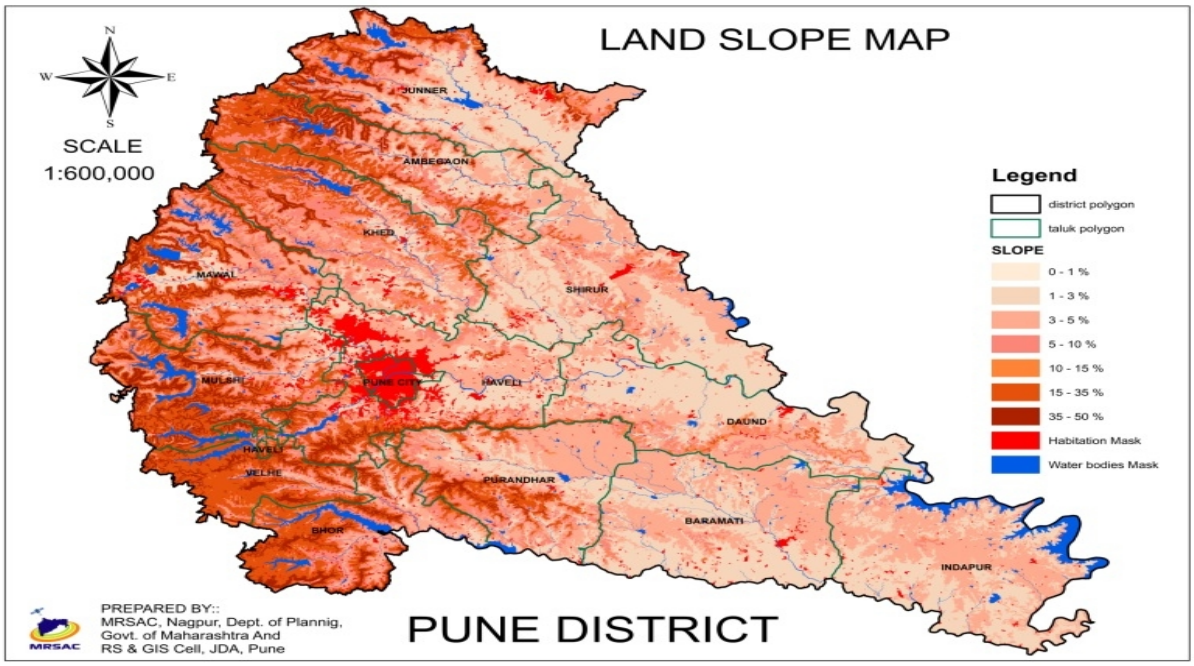


Figure 1.5.2 Soil Slope of Pune District



Slope Map

Soil Depth :

Soil depth is one of the important characteristic of soil profile. This is significant in categorizing the soils into different capabilities classes. Along with soil depth, soil slopes and texture also considered for soil classification. The department of agriculture has started soil survey and so far accomplished 102389 ha area. The findings of the said survey is presented in table no 5.2.1

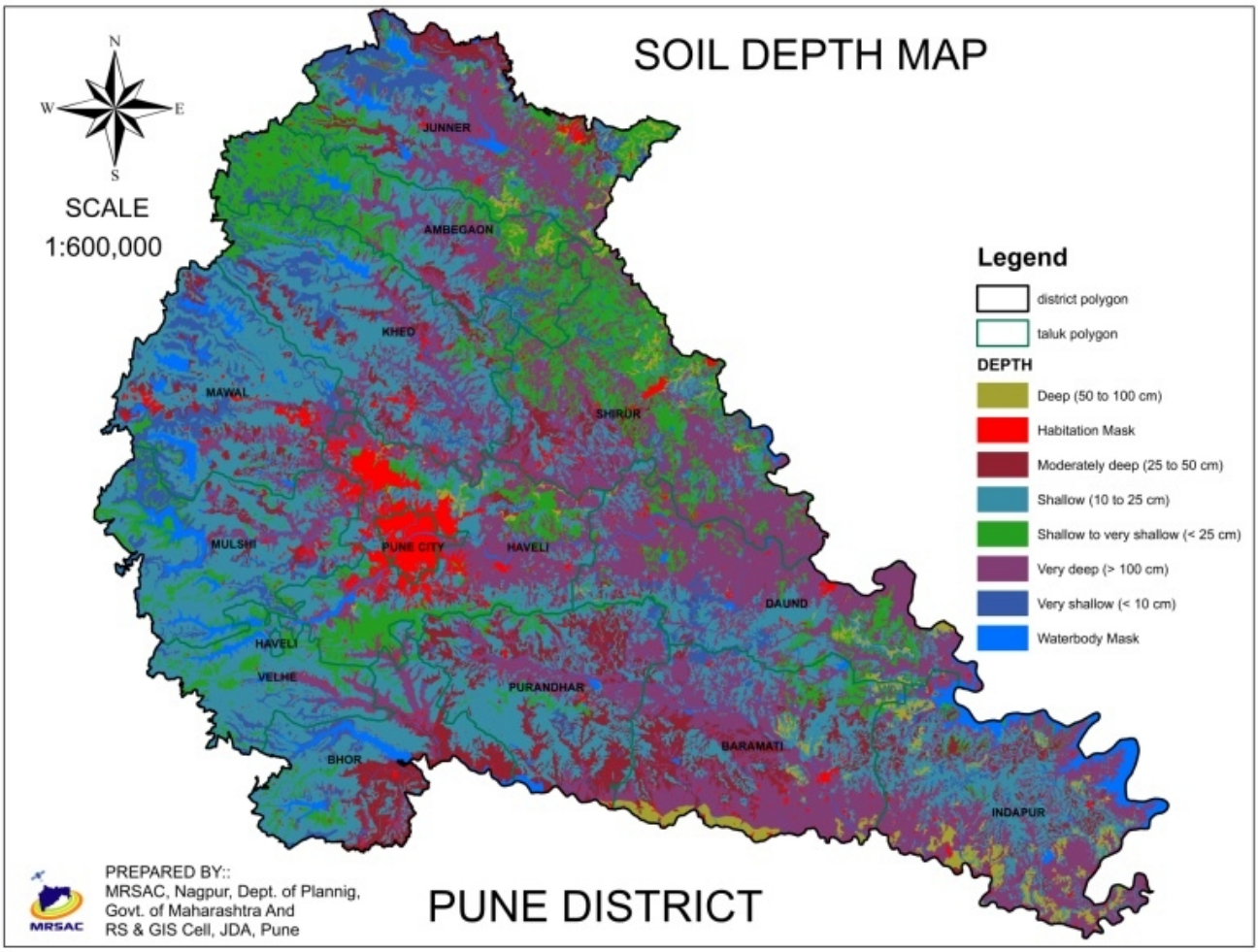


Figure 5.2.1 Soil Depth Map of Pune

There are total 8 capabilities classes viz class-I to class-VIII. Out of these 8 classes, first 3 classes i.e., class-I, II and III are suitable for farming. The class-IV lands are slopy and with lower soil depth. These soils are preferably kept for horticulture plantation. The class-v soils are salines or alkaline with drainage problems, these soils are used for salt tolerant crops. The class-vi and class-vii land are more slopy and normally used for plantation of fodder, fuel and timber trees. The class-viii land are exposed to natural vegetation and wild life century. that will justify required recurrent inputs without unacceptable risk to land resources.

The land capabilities survey carried out in the district with partial area covered not the whole district and the data compiled through the said survey till the present date is presented in the following table,

Table 1.5 Soil Profile								Source : Dept of Agriculture
Name of the state : Maharashtra								
Name of the disrict :Pune								
Soil Type		Land Slope					Depth (cm)	Area (Ha)
Sr.No	Block	Class	0-3 % (ha)	3-8% (ha)	8-25 % (ha)	> 25% (ha)		
1	Haveli	II	9568	0	0	0	45-90cm	
		III	0	10615	0	0	22.5-45cm	9925
		IV	0	14726	0	0	10-22.5cm	24539
		VI	0	0	11035	0	0-10cm	11681
		VII	0	0	3313	0		3881
		VIII	0	0	0	6456		5687
Total Area - 129574	Area surveyed - 55713		9568	25341	14348	6456	Total Area	55713
2	Maval	II	1155	0	0	0	45-90cm	
		III	0	3226	0	0	22.5-45cm	1768
		IV	0	10163	0	0	10-22.5cm	10231
		VI	0	0	3729	0	0-10cm	4614
		VII	0	0	525	0		1350
		VIII	0	0	0	618		1453
Total Area - 112796	Area surveyed - 19416		1155	13389	4254	618	Total Area	19416
3	Mulshi	II	6988	0	0	0	45-90cm	
		III	0	10181	0	0	22.5-45cm	391
		IV	0	11310		0	10-22.5cm	49221
		VI	0	0	38430	0	0-10cm	10905
		VII	0	0	463	0		4230
		VIII	0	0	0	212		2837
Total Area - 103350	Area surveyed - 67584		6988	21491	38893	212	Total Area	67584
4	Bhor	II	365	0	0	0	45-90cm	
		III	0	538	0	0	22.5-45cm	77
		IV	0	754	0	0	10-22.5cm	4469
		VI	0	0	4016	0	0-10cm	980
		VII	0	0	2117	0		2192
		VIII	0	0	0	85		157
Total Area - 87526	Area surveyed - 7875		365	1292	6133	85	Total Area	7875
5	Velha	II	626	0	0	0	45-90cm	
		III	0	2076	0	0	22.5-45cm	2102
		IV	0	2100	0	0	10-22.5cm	4069
		VI	0	0	4085	0	0-10cm	2856
		VII	0	0	1615	0		3057
		VIII	0	0	0	1897		315
Total Area - 52026	Area surveyed - 12399		626	4176	5700	1897	Total Area	12399
6	Khed	II	10267	0	0	0	45-90cm	
		III	0	18224	0	0	22.5-45cm	6818
		IV	0	26306	0	0	10-22.5cm	45389
		VI	0	0	15496	0	0-10cm	22123
		VII	0	0	14723	0		8973
		VIII	0	0	0	1502		3215
Total Area - 135907	Area surveyed - 86518		10267	44530	30219	1502	Total Area	86518
7	Juunar	II	11687	0	0	531	45-90cm	
		III	0	22498	0	0	22.5-45cm	8221
		IV	0	24169	0	0	10-22.5cm	43944
		VI	0	0	18258	0	0-10cm	23799
		VII	0	0	11050	0		6175
		VIII	0	0	0	531		7054
Total Area - 138211	Area surveyed - 88193		11687	46667	29308	531	Total Area	89193
8	Ambegaon	II	4941	0	0	0	45-90cm	
		III	0	10773	0	0	22.5-45cm	5364
		IV	0	13921	0	0	10-22.5cm	31036
		VI	0	0	18242	0	0-10cm	18074
		VII	0	0	10624	0		4418
		VIII	0	0	0	1061		670
Total Area - 102389	Area surveyed - 59562		4941	24694	28866	1061	Total Area	59562

Table 1.5

Table 1.5 Soil Profile								
Name of the state : Maharashtra								
Name of the disrict :Pune								
Soil Type		Land Slope					Depth (cm)	Area (Ha)
Sr.No	Block	Class	0-3 % (ha)	3-8% (ha)	8-25 % (ha)	> 25% (ha)		
9	Shirur	II	17350	0	0	0	45-90cm	
		III	0	32867	0	0	22.5-45cm	13096
		IV	0	53333	0	0	10-22.5cm	54432
		VI	0	0	11240	0	0-10cm	29870
		VII	0	0	2020	0		10808
		VIII	0	0	0	0		8604
Total Area - 158888	Area surveyed - 116810		17350	86200	13260	0	Total Area	116810
10	Baramati	II	10212	0	0	0	45-90cm	
		III	0	32725	0	0	22.5-45cm	12581
		IV	0	29514	0	0	10-22.5cm	29889
		VI	0	0	12281	0	0-10cm	20614
		VII	0	0	1570	0		5214
		VIII	0	0	0	0		17904
Total Area - 138496	Area surveyed - 86302		10212	62239	13851	0	Total Area	86302
11	Indapur	II	15574	0	0	0	45-90cm	
		III	0	43083	0	0	22.5-45cm	8621
		IV	0	36895	0	0	10-22.5cm	39339
		VI	0	0	10172	0	0-10cm	42179
		VII	0	0	850	0		11479
		VIII	0	0	0	0		4956
Total Area - 146666	Area surveyed - 106574		15574	79978	11022	0	Total Area	106574
12	Daund	II	17075	0	0	0	45-90cm	
		III	0	32354	0	0	22.5-45cm	11361
		IV	0	27489	0	0	10-22.5cm	25927
		VI	0	0	12183	0	0-10cm	20970
		VII	0	0	0	0		9353
		VIII	0	0	0	0		21490
Total Area - 125153	Area surveyed - 89101		17075	59843	12183	0	Total Area	89101
13	Purandar	II	13991	0	0	0	45-90cm	
		III	0	27629	0	0	22.5-45cm	11100
		IV	0	29102	0	0	10-22.5cm	34102
		VI	0	0	16748	0	0-10cm	24275
		VII	0	0	1866	0		13314
		VIII	0	0	0	0		6545
Total Area - 109658	Area surveyed - 89336		13991	56731	18614	0	Total Area	89336
Total Area	Surveyed Area	II	119799	0	0	531	45-90cm	
		III	0	246789	0	0	22.5-45cm	90425
		IV	0	279782	0	0	10-22.5cm	396687
		VI	0	0	175915	0	0-10cm	232940
		VII	0	0	51639	0		84444
		VIII	0	0	0	11459		80887
1540640	885383		119799	526571	227554	11459	Total Area	885383

Table 1.5 Soil Slope and Depth Map

The western part of Pune district comprising of Junner, Ambegaon, Maval, Mulshi, Velha, Bhor etc have more slopy and undulated lands as compare to Eastern part of district comprising of Indapur, Baramati, Daund, Shirur etc. Naturally the slope of land influences on intensity of runoff. Higher the slope greater be the runoff.

The excess runoff causes more erosion of fertile layer of soils which hampers the yield level of crops. Also the slope affects on infiltration rate of precipitation and consequently water retention capacity of the soils. The land with more slope has lesser soil retention capacity. The slopy land also enhances the subsurface runoff since such soils are sandy soils with greater porosity. Thus the slope of land manipulates erosion, water infiltration rate, water retention capacity, subsurface runoff etc key factors that impinge on crop productivity and production.

Here attempt would be required for minimizing slope of land by applying certain land measures like Continous contour Trenches (CCT) Terracing, Compartment bunding etc. This will help to check the runoff from the area and reduce the erosion of fertile soil of cultivable land. Also these measures will help to increase water holding capacity of arable lands. Such slopy area needs some drainage line measures like live check dam, gully control measures, loose boulder structures, earthen checks etc which help to check the velocity of water flowing through the drainage line and also resist the soil erosion.

The geomorphology of the district shows that the Western part of the district with greater slope will require soil erosion resisting interventions whereas the Eastern part needs water harvesting and moisture conservation measures on large scale.

1.7 Land Use Pattern

Land use involves management and modification of natural environment or wilderness into built environment such as settlements and semi-natural habitats such as arable fields, pastures forest etc., It also has been defined as "the total of arrangements, activities, and inputs that people undertake in a certain land cover type.

In the context of agriculture the landuse pattern is regrouped in different classes namely Agriculture area and Non-Agricultural area which consist of Area under forest, wasteland and other miscellaneous use. Pune has having 13 blocks covering 1877 villages with total geographical area of 1562018 ha . It

has gross agricultural area of 1172951 ha and net sown area of 1042282 ha. The district comprises of forest on 171809 ha and wasteland of 168696 ha. The area under other use is 179231 ha. This area is mainly used for rural and urban habitation, industries, roads and area under rivers and nallas etc

1.7 Land use pattern										Source : Dept of Agriculture
Name of the state : Maharashtra										
Name of the disrict : Pune										
Sr.no.	Name of the Block	No of the villages	Total Geogaphical area (ha)	Area under agriculture				Area in Ha		
				Gross cropped area (1)	Net sown area (2)	Area sown more than once (1-2)	Cropping Intensity (%)	Area under Forest (ha)	Area under Wasteland	Area under other use
1	Haveli	123	145617	70935	68310	2625	104	8099	29170	40038
2	Maval	187	113135	33964	33579	385	101	32736	7926	38894
3	Mulshi	145	103931	41036	40000	1036	103	16726	17676	29529
4	Bhor	195	89234	43624	42320	1304	103	12260	17375	17279
5	Velha	130	49955	19781	19469	312	102	10796	4863	14827
6	Juunar	183	138452	111126	105763	5363	105	20876	9855	1958
7	Khed	186	137354	107947	103480	4467	104	20079	9530	4265
8	Ambegaon	143	104275	81604	71586	10018	114	24300	6707	1682
9	Shirur	115	155727	143948	132407	11541	109	6019	15617	1684
10	Baramati	116	138248	136970	112574	24396	122	4929	13665	7080
11	Indapur	143	146791	146547	118000	28547	124	7864	8768	12159
12	Daund	102	128986	127469	103700	23769	123	4904	19336	1046
13	Purandar	108	110313	108000	91094	16906	119	2221	8208	8790
	Total	**1877**	**1562018**	**1172951**	**1042282**	**130669**	**113**	**171809**	**168696**	**179231**

Table 1.7 Landuse Pattern

The above table shows that Shirur block has highest geographical area of 155727ha whereas the Velhe has least geographical area of 49955 ha. Total 146547 ha area of Indapur block seems to be under

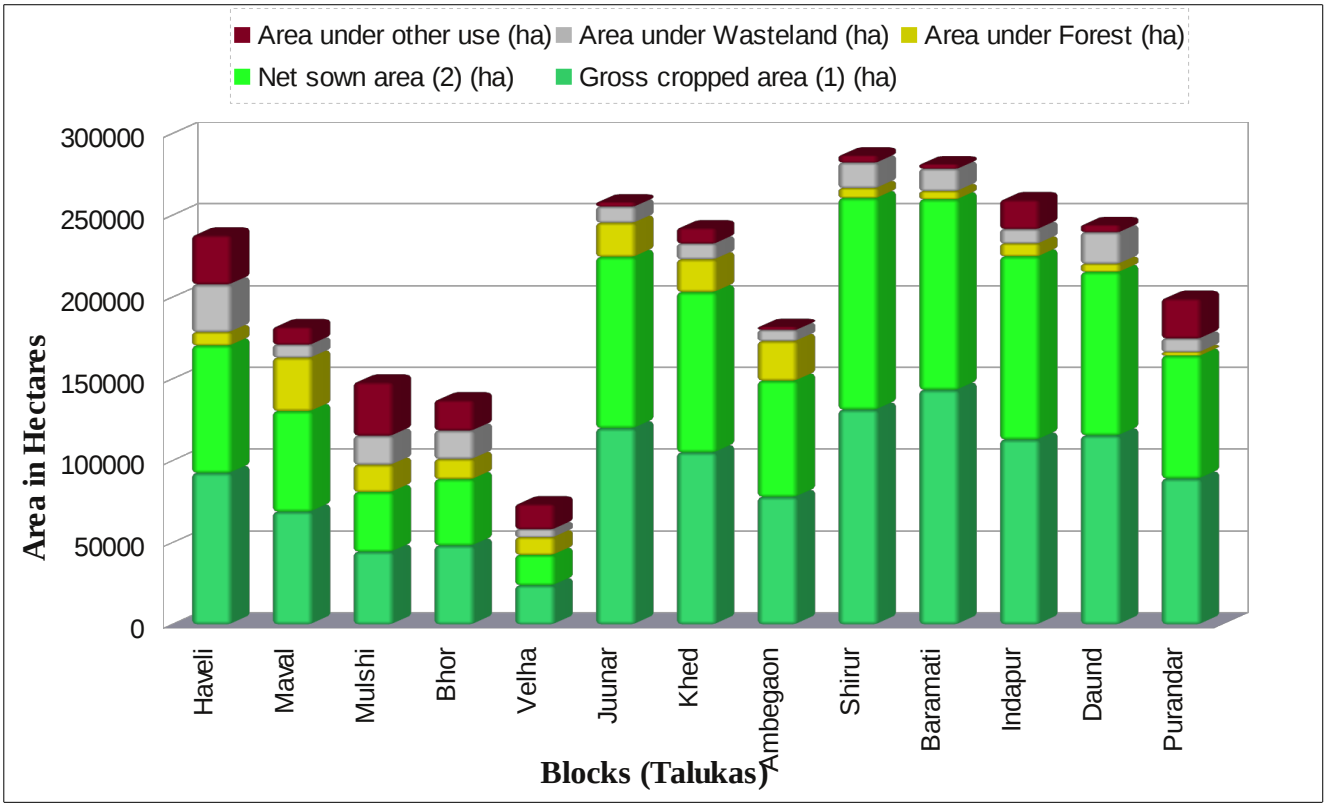


Figure 1.7 Landuse Pattern

gross cropped area which is highest in the district whereas 19781 ha of Velha block is under gross cropped area. This is lowest gross-cropped area of among all 13 blocks of the district. The net sown area of the district is 1042282 ha of which maximum i.e., 132407 ha seen Shirur block. The average cropping intensity of the district seems to be 113 percent and it is maximum in Indapur 124, Daund 123 and Baramati 122 etc. The least cropping intensity is in Maval block and is 101 percent.

Cropping intensity refers to raising of a number of crops from the same field during one agriculture year. It can be expressed as

Cropping intensity = (Gross cropped area / Net sown area) x 100

Thus, higher cropping intensity means that a higher portion of the net area is being cropped more than once during one agricultural year. This also implies higher productivity per unit of arable land during one agricultural year.

The cropping intensity is highest in Velhe taluka since the horticulture is more compared to other crops which is why the Velha is having more intensity, since horticulture gives bi-seasonal output for the same sown area.

The district occupies total 171809 ha forest. This mainly covered western part i.e., Shayadri ranges of the district. The blocks like Maval (32736 ha), Junnar (20876 ha), Khed (20079 ha), Ambegaon (24300 ha) have more than 20,000 ha forest whereas the blocks Mulshi (16726 ha), Bhor (12260 ha) and Velha(10796 ha) consist of more than 10000 ha forest. The Purandar has least forest area of 2221 ha.

The district consist of remarkable area under waste land i.e., 168696 ha of which maximum is seen maximum in Haveli block (29170 ha) and minimum 4863 ha in Velha block. Out of total 179231 ha area under other use, maximum 40038 ha and 38894 ha are seen in Haveli and Maval blocks of the district respectively. This is followed by Mulshi block with 29529 ha. All these 3 blocks comparizes of big urban habitations like Pune and Pimpri Chinchawad and major industrial belt of the district.

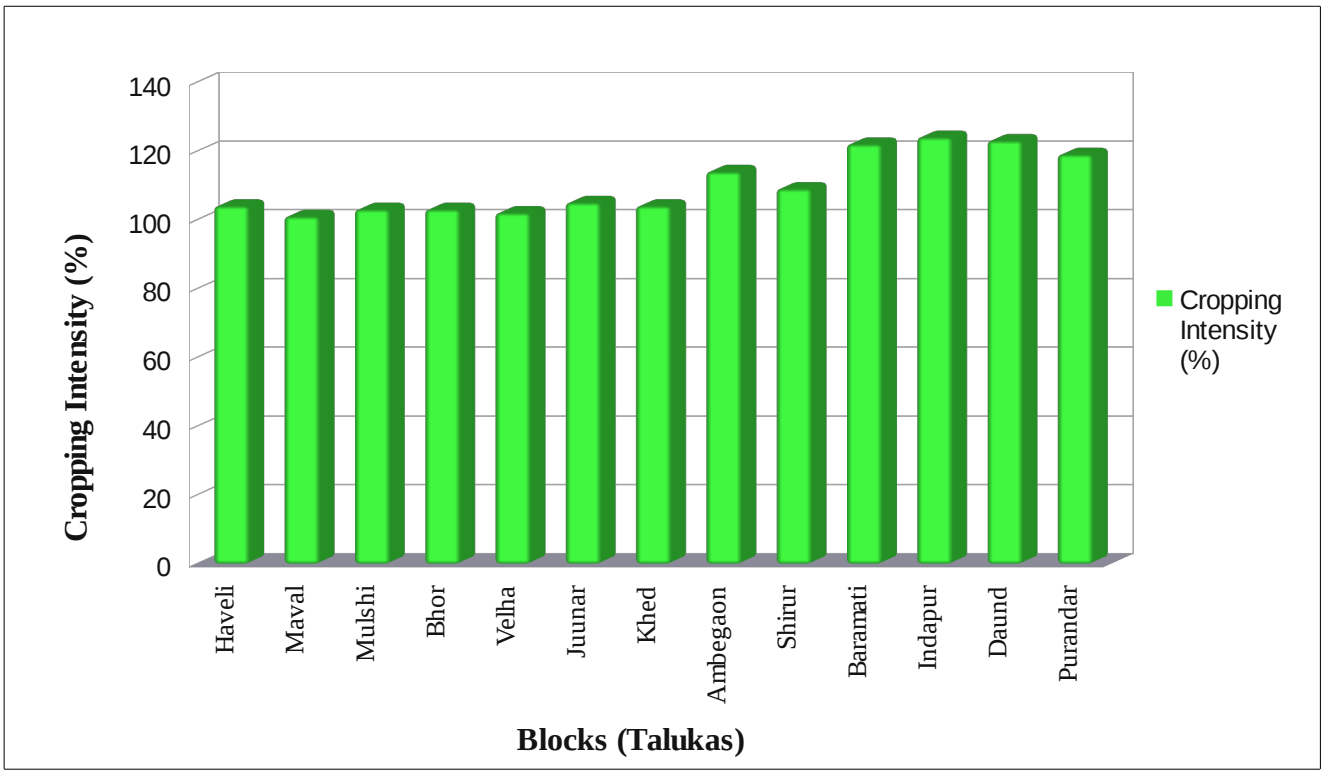


Figure 1.7 Cropping Intensity Pune District

The average cropping intensity of the district seems to be 113 whereas the intensity of the state Maharashtra and India is 127% and 130% respectively. In case of India it shows great spatial variations with 'higher levels in northern plains. Punjab has the highest cropping intensity of 176 per

cent, followed by Himachal Pradesh (169 per cent), West Bengal (157 per cent), Haryana (145 per cent) and Uttar Pradesh (143 per cent). The intensity is low in dry, rainfed regions of Rajasthan, Gujarat, Maharashtra and Karnataka (110-125 per cent).

Here soil and water conservation measures plays significant role in enhancing cropping intensity of the area. The proposed action plan of the district emphasizes on typology of soil and water conservation works which will support to increase cropping intensity of the district.

Chapter-II: District Water Profile

2.1 Area Wise, Crop-Wise irrigation Status:

The maximum area of Pune district is categorized as scarcity zone and agriculture is dependent mainly on monsoons. Hence rabbi crops dominate the cropping pattern. The area under Rabi crops is 60 per cent, while that under Kharif crops is 26 percent and 11 percent area is under sugarcane. The area under summer crops is negligible while area under fruit and vegetables crops is approximately 3 and 10 percent respectively.

The district comprises of three cropping seasons, which includes Kharif, rabi and summer/annual.
Of which the Kharif season begins in June or July and ends in September or October whereas Rabbi starts from September or October and ends in February or March.
The area under Rabi crops is dominated by food grains and is under cereals. The main food-grain crop in the Rabi season is Jowar and 60 percent of area under rabi cereals is under Rabi Jowar and 19 percent of area is under Wheat crop. The main pulse crop is gram which constitutes 12 percent of area in Rabi crops. Rabi oil-seeds are negligible.

The cropping pattern of the district consists of four major types.

A) Paddy based
B) Bajara Based
C) Sugarcane belt
D) Fruits and vegetables

The Western part of the comprising of Junner, Ambegaon, Mulshi, Maval, Velha and Bhor especially during Kharif season has paddy as a predominant crop. This part of the district has major kharip agriculture season and it also grows types of vegetables too. The part of the district covering blocks Shirur, Daund, Purandar, Eastern part of Bhor and Velha, Baramati, Indapur etc has Bajara as a main Kharip crop. As mentioned above around 10 percent area of the district grows sugarcane across the year. This area mainly covers blocks like Indapur, Daund, Baramati, Shirur,Khed, Junner, Ambegaon etc. This belt has assured source of irrigation and hence go for sugarcane cultivation. There are some pockets in the district which are famous for vegetable and fruit production. It mainly comprises of Junner, Narayangaon, Khed, Shirur, Daund, Baramati, Indapur etc. The vegetable and sugarcane

producing belt of the district is placed along the main river basin of Bhima. This belt has also very good access to markets like Pune, Mumbai, Nashik, Ahmednagar etc.

The district grows wide range of crops viz cereals, course cereals, pulses, oil seeds, fruit and vegetables etc across all seasons of the year. The crop category and types of crop grown in the district is presented below.

Sr. No.	Crop Category	Type of Crops
1	Cereals	Wheat, paddy
2	Course Cereals	sorghum, pearl millet, maize etc
3	Pulses	pigeon pea, *green gram (moong)*, *udid*, gram etc
4	Oil Seeds	Sunflower, ground nut etc
5	Fiber	cotton
6	Vegetables	tomato, onion, potato, okra, cabbage, cauliflower, peas and beans, cucurbits, leafy vegetables like coriander, Fenugreek etc.
7	Fruits	mango, banana, grapes, sapota (*Chiku)* and pomegranate etc.
8	Other	Sugarcane

The cereals mainly grown in the district are wheat and paddy. The blocks like Mulshi, Maval, Bhor, Velha, Junner, Khed etc grow cereals on large scale in Kharif season whereas blocks like Purandar, Daund, Baramati, Indapur grow it in Rabi season. During summer none of the block go for cereals. In case of course cereals the district grows bajara, sorghum, maize etc crops. The Junner, Khed, Shirur, Ambegaon, Haveli occurred to be highest grower blocks of course cereals in Kharif season and Indapur, Shirur, Baramati, Daund, Purandar are frontier in cultivating course cereals in Rabi season. Area under course cereals during summer season is negligible.

The pulses are grown in Shirur, Junner, Purandar, Baramati, Indapur, Daund etc blocks on large scale in both Kharif and Rabi seasons. During summer season none of the block cultivates pulses. The key pulse crops grown in the district are pigeon pea, *green gram (moong), udid*, bengal gram etc

The Junner, Ambegaon, Shirur, Khed, Indapur etc are major vegetable growing blocks in the district. These blocks take up vegetables in both Kharif and Rabi seasons. The sugarcane is also cultivated on substantial area in the district and blocks like Indapur, Baramati, Daund, Shirur are the prominent areas in sugarcane cultivation.

2.1 Area-wise Crop-wise irrigation Status															Source: Dept of Agriculture
Name of the State : Maharashtra															
Name of the District : Pune															
Name of the Block:															
Crop Type	Kharif (Area in ha)			Rabi (Area in ha)			Summer Crop (Area in ha)			Total (Area in ha)			Horticulture & Plantation Crop (Area in ha)		
	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total
A) Cereals	10471	53822	64293	39135	27255	66390				49606	81077	130683			
B) Coarse Cereals	26092	71757	97849	76779	365551	442330	1240	0	1240	104111	437308	541419			
C) Pulses	6864	14462	21326	20199	42961	63160				27063	57423	84486			
D) Oil Seeds	9746	36602	46348	2815	5925	8740	5220	0	5220	17781	42527	60308			
E) Fibre	0	60	60		0					0	60	60			
F) Any Other Crops					0		113497	0	113497	113497	0	113497	62995	48834	111829
Total	**53173**	**176703**	**229876**	**138928**	**441692**	**580620**	**119957**	**0**	**119957**	**312058**	**618395**	**930453**	**62995**	**48834**	**111829**

Table 2.1 Area-wise Cropwise Irrigation Status

The cereals, pulses and oil seed crops are mainly depending on precipitation and hence are rainfed crops. Sometimes these crops are supported with protective irrigation during dry spell of the season. The crops like sugarcane, vegetables and fruits are provided with regular watering during its life cycle. The average frequency of irrigations is 10 to 12 days. However, it comes down to 6 to 8 days during summer season. The fruit crops especially in its early age and then during flowering and fruiting stages of productive age are supplied with frequent watering. The district uses both ground water and surface water sources for irrigation.

The overall picture of the district shows that out of total 1042282 ha cultivated area 375053 ha i.e. 35.98 % is under irrigated crops whereas 667229 ha i.e. 64.02 % is grown on rain water. Of this 62965 ha area is under irrigated horticultural crops or sugarcane and 48834 ha area is under rain-fed horticulture.

Out of total 618395 ha seasonal rain-fed area highest area i.e. 441692 (71.42%) seen during Rabi season followed by Kharif season with 176703 ha (28.58%) area. The summer season does not grow any rain-fed crop in the district. In case of total 312058 ha seasonal irrigated crops maximum 138928 ha (44.52%) are seen in Rabi season, followed by 119957 ha (38.44%) and 53173 (17.03%) in Summer and Kharif seasons respectively.

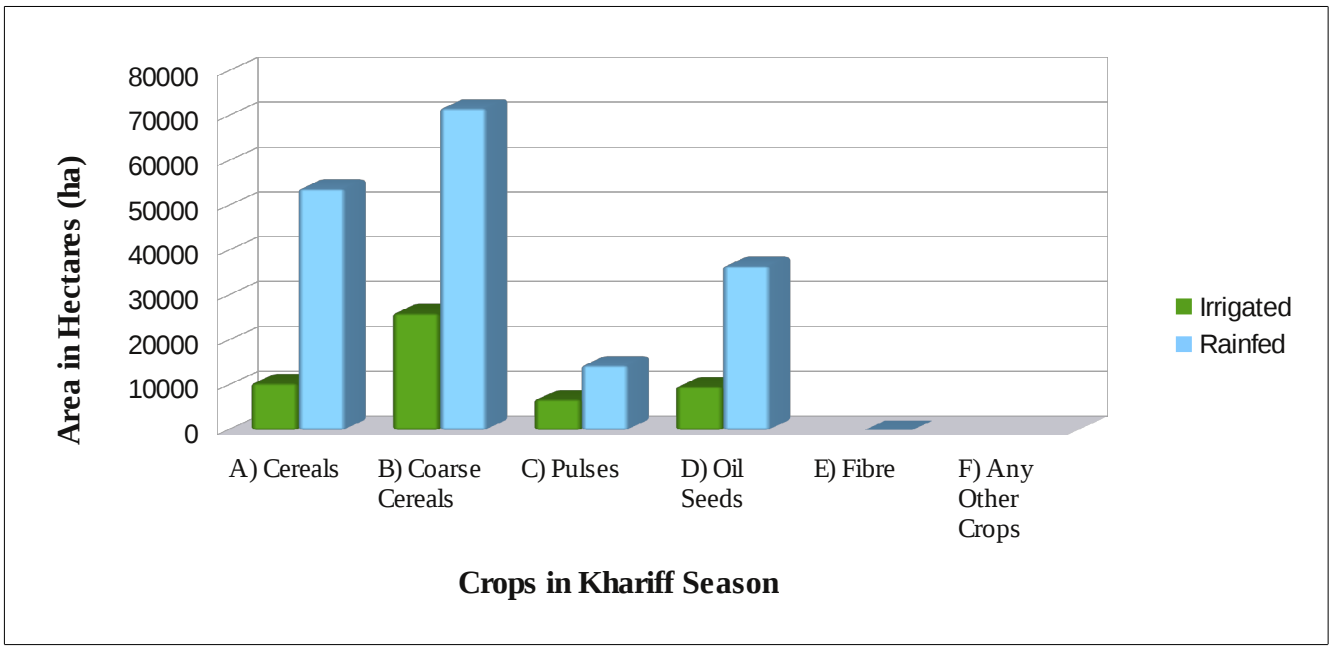


Figure: 2.1 Cropwise Irrigation Distribution - Khariff

In Rabi season out of total 138928 ha irrigated area maximum i.e.76779 ha (52.26%) area is seen under course cereals whereas only 2815 ha (2.03%) area is under oil seed crops. The proportion of irrigated cereals and pulses in Rabi season found to be 39135 ha (28.17%) and 20199 ha (14.53%) respectively.

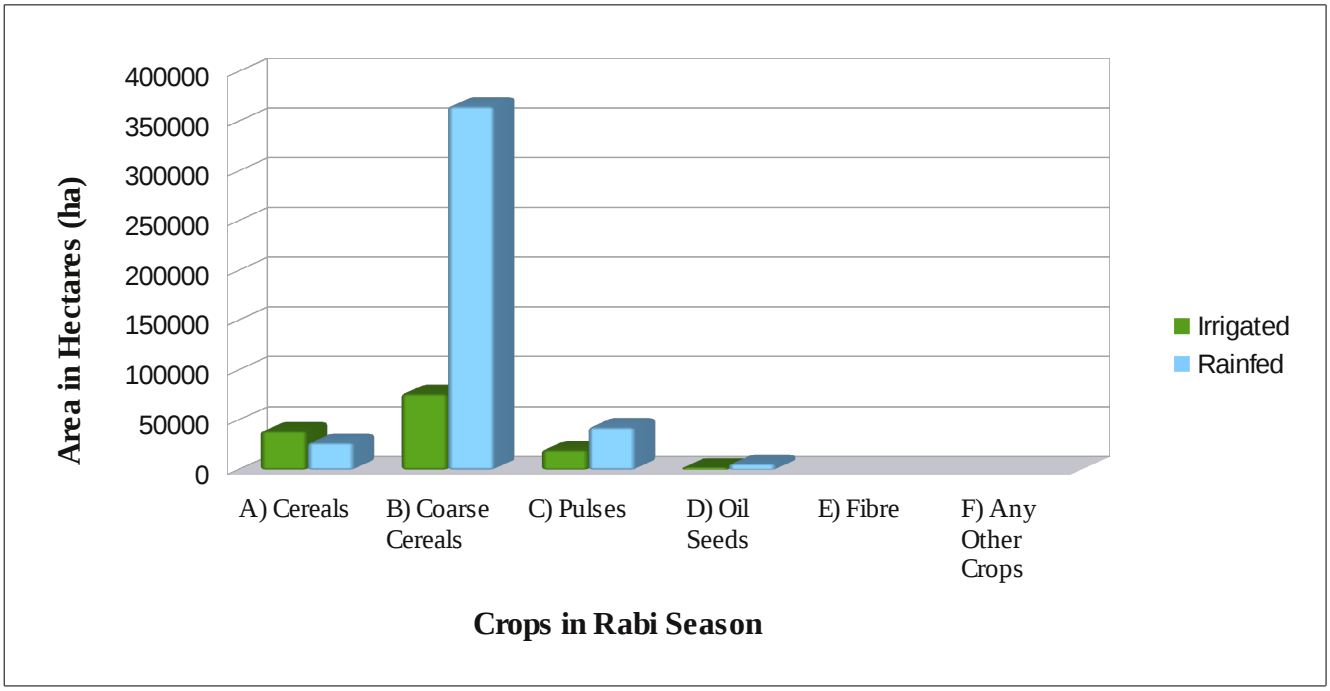


Figure : 2.1.1 Cropwise Irrigation Distribution – Rabi

Similarly in Summer season out of total 119957 ha irrigated area 113497 ha (94,61%) area seems to be under other crops like sugarcane followed by only 5220 ha (4.35%) and 1240 ha (1.03%) area is under

oil seed crops and course cereals respectively. In Kharif season out of total 53173 ha irrigated area maximum i.e.26092 ha (49.07%) area is seen under course cereals whereas only 6864 ha (12.91%) area is under pulses. The proportion of irrigated cereals and oil seed crops in Kharif season found to be 10471 ha (19.69%) and 9746 ha (18.32%) respectively.

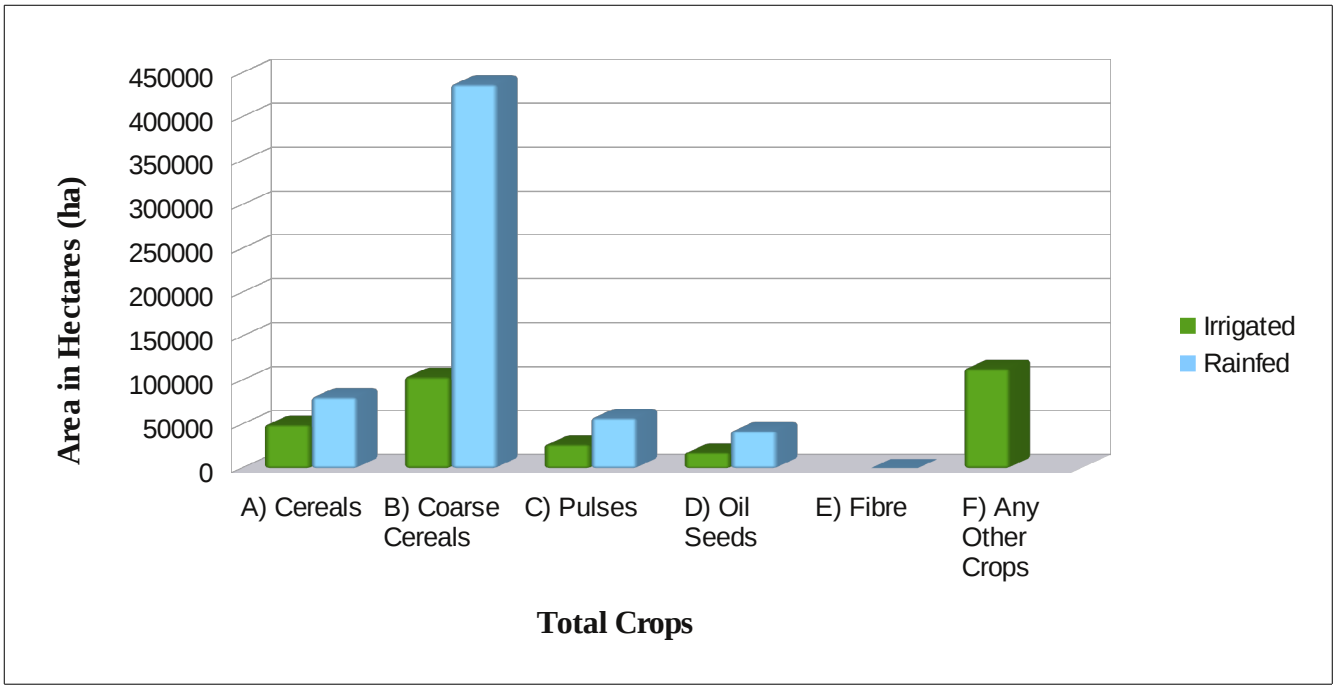


Figure : 2.1.2 Cropwise Irrigation Distribution - Total

2.2 Production and Productivity of Major Crops:

The Agricultural productivity is a ratio of agricultural outputs to agricultural inputs. The productivity of the district is mainly significant for district in providing more food, increasing growth, income distribution, savings etc and reducing migration etc. The adoption level of new technologies by farmers helps to enhance the productivity and production of the district.

The agriculture production is nothing but the technical relationship that transforms inputs (resources) into output (commodities). Therefore optimum utilization of input by scientific manner results into better crop production. One of the key factors that influence the crop productivity and production is irrigation. It is experienced that provision of irrigations during critical stages of crop growth enhances approximately doubled yield of agricultural crops.

Cost of cultivation of any crop is the sum total of several components on which the farmer has to spent the cost from land preparation to harvesting of the crop. These costs incurred on farm can be classified as cash cost or non-cash cost. Cash Costs are the costs for which farmer spends money for acquisition

of material inputs like seeds, fertilizer, chemicals or labour (hired) etc. On the other hand, non-cash costs are attributable to items of cost, which do not require spending money. These may be items of cost like family labour, payments made in kind, home grown seeds, manure etc, exchange labour, depreciation, interest on operating capital etc.

2.2 Production and Productivity of major Crops												
Name of the State : Maharashtra												
Name of the District :Pune												
Source : Dept of Agriculture, Agriculture Statistics of State Agristat												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	53822	809483	1504	67000	10471	230781	2204	97427	1040264	1854	82214
	Coarse Cereals	71757	798655	1113	45337.75	26092	359156	1376.5	60043	1157812	1245	52690
	Pules	14462	82542	570.75	38558.25	6864	39674	578	55337	122216	574	46948
	Oil Seeds	36602	240962	658.33	37686.83	9746	100871	1035	55908	341833	847	46797
	Cotton	60	307	511	69398	0	0	0	0	307	256	34699
	Vegetables	19387	14540250	75000	70625	20105	18094500	90000	85000	32634750	82500	77813
B) Rabi	Cereals	27255	408007	1497	62084	39135	776438	1984	67084	1184446	1741	64584
	Coarse Cereals	365551	3202227	876	30592	76779	1676853	2184	65256	4879080	1530	47924
	Pules	42961	252611	588	37727	20199	133111	659	52770	385722	624	45249
	Oil Seeds	5925	24885	420	28038	2815	14920	530	35200	39805	475	31619
	Vegetables	16603	12950340	78000	75625	23913	21999960	92000	98000	34950300	85000	86813
C) Summer	Cereals	0	0	0	0		0			0	0	0
	Coarse Cereals	0	0	0	0	1240	11482	926	40000	11482	1240	11482
	Pules	0	0	0	0		0			0	0	0
	Oil Seeds	0	0	0	0	5220	50112	960	60000	50112	5220	50112
	Sugarcane	0	0	0	0	113497	116902	103	120540	116902	113497	116902
	Vegetables	0	0	0	0	1498	1213380	81000	90000	1213380	1498	1213380
	Fruit crop	12844	7669152	59710	98000	17479	15072142	86230	313520	22741294	72970	205760
Total	Cereals	81077	1217490	1501	64542	49606	1007219	2094	82256	2224709	1797	73399
	Coarse Cereals	437308	4000882	995	37965	104111	2047492	1496	55100	6048374	1245	46532
	Pules	57423	335153	579	38143	27063	172785	619	54054	507938	599	46098
	Oil Seeds	42527	265847	539	32862	17781	165903	842	50369	431750	690	41616
	Cotton	60	307	511	69398	0	0	0	0	307	511	69398
	Sugarcane	0	0	0	0	113497	116902	103	120540	116902	103	120540
	Vegetables	35990	27490590	76500	73125	45516	41307840	87667	91000	68798430	82083	82063
	Fruit crop	12844	7669152.4	59710	98000	17479	15072142	86230	313520	22741294	72970	205760

Note : Productivity and Prodcution of Sugarcane is mentioned in tonnes

Table 2.2 Production and Productivity of Major Crop

The district grows both rainfed and irrigated crops in all three agriculture seasons. The productivity, production and cost of cultivation of cultivated crops varies from crop to crop and also season to season, since some cost components are crop and season specific. The cost of cultivation of irrigated crops is greater than the rainfed crops as the rainfed crops need limited inputs and resources that comprise the cost factor.

The above table indicates that the productivity, production and cost of cultivation differs from crop to crop category and even season to season. The average productivity of crops ranges from 511 kg/ha (of cotton) to 82083 kg/ha (of vegetables). The average crop productivity of cereals, course cereals, pulses ranges for half quintal to around two quintal per ha. The productivity of rainfed cereals and course cereals is 1501 kg/ha and 995 kg/ha respectively and that of irrigated cereals and course cereals is 2094 kg/ha and 1496 kg/ha respectively. It means the productivity of irrigated cereals and course cereals is 140% and 150% more than rainfed cereals and course cereals respectively. The productivity of rainfed and irrigated pulses does not show much difference. It is more by just 39 kg/ha than rainfed pulse crop. The difference between rainfed and irrigated vegetables 11167 kg/ha i.e. the productivity of rainfed vegetables is 76500 kg/ha and it is 87667 kg/ha in case of irrigated vegetables. It means productivity of irrigated vegetables is 115% more than rainfed vegetables.

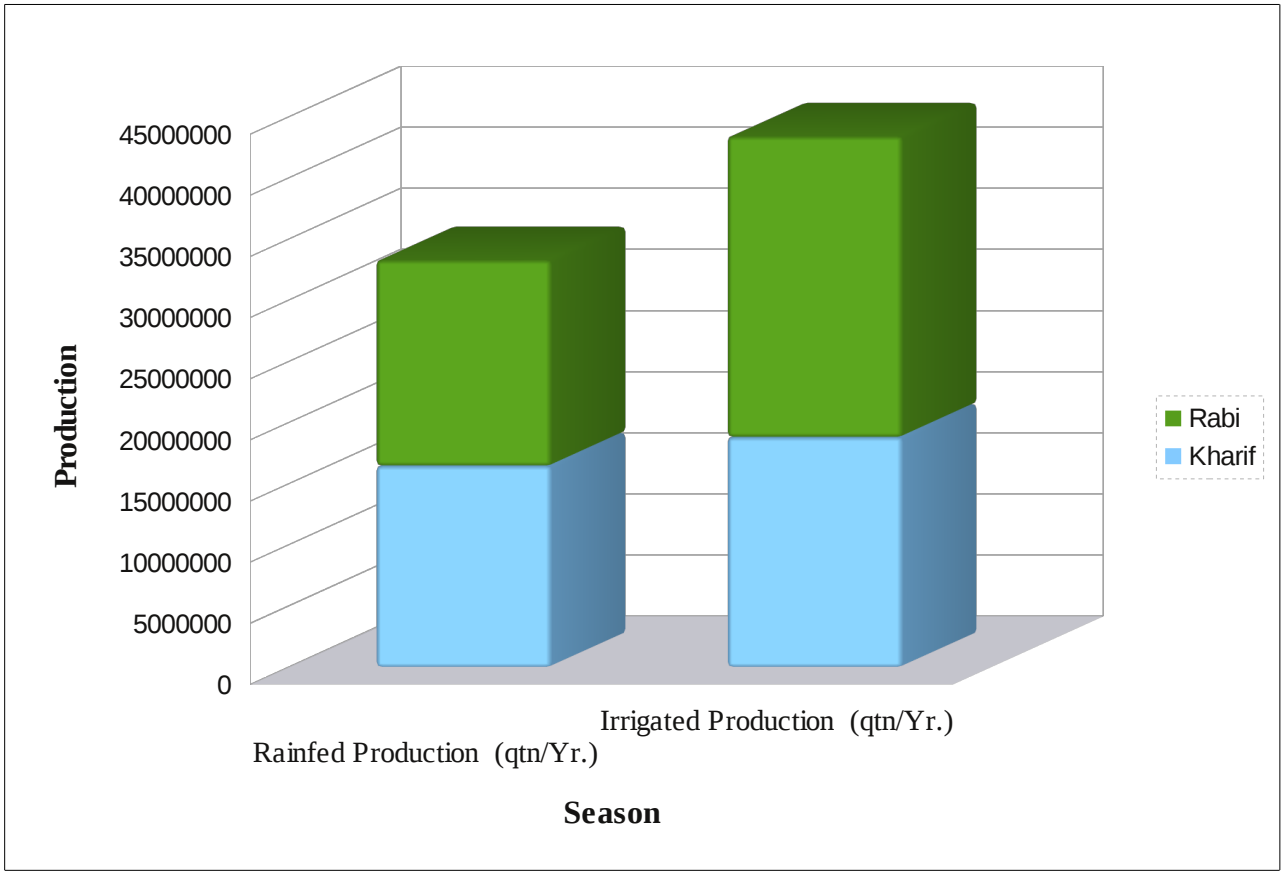


Figure 2.2 Production of Rainfed and Irrigated crops

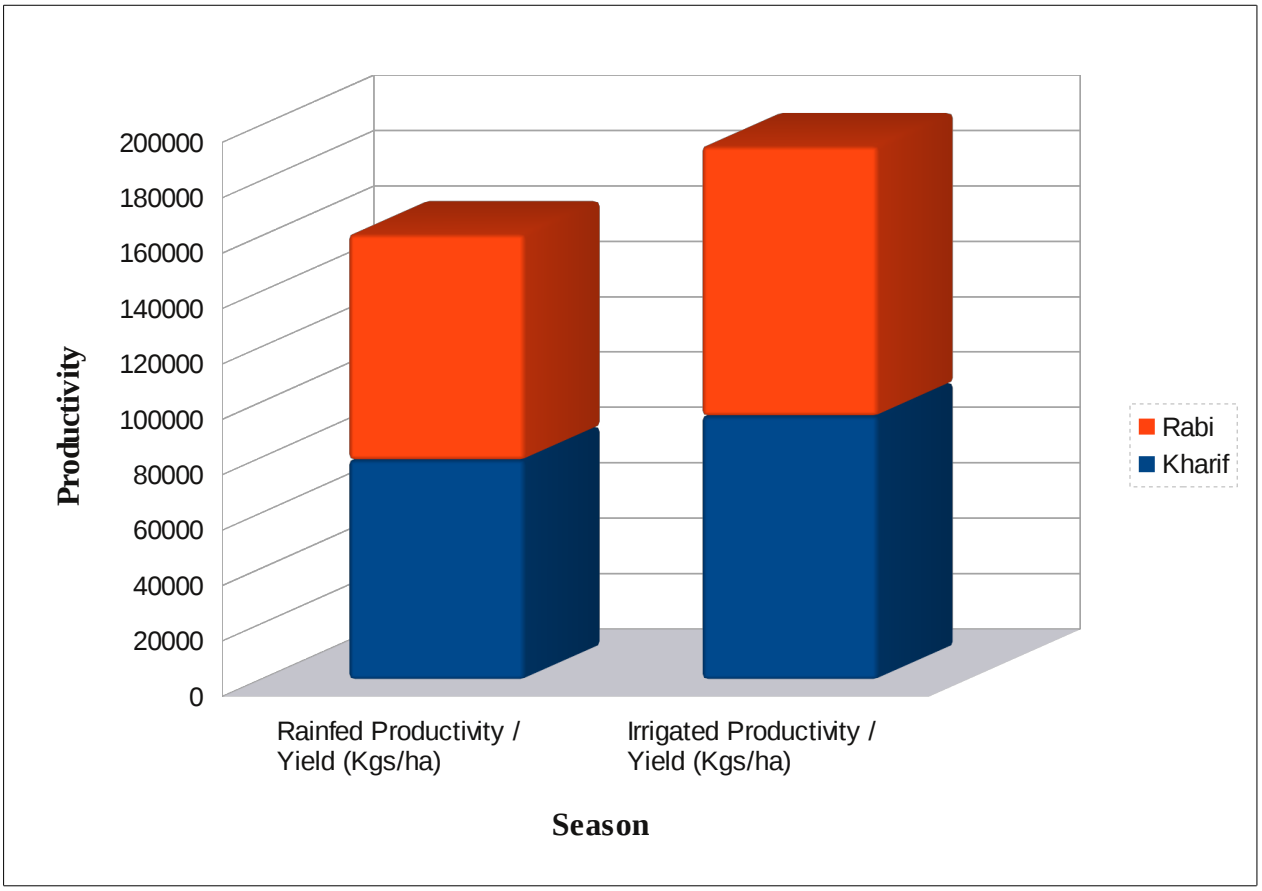


Figure 2.1 Productivity of Rainfed and Irrigated crops

The Total agriculture production of the district is estimated to be 123610998 quintal in which vegetables contribute highest share of 68798430 quintals per year followed by fruits i.e. 22741294 quintals. On the other side the district produces least quantity of cotton and is only 307 quintal per year. Total 8273084 quintals of cereals and course cereals and 939687 quintals of pulses and oil seeds are produced in the district.

The average crop cultivation costs varies from crop to crop. It is seen minimum in oil seeds i.e. Rs 41616/- per ha and maximum in fruit crops i.e. Rs 205760/- per ha. It is Rs 68399/- and Rs 46532/- per ha in case of cereals and course cereals respectively. The per ha cost of cultivation of pulse and oil seed crops is about Rs 46098/- and Rs 41616/-.

The cost also differs in rainfed and irrigated crops. The table shows the difference in cost of cultivation of rainfed and irrigated cereals and course cereals is Rs 27714/- and Rs 17135/- per Ha respectively. This means the cost of cultivation of irrigated cereals and course cereals is 151% and 145% more than rainfed cereals and course cereals.

2.3 Irrigation based classification:

The district consist of highest number of dams in the state with greater storage capacity. However now a days the district is facing acute shortage of water resource for all purposes viz domestic, agricutlure, industries etc. The reason behind may be increasing Urbanization, Industralization and rising area of water loving crops like sugarcane.

Gross irrigated area is nothing but total area under crops, irrigated once and/or more than once in a year. It is counted as many times as the number of times the areas are cropped and irrigated in a year. The net irrigated area means a particular crop area irrigated through any source once in a year.

The area under irrigated and rainfed crops of the district is tabulated in following table

2.3 Irrigation based Classification				**Source : Dept of Agriculture**
Name of the state : Maharashtra				
Name of the District : Pune				
Name of Block / Taluka	**Irrigated (area in ha)**		**Rainfed (area in ha)**	
	Gross irrigated area	**Net Irrigated area**	**Partially irrigated protective/protective irrigation**	**Un-irrigated**
Bhor	12567	9179	2748	24393
Purandar	22658	22547	7152	43438
Baramati	34422	29073	18154	73679
Indapur	59602	59581	26469	61470
Mulshi	20999	20977	2216	13343
Mawal	13767	13749	2314	7516
Khed	32010	21771	9845	63864
Haveli	28834	22247	14487	10420
Shirur	77980	69755	11656	74187
Velha	1774	1774	1145	12550
Ambegaon	25830	24480	8459	28719
Daund	54438	46601	24419	46722
Junnar	37771	33319	16479	61385
Total	**422652**	**375053**	**145543**	**521686**

Table 2.3 Irrigation based Classification

As indicated in the above table district has 422652ha gross-irrigated area of which maximum 59602 ha area belongs to Shirur 77980 ha followed by Indapur, 54438 ha Daund, 37771 ha Junnar block etc. The district comparises of total 375053 ha net irrigated area of which maximum in Shirur block i.e., 69755 ha and minimum in Velha block 1774 ha. The partially irrigated area of the district is 145543 ha of which maximum 26469 ha is in Indapur block. The total unirrigated area of the district is 528686 ha, the blocks like Shirur, Khed, Baramati, Indapur have maximum unirrigated area.

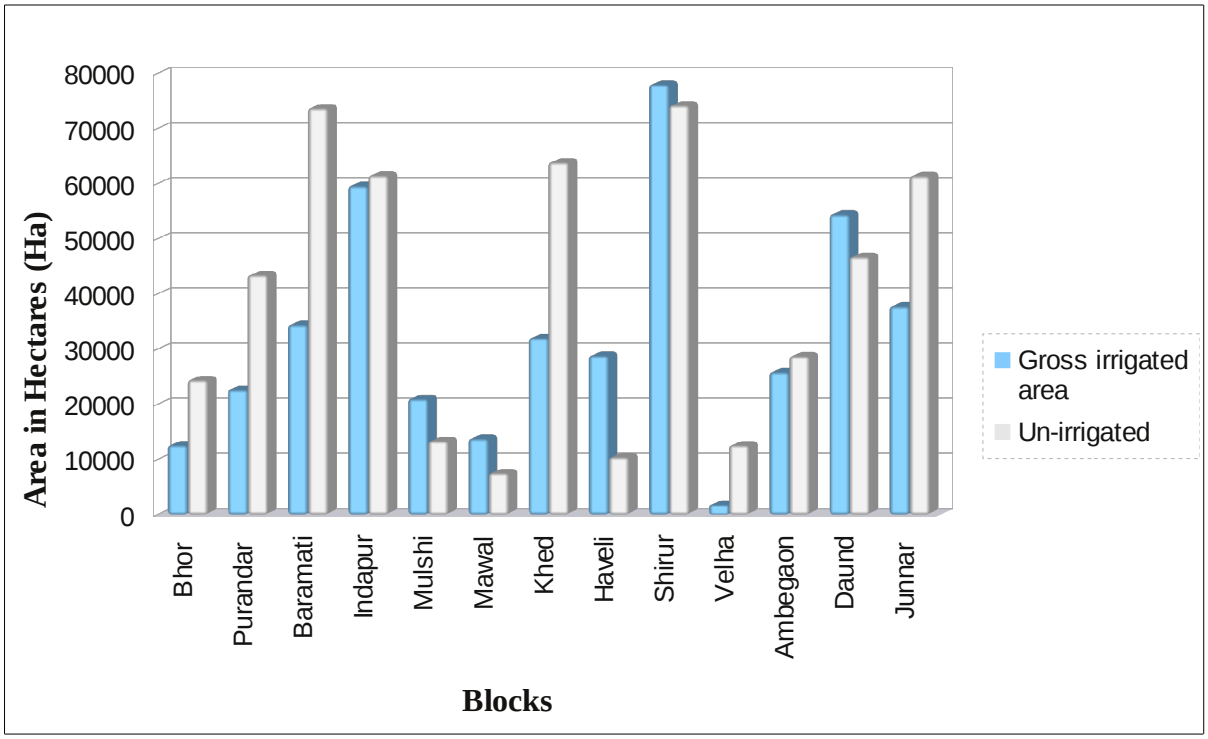


Figure : 2.3 Gross Irrigated Area v/s UN-irrigated Area

The district irrigation plan provides attention towards making the whole district sufficient from water requirement point of view. The rainfed area will emphasize on such interventions which will harness maximum rain water through water conservation and harvesting measures. The area with assured irrigation source will be promoted for protecting their water resources and for efficient use of available water resources by adopting micro-irrigation systems.

Chapter III: Water Availability

The Integrated Water Resource Development and Management Plan either for basin or for district is prepared for optimal use of available water resources. The main components are:

- Estimation of surface water
- Estimation of ground water
- Estimation of demand for various purposes
- Water budgeting of the district
- Strategic Plan for Integrated Water Resource Development and Management

In the Pune District water availability is very large as the district has substantial area with high rainfall. The district has also favorable site conditions for constructing the reservoirs and storages especially in western belt of the District. Therefore many major irrigation projects such as Bhatghar, Veer, Nira-Deoghar, Gunjavani in Nira sub-basin, Varasgaon, Panshet, Khadakwasla, Mulshi in Upper Bhima sub basin, Chaskaman and Ujani in Middle Bhima Basin and Dimbe, Manikdoh, Wadaj, Pimpalgaon Joge and Ghod in Kukadi basin have been constructed.

Sr. No.	Sub Basin	Reservoir
1	Neera	Bhatghar, Veer, Neera-Deoghar, Gunjavani
2	Upper Bhima	Panshet, Khadakwasla, Varasgaon Mulshi
	Middle Bhima	Chaskaman, Ujani
3	Kukadi	Dimbe, Manikdoh, Pimpalgaon Joge, Wadaj, Ghod

Moreover, Nira Left Bank Canal, Khadakwasla/Mutha Right Bank Canal, Chaskaman Canal and Kukadi Left and Right bank Canal System along with its distribution network have been constructed in Pune District. However, the water available and stored in these reservoirs and canal network system is not only meant for Pune but Pune, Satara, Solapur and Ahmednagar Districts. The inter distribution of available and stored water among these districts have been planned by the State on the basis of basin as unit. The water allocated to Pune District as a riparian right in integrated planning of these sub basins is binding and needs to be considered accordingly instead of total available and stored water in Pune District.

The overwhelming rural and urban population of the district to some extent dependent on groundwater for drinking purposes. It is observed that prior to the year 1972 the use of groundwater was relatively insignificant in the district. Subsequently due to frequent occurrence of droughts, limitations of the availability of surface water, development of low cost drilling devices,easy availability of institutional finance and energization etc led to proliferation of irrigation wells. Pune is one of the most well endowed District in the state in respect of rainfall. But, it may soon become a district where large parts of it face perennial water shortage, if urgent institutional, policy and technological initiatives are not taken to address both quantity and quality issues of groundwater. Most of the discussions at National and State level as well as local level are centered on demand-side projections, supply-side shortfalls and large scale investment estimates for resource development. The fact is that good governance and better management practices along with partnership with the community can bring better and sustainable benefits. Efforts are need to be taken to empower local communities to manage their own water resources and rebuild a new relationship with water in the society.

Groundwater is a natural resource with both ecological and economic value and is of vital importance for sustaining life, health and integrity of ecosystems. This resource is increasingly threatened by over-extraction which has insidious long-term effects. Scarcity and misuse of groundwater pose a serious threat to sustainable development and livelihood.

The availability of groundwater is extremely uneven, both in space, time and depth and so will be the case in future. The uneven distribution of groundwater in the district can be mainly attributed to highly heterogeneous lithology and regional variation of rainfall.

There area limitations on the availability of groundwater due to area occupied by hard rocks and variations in their basic characteristics, physiography and variability in rainfall etc.

In order to assess the availability of groundwater and to ensure maximum accuracy in groundwater estimates, the Central Government has, from time to time, appointed committees comprising groundwater experts and has laid down guidelines for this purpose. Estimates of groundwater in Maharashtra are being revised from time to time on the basis of these guidelines. The total demand for water from the groundwater domain is increasing day by day. The main reason for this is self reliance being experienced by users of groundwater. But, as this is leading

to inexorable withdrawal, and as the status regarding total availability of groundwater is of uncertain nature, it is imperative to give more serious thought and a new direction to groundwater planning and management.

While first major but adhoc attempt of estimating groundwater resources was made in 1973, the first scientific groundwater resource assessment of Maharashtra was made during 1979 on the basis of Overexploitation Committee Report. Since then various committees constituted by Government of India and Government of Maharashtra to estimate the groundwater resources have made the attempts from time to time. After due consideration of the limitations in the earlier methodologies, the Groundwater Estimation Committee (GoI), during 1997, proposed the revised methodology known as GEC 97. Little modifications have been suggested in this methodology by the R & D Advisory Committee of GoI.

3.1: Status of Water Availability

Surface Water

The surface irrigation in Pune District is provided through major, medium and minor irrigation projects through canal and distribution network. Moreover, surface irrigation is provided through minor irrigation tanks and K. T. Weirs. Across the streams surface irrigation is also provided through lift irrigation diversions from rivers and canals of major projects. Various water bodies such as village tanks, percolation tanks and farm tanks are also used for surface irrigation. Treated effluents received from STPs of Pune, Pimpri-Chinchwad corporation area is also available for surface irrigation. Moreover, untreated effluence from different municipalities is also available for surface irrigation.

The surface irrigation is provided in kharif, rabi and summer seasons. The kharif irrigation is provided as a protective one in absence of adequate rainfall. Rabi irrigation is provided predominantly in Pune district.

The surface irrigation as provided in Pune district through above various sources in different seasons have been estimated as BCM/Ha and these are shown in Table 3.1 (1).

3.1 Status of water availability					
					Source: Dept of Irrigation, GSDA
District: Pune					BCM Per ha.
Sr.No.	Sources	Kharif	Rabi	Summer	Total
1	Surface Irrigation as per project				
i)	Canal (Major and Medium Irrigation)	0.000003792	0.000002096	0.000001672	0.000002473
ii)	Minor irrigation Tanks and K.T.Weirs	0.000000154	0.000000992	0.000000174	0.000000656
iii)	Lift Irrigation /diversion	0.000000168	0.000000058	0.000000100	0.000000093
iv)	Various water bodies including rain water harvesting	0.000000000	0.000000000	0.000000000	0.000000000
v)	Treated effluent Received from STP	0.000000198	0.000000103	0.000000438	0.000000176
vi)	Untreated effulent	0.000000521	0.000000222	0.000000813	0.000000385
vii)	Perenial sources of water	0.000000000	0.000000000	0.000000000	0.000000000
	Total Surface Irrigation as per project	**0.000004834**	**0.000003471**	**0.000003198**	**0.000003783**
2	Ground Water				
i)	Open well	0.0000000110	0.0000000096	0.0000000196	0.0000000114
ii)	Deep/medium Bore well	0.0000000023	0.0000000020	0.0000000041	0.0000000024
	Total Ground Water	**0.0000000134**	**0.0000000116**	**0.0000000237**	**0.0000000138**
	Total	**0.0000048469**	**0.0000034823**	**0.0000032218**	**0.0000037970**

Table 3.1(a) Status of Water Availability

The highest water utilization rate for surface irrigation is through canals and distribution network and is 0.000002473 BCM/Ha. The next water utilization rate for surface irrigation is 0.000000656 BCM/Ha through minor irrigation tanks. The water utilization rate for surface irrigation through untreated and treated effluents are also substantial and are 0.000000385 BCM/ha and 0.000000176 BCM/Ha respectively, which is mainly because of Pune and Pimpri-Chinchwad metropolitan cities and industries. Water utilization rate for surface irrigation through lifts on rivers, streams and canals is 0.000000093 BCM/Ha as substantial area is at higher altitude than the canal and distribution network. Thus, the total water utilization rate as a surface

irrigation as whole is 0.000003783 BCM/Ha.

The water utilization rate through ground water sources through open wells is 0.0000000114 BCM/Ha. The water utilization rates through other ground water sources is 0.0000000024 BCM/Ha for medium and bore deep bore well. The availability of ground water is itself marginal as the terrain is mostly basaltic in origin and very low porosity and permeability.

The quantity of surface irrigation used through above sources and for each season has been also estimated and details are given Table 3.1 (a).

Status of Water Availability					**Source: Dept of Irrigation, GSDA**
Sr. No.	**Sources**	**Kharif**	**Rabi**	**Summer**	**Total BCM**
1	**Surface Irrigation as per project**				
i)	Canal (Major and Medium Irrigation)	1.021314200	1.302006200	0.253846378	2.577166778
ii)	Minor irrigation Tanks and K.T.Weirs	0.041598522	0.616164886	0.026385000	0.684148408
iii)	Lift Irrigation /diversion	0.045360000	0.035960000	0.015176742	0.096496742
iv)	Various water bodies including rain water harvesting	0.000000000	0.000000000	0.000000000	0.000000000
v)	Treated effluent Received from STP	0.053424000	0.064008000	0.066528000	0.183960000
vi)	Untreated effluent	0.140307159	0.137629374	0.123456784	0.401393317
vii)	Perennial sources of water	0.000000000	0.000000000	0.000000000	0.000000000
	Total Surface Irrigation as per project	**1.302003880**	**2.155768461**	**0.485392905**	**3.943165246**
2	**Ground Water**				
i)	Open well	0.0029750000	0.0059500000	0.0029750000	0.0119000000
ii)	Deep Bore well	0.0001664000	0.0003328000	0.0001664000	0.0006656000
iii)	Medium /Shallow Bore well	0.0004625000	0.0009250000	0.0004625000	0.0018500000
	Total Ground Water	**0.0036039000**	**0.0072078000**	**0.0036039000**	**0.0144156000**
	Total	**1.3056077803**	**2.1629762607**	**0.4889968048**	**3.9575808458**

Table 3.1(b)

The water utilization for the surface irrigation is mainly from canal and distribution network and it is 2.577166778 BCM for Pune District as a whole in a year. It is predominantly for Rabi season 1.302006200 BCM for crop production and in Kharif 1.021314200 BCM for protective irrigation in support of late monsoon. The summer season utilization 0.253846378 BCM is mostly for perennial crops, fruit gardens and vegetables.

The pattern for minor irrigation tanks for utilization is predominantly for Rabi season as 0.616164886 BCM as against Kharif and summer seasons are 0.041598522 BCM and 0.026385000 BCM respectively as a marginal one. The contribution from untreated and treated effluents from HTP is 0.401393317 BCM and 0.183960000 BCM respectively. The total surface water availability for irrigation in a year is estimated as 3.943165246 BCM with distribution as 2.155768461 BCM, 1.302003880 BCM and 0.485392905 BCM, predominantly in Rabi season for crop production, in Kharif for crop protection and in summer for perennial crops, fruit gardens and vegetables.

The total ground water availability as estimated is 0.0144156000 BCM which is mainly from open well 0.0119000000 BCM and 0.0006656000 BCM which is marginal as a terrain is mostly basaltic with low permeability.

3.2: Status of Ground Water Availability

3.2 Status of Ground water Availability					Source:GSDA
Name of the state : Maharashtra					
Name of the disrict : Pune					
Name of the Block:					
Status of Block as per central Ground water Board notification			Ground water (BCM)		
Critical	Semi critical	safe	Recharge	Draft	GAP
--		Ambegaon	0.11207	0.09444	0.01763
--	Baramati	---	0.18620	0.17288	0.01332
--		Bhor	0.04022	0.01881	0.02141
--		Daund	0.15545	0.11416	0.04129
--		Haveli	0.13746	0.07208	0.06538
--		Indapur	0.26281	0.22121	0.04160
--		Junnar	0.17617	0.15053	0.02564
--		Khed	0.14961	0.11263	0.03698
--		Maval	0.08967	0.01719	0.07248
--		Mulshi	0.08706	0.01208	0.07498
--		Shirur	0.17062	0.15407	0.01655
--	Purandhar	--	0.13108	0.12184	0.00924
---		Velha	0.04340	0.00392	0.03948

Table 3.2 Status of Ground water Availability

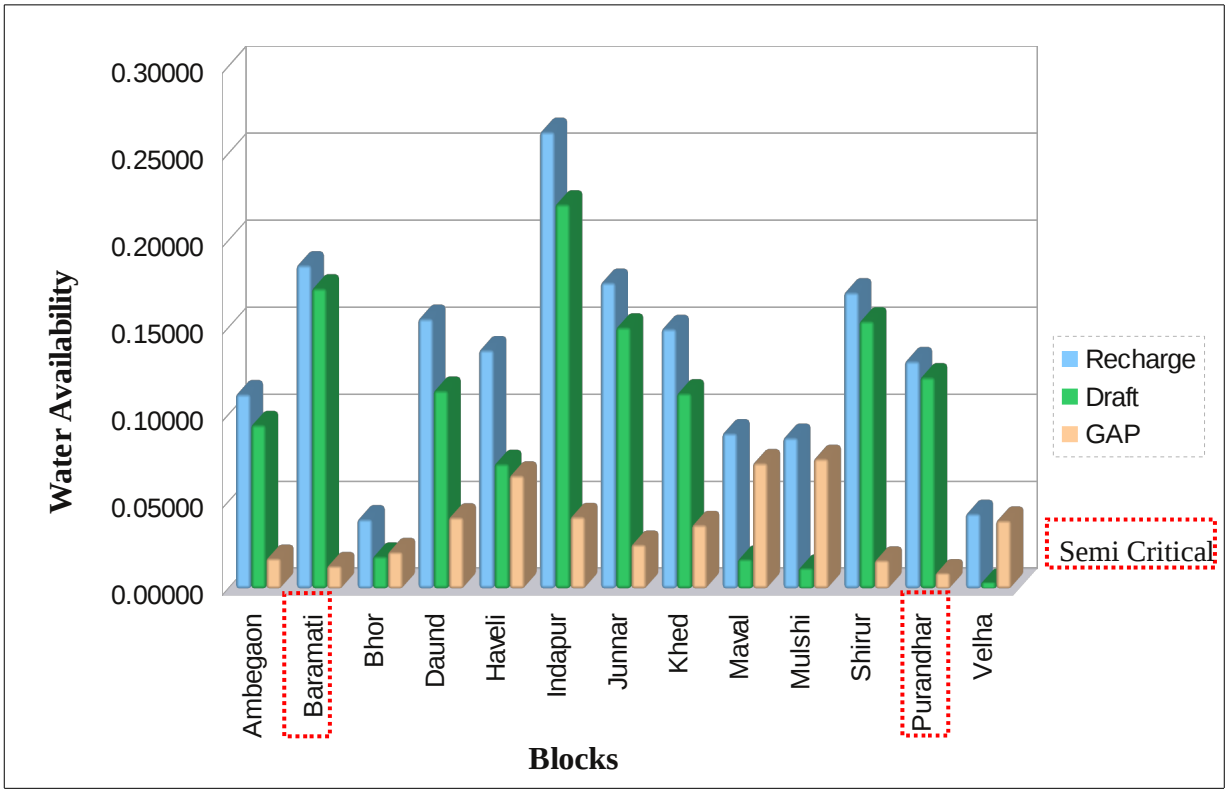


Figure 3.2.1 Status of Groundwater Availability

3.2.1 Status of Ground water Availability							Source : GSDA
Name of the state : Maharashtra							
Name of the disrict :Pune							
Sr. no.	Name of Watershed	Status of Watershed as per central Ground water Board notification			Ground water (BCM)		
		Over Expoited	Semi critical	safe	Recharge	Draft	Gap
1	2	3	4	5	6	7	8
1	BM-1	Over Expoited			0.01398	0.01424	-0.00026
2	BM-2			Safe	0.01508	0.00926	0.00582
3	BM-3			Safe	0.02813	0.02072	0.00741
4	BM-4		Semi critical		0.02556	0.02517	0.00039
5	BM-5	Over Expoited			0.01098	0.01305	-0.00207
6	BM-6			safe	0.02642	0.02240	0.00402
7	BM-7		Semi critical		0.03057	0.02689	0.00368
8	BM-8			safe	0.03200	0.02323	0.00877
9	BM-9		Semi critical		0.29730	0.02806	0.26924
10	BM-10			safe	0.11690	0.00266	0.11424
11	BM-11			safe	0.14420	0.00975	0.13445
12	BM-12	Over Expoited			0.03826	0.04000	-0.00174
13	BM-13			safe	0.01317	0.01059	0.00258
14	BM-17			safe	0.03060	0.02104	0.00956
15	BM-18	Over Expoited			0.02839	0.04486	-0.01647
16	BM-19			safe	0.02410	0.01963	0.00447
17	BM-20		Semi critical		0.02891	0.02399	0.00492
18	BM-21			safe	0.02959	0.01976	0.00983
19	BM-22			safe	0.25520	0.01818	0.23702
20	BM-23			safe	0.02614	0.02030	0.00584
21	BM-24			safe	0.03144	0.00496	0.02648
22	BM-25			safe	0.02358	0.01348	0.01010
23	BM-26			safe	0.03314	0.02845	0.00469
24	BM-27			safe	0.02169	0.17500	-0.15331
25	BM-34			safe	0.01811	0.01609	0.00202
26	BM-35			safe	0.02916	0.02157	0.00759
27	BM-36			safe	0.02025	0.01486	0.00539
28	BM-37			safe	0.01447	0.00450	0.00997
29	BM-38			safe	0.01089	0.00109	0.00981
30	BM-39			safe	0.01494	0.00414	0.01080
31	BM-40			safe	0.01097	0.00046	0.01051
32	BM-41			safe	0.01547	0.00351	0.01196
33	BM-42			safe	0.00975	0.00115	0.00860
34	BM-43			safe	0.02032	0.00352	0.01680
35	BM-44			safe	0.01350	0.00241	0.01109
36	BM-45			safe	0.02196	0.00641	0.01555
37	BM-46			safe	0.04385	0.01959	0.02426
38	BM-48			safe	0.03407	0.02484	0.00923
39	BM-49		Semi critical		0.02864	0.02755	0.00109
40	BM-50			safe	0.03204	0.02404	0.00800
41	BM-51		Semi critical		0.33820	0.02740	0.31080
42	BM-52			safe	0.02159	0.01612	0.00547
43	BM-53			safe	0.02093	0.00155	0.01939
44	BM-54			safe	0.03490	0.00035	0.03455
45	BM-55			safe	0.01258	0.00495	0.00763
46	BM-56			safe	0.02621	0.00106	0.02515
47	BM-57			safe	0.01738	0.01504	0.00234
48	BM-58		Semi critical		0.03770	0.03623	0.00147
49	BM-59	Over Expoited			0.02981	0.34687	-0.31706
50	BM-60			safe	0.02305	0.01864	0.00441
51	BM-61			safe	0.03861	0.02547	0.01314
52	BM-66			safe	0.03850	0.01622	0.02228
53	BM-67			safe	0.04339	0.01992	0.02347
54	BM-68			safe	0.02260	0.01918	0.00342
55	BM-69	Over Expoited			0.04436	0.04933	-0.00497
56	BM-70		Semi critical		0.02737	0.02148	0.00589
57	BM-71			safe	0.00933	0.00159	0.00774
58	BM-72			safe	0.01988	0.01740	0.00248
59	BM-73			safe	0.04372	0.03371	0.01001
60	BM-74		Semi critical		0.01955	0.01884	0.00071
61	BM-75		Semi critical		0.05724	0.05378	0.00346
62	BM-76	Over Expoited			0.08874	0.10200	-0.01326
63	BM-77	Over Expoited			0.03967	0.04524	-0.00557
64	BM-78			safe	0.03602	0.02377	0.01225
65	BM-81			safe	0.01707	0.00323	0.01385
66	GV-114			safe	0.00449	0.00358	0.00092
67	WF-38			safe	0.00021	0.00010	0.00011
68	WF-41			safe	0.00250	0.00048	0.00202
69	WF-45			safe	0.00360	0.00013	0.00347
70	WF-47			safe	0.00120	0.00011	0.00109
71	WF-51			safe	0.00420	0.00011	0.00410

Table 3.2.1 Status of Groundwater Availability

The District is divided in 71 elementary watersheds of Bhima river basin.As per revised

assessment of 2011-2012, out of 71 watersheds 8 becomes Over-exploited (BM-5,12,18,59,69,76,77,) and 10 watersheds becomes Semicritical (BM-4,7,9,20,49,51,58,70,74,75) Remaining 53 watersheds are in safe category. As per this assessment we feel there is ample balance of ground water for further development. But the data used for assessment is of 2010, and normal rainfall is considered for recharge purpose. After 2011-2012 there was rainfall deficiency and large number of deep bore wells are drilled for irrigation purpose which is not accounted .The result of continuous rainfall deficiency and large quantity of ground water withdrawal .The drinking water and irrigation sources has gone dry and large number of tankers are deployed for drinking water purpose. The district is facing severe scarcity.

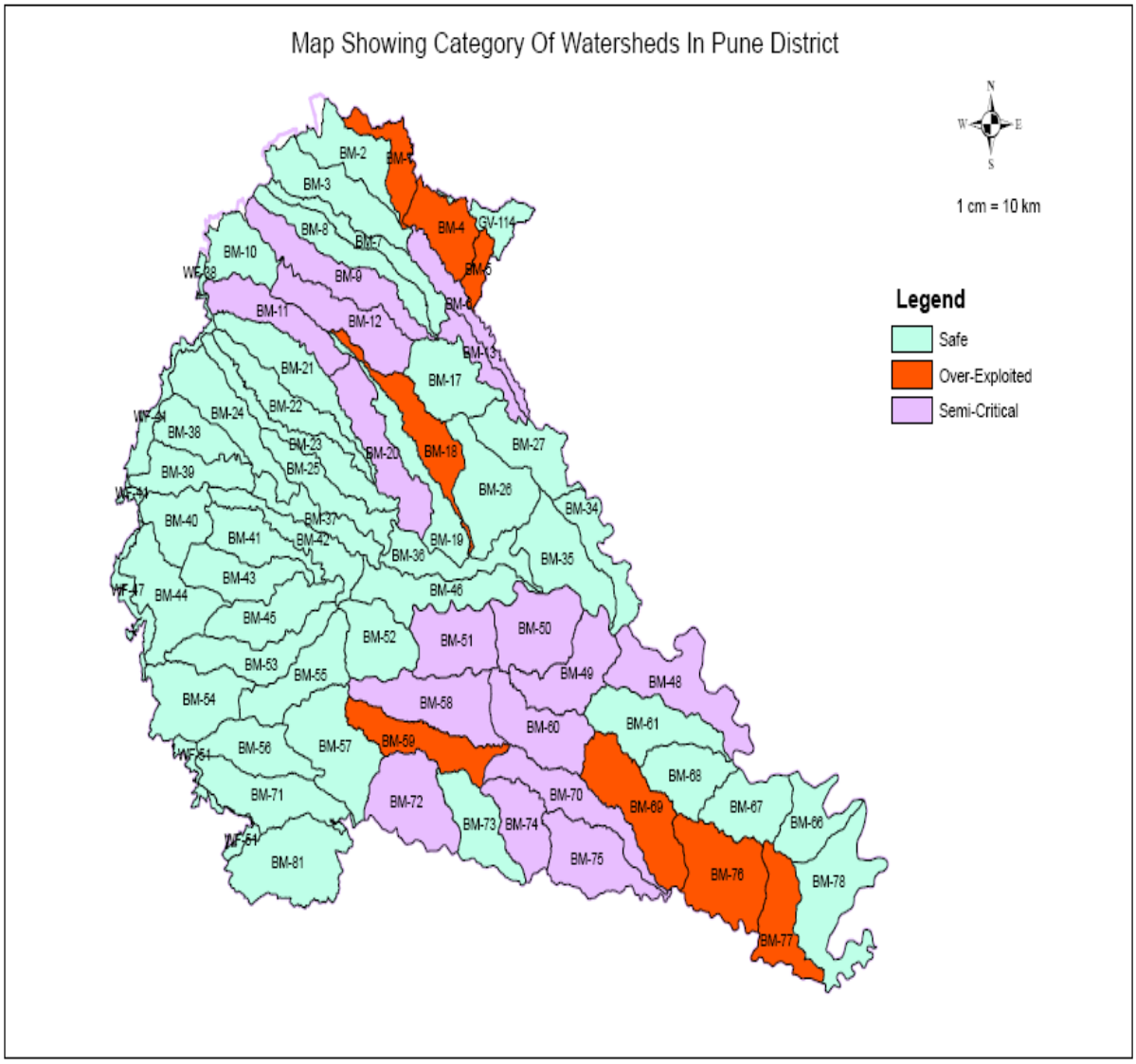


Figure 3.2.2 Map showing categories of watersheds

In all Over exploited watersheds long term water level trend (pre and post monsoon) is falling due to large no of irrigation bore wells. In these watersheds there is no scope for further ground water development. Only ground water recharge structures are to be proposed and irrigation system to be changed from flow irrigation to drip irrigation without increasing irrigated area.

The 10 watersheds are semi critical, Talukawise distribution is as follows

1)Junnar-BM-4, 2)Ambegaon-BM-7,BM-9, 3)shirur-BM-20 4)Daund-BM-49 ,BM-51 5)Purandar-BM-58 6)Baramati-BM-70 BM-74 BM-75

In semi critical watersheds ground water development is limited. Remaining 53 watersheds are safe in nature and there is scope for further development.

3.3: Status of Command Area

The command area for surface irrigation is mainly through canal and distribution network and also from other sources such as minor irrigation tanks, village tanks, percolation tanks etc. Projected command area of the project is the ultimate command that is to be developed and is being achieved by water Resources Department, Government of Maharashtra. The block wise command area developed so far under canal system and other sources have been estimated. There are some technical socio-economical and financial limitations causing delay in developing the projected command area.

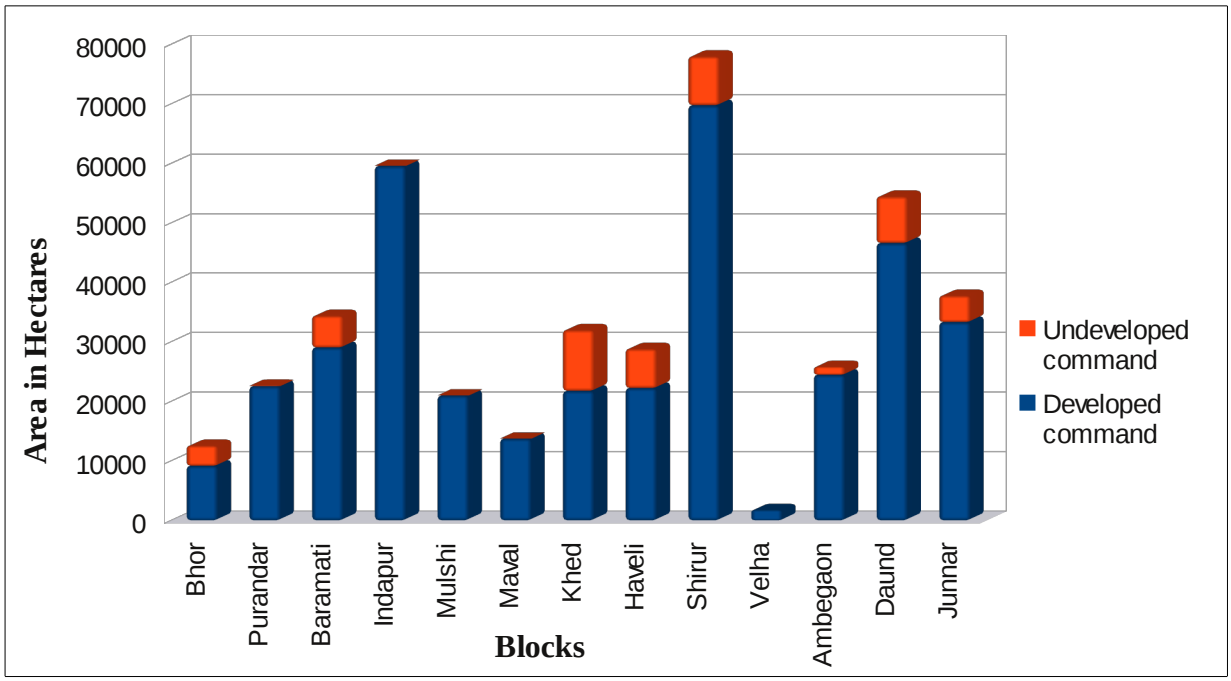


Table 3.3 Status of Command area – Developed and Underdeveloped Land

The block wise command area developed so far under canal system and other sources have been also estimated and efforts are being done for completing the same at an earliest. The details for block wise projected command area developed so far and yet to be developed through canal system and other sources has been given in table 3.3.

3.3 Status of Command area									Source : Dept of Irrigation
Name of the State : Maharashtra									
Name of the District : Pune									
Name of the Block									
									Area in Ha.
Sr. No.	Name of the Taluka	Information of canal Command			Information on the other Services			Total Area	
		Total area	Developed Area	Undeveloped Area	Total area	Developed Area	Undeveloped Area	Developed command	Undeveloped command
1	2	3	4	5	6	7	8	9	10
1	Bhor	6670	3295	3375	5897	5884	13	9179	3388
2	Purandar	2320	2320	0	20338	20227	111	22547	111
3	Baramati	16055	16055	0	18367	13018	5349	29073	5349
4	Indapur	52116	52116	0	7486	7465	21	59581	21
5	Mulshi	5345	5345	0	15654	15632	22	20977	22
6	Maval	8001	8001	0	5766	5748	18	13749	18
7	Khed	19900	9676	10224	12110	12095	15	21771	10239
8	Haveli	13732	7145	6587	15102	15102	0	22247	6587
9	Shirur	65603	57507	8096	12377	12248	129	69755	8225
10	Velha	0	0	0	1774	1774	0	1774	0
11	Ambegaon	14481	13214	1267	11349	11266	83	24480	1350
12	Daund	35490	28055	7435	18948	18546	402	46601	7837
13	Junnar	25805	21490	4315	11966	11829	137	33319	4452
	Total	**265518**	**224219**	**41299**	**157134**	**150834**	**6300**	**375053**	**47599**

Table 3.3 Status of Command Area

The projected command area from canal and other sources are 265518 Ha and 157134 Ha respectively totaling to 422652 Ha. The command area developed so far under canal and other sources are 224219 Ha and 150834 Ha respectively, thus totaling to 375053 Ha. The command area to be developed under canal and other sources are 41299 Ha and 6300 Ha totaling to 47599 Ha respectively.

The reasons for non development of command for various Projects have been identified and the

remedies for rectifying the same so as to develop the command at its projected area are summarized as below:

Sr. No.	Project	Reasons	Remedies
1	Khadakwasla Project	Leakages through structures & canal beds being old canal Silting, dilapidated structures, disturbance in canal section and garbage from city area	Extension & Improvement for old systems main canal, distributaries and minors Handing over command to farmers through participatory Irrigation Management & formation of Water Users Associations (WUAs)
2	Nira Left Bank Canal	Pretty old structure and disturbed Earthworks of main canal, distributaries and minors Seepage and leakages	Renovation of 100 year old structures & disturbed Earthworks of main canal, distributaries and minors to reduce leakages and improve efficiency Formation of Water Users Associations (WUAs) and handing over command area to WUAs for participatory irrigation management
3	Chaskaman Project	Inadequate capacity (550 cusecs against 950 cusecs) as lining is incomplete Seepages and leakages as canal construction was done in 1972 famine under scarcity work	Lining of canals, distributaries and field channel Establishment of Water Users Association for participatory irrigation management
4	Purandar Lift Irrigation Scheme	Non completion of OFD work for planned chak of 100 Ha	Distribution system of closed conduit below outlet so as to receive water to farm without conveyance losses
5	Kukadi	Inadequate field channels below outlets by farmers	Construction of field channels as CAD works
6	Nazare Project and Veer nalla, Pilanwadi, Garade Mahur M.I. Tanks	Seepage and percolation losses	Close pipe distribution system
7	200 K. T. Weirs	Substantial damages due to old structure	Repairs, Rehabilitation & Renovation of old structure
8	Scheme for Use of Recycled Waste Water	Non planned water use of urban area reducing water allotted for agriculture purpose	Renovation of Old Mutha Canal Timely completion of a scheme for use of recycle waste water to agricultural land

3.4 Existing type of irrigation

The Water Resource Department, Government of Maharashtra is taking care of surface irrigation through major, medium irrigation projects. The command area under major/medium projects is through canal and network system. Moreover, substantial command from other sources such as minor tanks, village tanks, percolation tanks, other reservoir bodies is also available in Pune District. The surface irrigation is provided in kharif, rabi and summer seasons. The block wise area irrigated under government system and other sources have been estimated and shown in table below.

3.4 Existing Type of Irrigation																			Source : Dept of Irrigation, GSDA
Name of the State : Maharashtra																			
Name of the District : Pune																			
Name of the Block																			
Sr No	Source of Irrigation / Name of Block	Surface irrigation (1)					Ground water Nos. (2)						Other sources including traditional WHS (3)	Treated effluent discharged form STP	Water extraction devices /lift			Total	
		Canal Based		Tanks/ponds/Reservoirs			Tube Wells		Open Wells		Bore well				Electricity pump (4)	Diesel pump (5)	Other (6)	Irrigation Sources (1+2+3)	water extracting units (4+5+6)
		Govt. Canal	Community/Pvt. canal	Community ponds including small	Individual/ Pvt. ponds	Govt. Reservoir /Dams	Govt. Canal	Community/Pvt. canal	Community ponds including small	Individual/ Pvt. ponds	Govt. Reservoir/Dams	Individual/ Pvt. ponds							
1	Bhor	1	0	0	0	24	0	0	0	3113	0	546	0	0	3659	0	0	3684	3659
2	Purandar	2	0	0	0	21	0	0	0	15405	0	2292	0	0	17697	0	0	17720	17697
3	Baramati	3	0	0	0	34	0	0	0	16599	0	3454	0	0	20053	0	0	20090	20053
4	Indapur	3	0	0	0	18	0	0	0	25705	0	3782	0	0	29487	0	0	29508	29487
5	Mulshi	2	0	0	0	28	0	0	0	1189	0	427	0	0	1616	0	0	1646	1616
6	Maval	4	0	0	0	13	0	0	0	1869	0	222	0	0	2091	0	0	2108	2091
7	Khed	4	0	0	0	28	0	0	0	8702	0	990	0	0	9692	0	0	9724	9692
8	Haveli	6	0	0	0	12	0	0	0	11657	0	3269	0	1	14926	0	0	14944	14926
9	Shirur	3	0	0	0	35	0	0	0	28380	0	5027	0	0	33407	0	0	33445	33407
10	Velha	0	0	0	0	8	0	0	0	129	0	2	0	0	131	0	0	139	131
11	Ambegao	1	0	0	0	33	0	0	0	12346	0	687	0	0	13033	0	0	13067	13033
12	Daund	3	0	0	0	16	0	0	0	13738	0	2877	0	0	16615	0	0	16634	16615
13	Junnar	5	0	0	0	48	0	0	0	15908	0	3002	0	0	18910	0	0	18963	18910
	Total	**37**	**0**	**0**	**0**	**318**	**0**	**0**	0	**154740**	0	**26577**	**0**	**1**	**181317**	**0**	**0**	**181672**	**181317**

Table 3.4 Existing Type of Irrigation

The data of above table reveals that, in the district surface irrigation is provided mainly through

government canal and government reservoirs/dams. In the district there are total 36 govt canal and 318 reservoirs being used for surface irrigation. Out of 318 reservoirs 48 occurs in Junnar followed by 35 in Shirur, 33 in Ambegaon and 28 in Mulshi and Khed each. In the district there are no private/community cannals or community community/private ponds being used for irrigation purpose.

The another source used for irrigation is groundwater resource. In the district there are 154740 open wells of which maximum i.e. 28380 are in Shirurs and only 129 Velha block. There are 26577 private borewells in the district of which 5027 are in Shirur and 2 in Velha. There are total 181317 electrical pumps in the district.

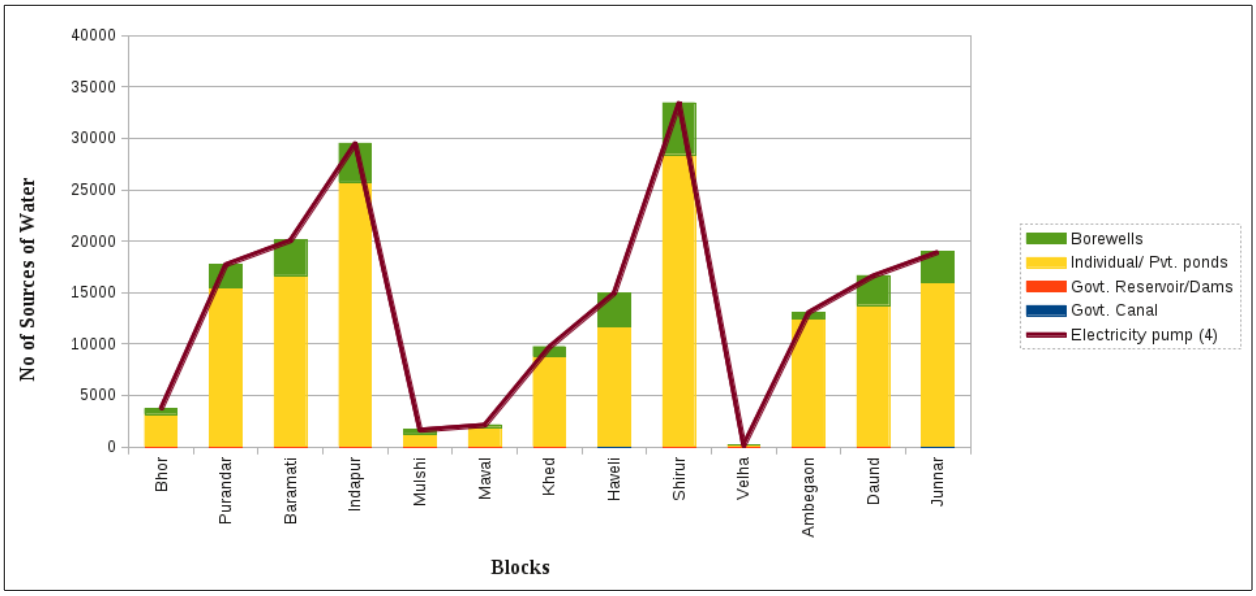


Figure 3.4 Types of Irrigation sources

The precision of groundwater resource estimation is mainly dependent on the quality of the basic data. This assessment is carried out with help of the secondary data. Despite the limitations of the secondary data, GSDA has taken efforts and has tried to collect more accurate well census, crop and surface irrigation data. The well census data have been cross verified with the electric connections given by Maharashtra State Electricity Distribution Company. This is followed by sample field confirmation by GSDA.

A note of caution is required to be placed on record, regarding the situation that may emerge from the statement in the preceding paragraph i.e. after subtracting the present groundwater draft

of district it appears that there is still a balance of 45714 ham. During the course of preparation of this Groundwater Assessment report it is observed that irrigation bore wells with pump sets have been accounted for the draft calculations. Around 26577 nos irrigation borewells data is used for GW assessment It is a well-known fact that the large number of irrigation borewells are the additional source of irrigation across the district and tentative number of these are not even on record for electricity connections. If actual draft from those borewells had been accounted for then the balance position would have certainly emerged as alarming.

Groundwater is one amongst the State's most important natural resources. It provides drinking water to rural as well as urban community, supports irrigation and industry, sustains the flow of streams and rivers and maintains wetland ecosystem. There is significant freshwater deficit in many areas of the State. Human health, welfare and food security are at risk unless the groundwater resources are managed more effectively and efficiently by the community. It is very much obvious that over extraction of ground water over years without any compensatory replenishment is affecting large tracts of land adversely. The non-replenishment of the shallow aquifers and depletion of the deeper aquifers on account of unregulated sinking of deep borewells almost amounting to "water-mining" unmindful of the adverse ecological effects is one of the contributory causes for recurring droughts. Concerted action, therefore, is needed to reverse the present trend of periodic occurrence of droughts.

Taking into consideration variation of rainfall in Maharashtra, which is the most influential factor in the dynamic groundwater estimation, it will be imperative henceforth to estimate freshly the groundwater resources every year. For this, it is necessary to make the method prescribed by the Central Committee much easier to operate and to publish in November every year the groundwater storage created in each watershed so as to apprise the community about the status of irrigation and drinking water. It would also be necessary to plan and control the use of groundwater under the prevailing conditions. Publication and distribution of annual reports and related programmes for creating awareness amongst the community and for educating them will have to be undertaken regularly. This will enable avoiding scarcity, as well as the hectic activity and excessive expenditure that has become characteristic of summer months.It is necessary to formulate and adopt a long-term policy to protect groundwater by preventing pollution and overuse. This policy should be comprehensive and implemented at all appropriate levels. It

should be consistent with other water management policies and be duly taken into account in other sectoral policies. **Priority is to be given for demand management measures supported by artificial groundwater recharge in all the over developed (over-exploited/critical/semi-critical) watersheds. Similarly the convergence of the GoI and GoM schemes of watershed development or artificial groundwater recharge need to be promoted in these areas.**

Unlike the land resource, groundwater is a dynamic resource. The groundwater flow cannot be measured like the flow through canals or pipe lines. The groundwater flows downward and spreads according to natural gradient and the permeability of the formations. Therefore, management of groundwater in fact, involves management of a dynamic, immeasurable and uncontrollable entity. While managing the groundwater resource, it is necessary to consider it to be a common property resource and is required to be controlled appropriately with the assistance of the community. Concepts like Village level Watershed Water Account, Village level Water Safety and Security, Basin/ Sub-basin Water Auditing, Aquifer delineation and its management etc will have to be popularized and made a basis for equitable distribution of ground water.

This Groundwater Assessment is very much useful to the planners, policy makers, farmers and other stakeholders etc. More weightage has to be given formeticulous implementation of Maharashtra Groundwater (Regulation for Drinking Water Purpose) Act, 1993 especially in Over-exploited and Critical watersheds. An appropriate policy will be in place as soon as the Maharashtra Groundwater (Development and Management) Act 2009, comes in force. The objective of it is to adopt the developmental measures along with the regulatory measures, till the over extracted areas become safe.The assessment will be beneficial particularly in drought management. The computation of groundwater assessment on annual basis only would be able to establish the relationship between the annual rainfall and annual groundwater recharge. This is important for water scarcity mitigation and management.

Chapter IV: Water Requirement /Demand

4.1: Domestic Water Demand

Demography of the region is changing rapidly with the increase in population. It is the need of the time to analyze the demographic projections with the changing projectile of water demand. Water consumption varies greatly from one region to the other. This is mainly due to the climatic, socio-economic and demographic characteristics of the district. The agricultural sector accounts for the majority of water consumption,followed by the domestic and industrial sectors. The computation of water requirement for present is done using the demographic data of the year 2015. The Pune district's present population is 10479610 and projected population of 13098006 for 2020. This shows that the projected growth of population over next five years would be 24.98 percent i.e., around 25 percent.

4.1 Domestic Water Demand				**Source :PMC,PCMC, Zilla Parishad Pune**	
Name of State : Maharashtra					
Name of District : Pune					
Sr.No.	**Block**	**Population in 2015**	**Existing water Demand (BCM)**	**Projected Population in 2020**	**Gross water Demand (BCM)**
1	Bhor	178495	0.003943	208055	0.0045965
2	Purandar	270133	0.006058	323769	0.0072610
3	Baramati	462428	0.01385	507517	0.0152000
4	Indapur	408736	0.008627	444035	0.0093725
5	Mulshi	165085	0.00287	178292	0.0031000
6	Maval	374451	0.018398	404304	0.0198650
7	Khed	505357	0.012893	602728	0.0153775
8	Haveli	6568492	0.517475	8740005	0.6885500
9	Shirur	409980	0.008481	453072	0.0093725
10	Velha	57584	0.001019	62191	0.0011000
11	Ambegaon	242052	0.004167	261416	0.0045000
12	Daund	399874	0.00834	442064	0.0092200
13	Junnar	436943	0.008209	470558	0.0088410
	Total	**10479610**	**0.614330**	**13098006**	**0.79635595**

Table 4.1 Domestic Water Demand

The current water demand is 0.614330 BCM and projected water demand is 0.79635595 BCM which is 29.62 percent more than present water demand.

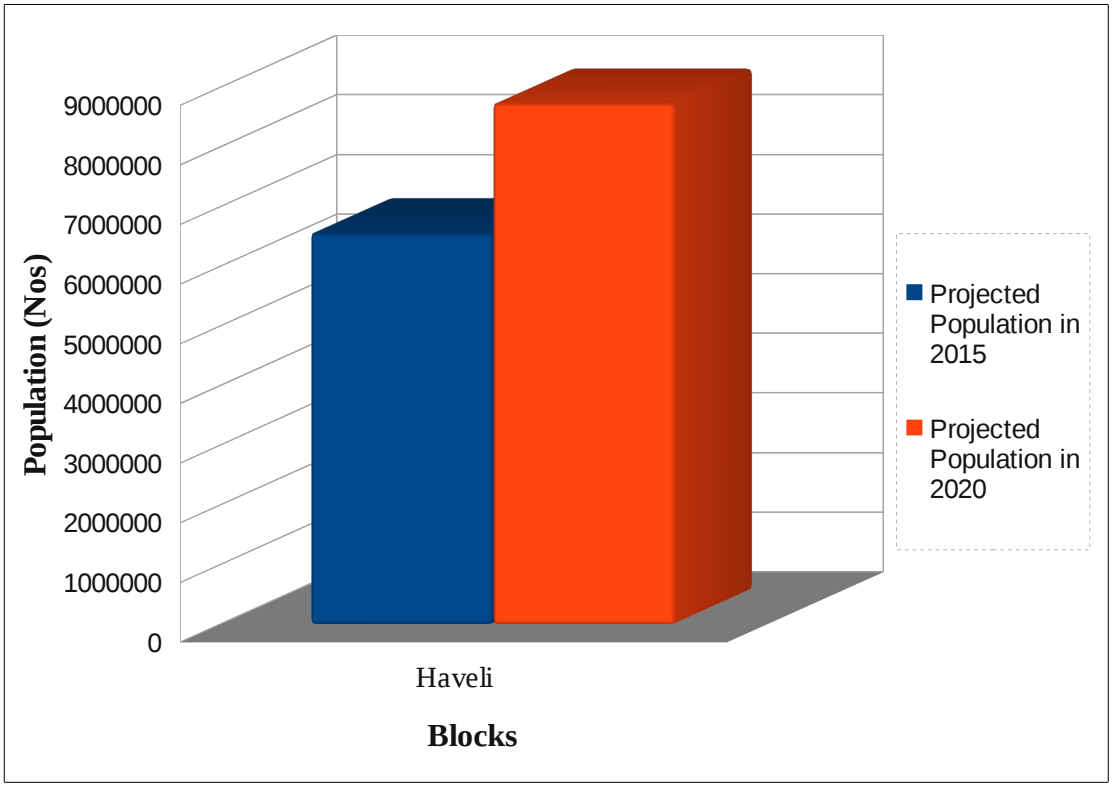


Figure 4.1 Urban Projected Population

The Pune and Pimpri Chinchwad Corporation cities are located in Haveli block therefore the domestic water requirement of Haveli block appears to be maximum. The present population of this block is 6568492 in 2015 and expected to be raised to 8740005 by year 2020. Obviously the present and projected water requirement of this block would be 0.517475 and 0.6885500BCM respectively.

The gross per capita water supply of Pune city is 321lpcd. As per the PMC data actual supply is 228 lpcd. So, while calculating domestic water requirement of Pune city, the above figures are considered.

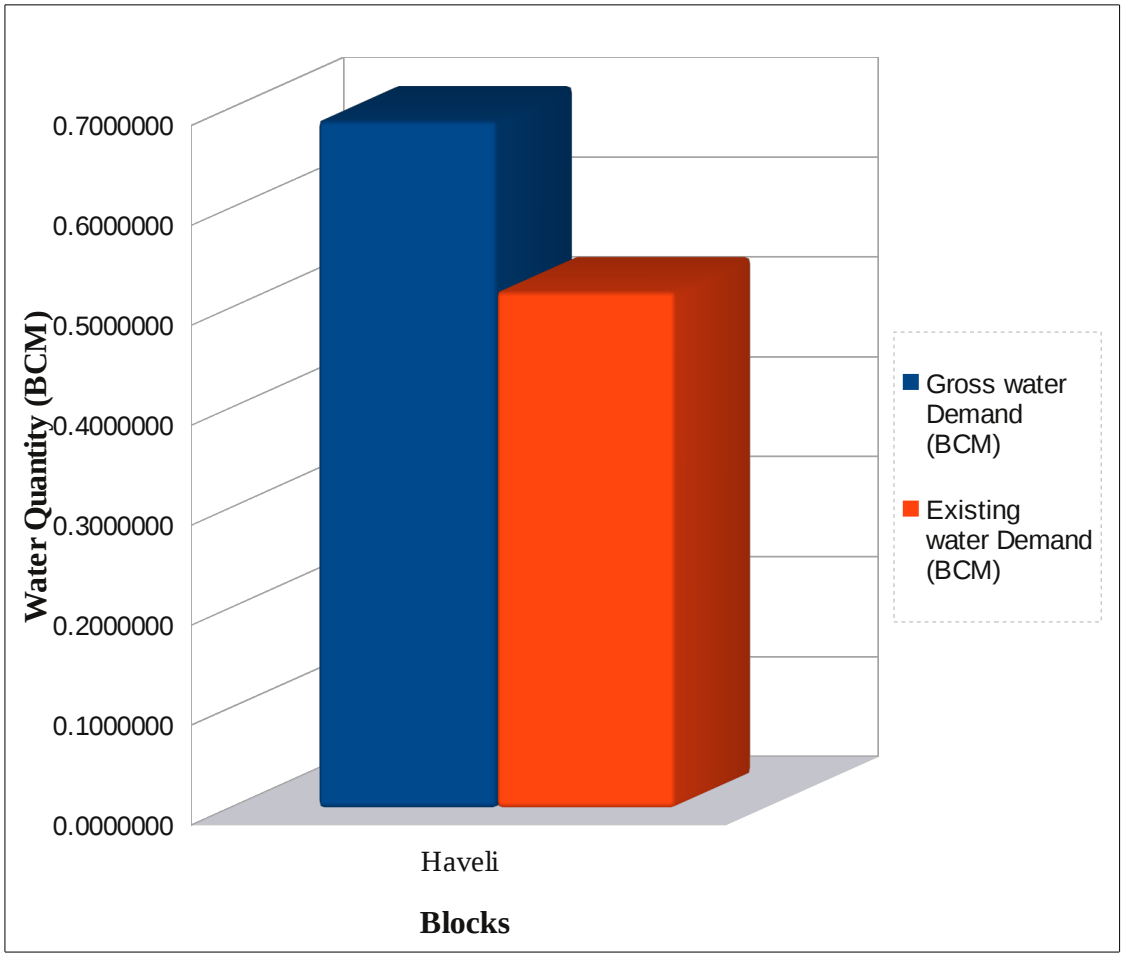


Figure 4.1.1 Urban Water Demand

The total population of the district excluding Haveli block is 3911118, which is expected to reach to 4358001 by 2020. The present and projected water demand of this much population is 0.096855 BCM and 0.10780595 BCM receptively.

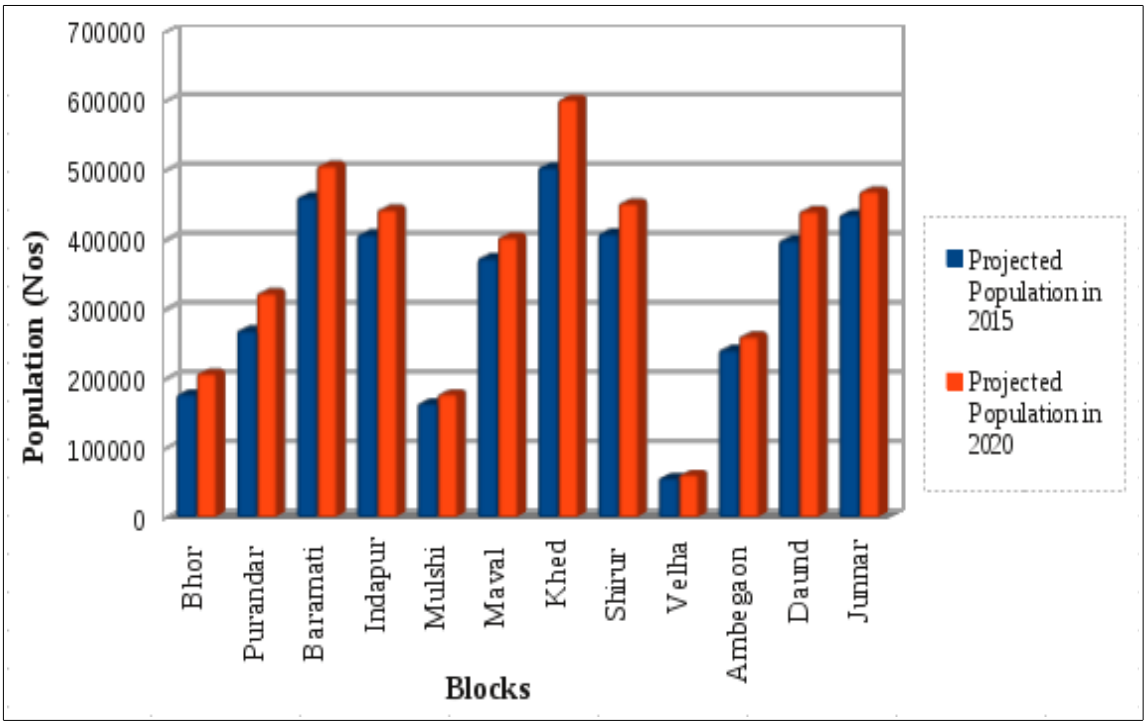


Figure 4.1.2 Rural Projected Population

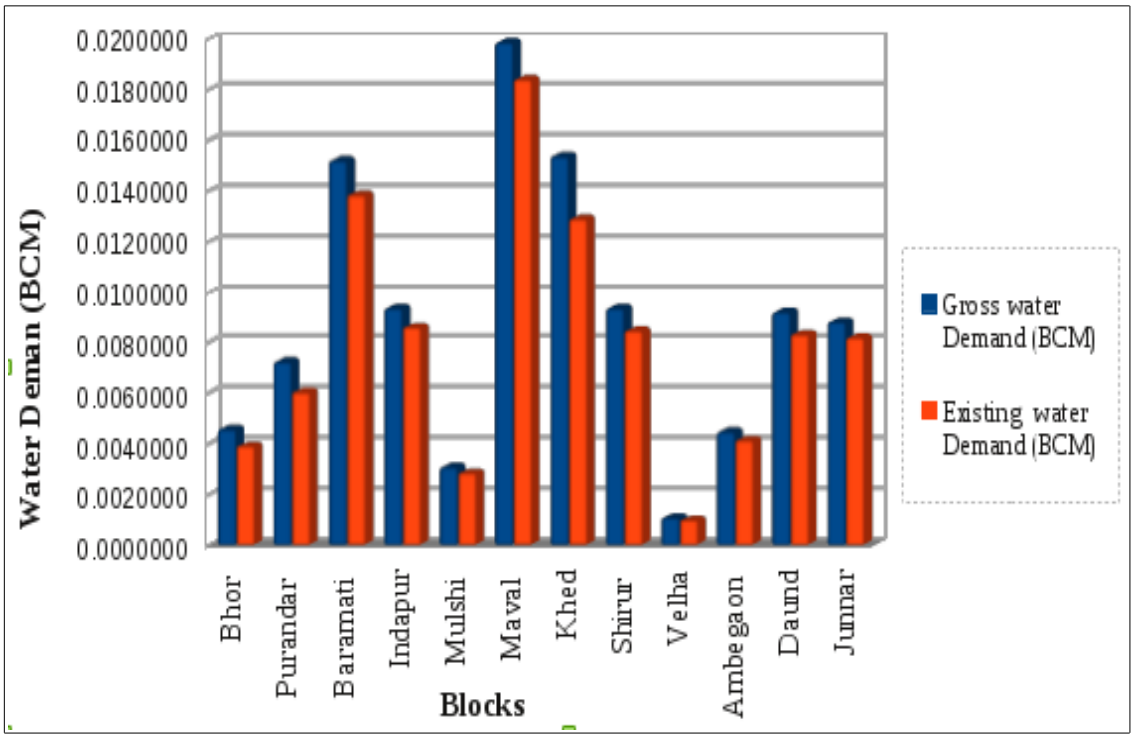


Figure 4.1.3 Rural Water Demand

4.2: Crop Water Demand

The crop water requirement is the depth (or amount) of water needed to meet the water loss through evapo-transpiration. In other words, it is the amount of water needed by the various crops to grow optimally. It always refers to a crop grown under optimal conditions, i.e. a uniform crop, actively growing, completely shading the ground, free of diseases, and favorable soil conditions. The crop thus reaches its full production potential under the given environment. The crop water requirement mainly depends on climate and crop type.

The crop water requirement works out on the basis of certain parameters viz. actual water required for vegetative and productive growth of crop, evaporation, evapo-transpiration etc. The key source of water for the crops is precipitation which would be supported by protective irrigation from surface storages or ground water resources.

The district grows variety of crops in different agriculture seasons. These crops are either rainfed, partially irrigated or irrigated using surface or ground water resource. The following table presents block wise crop water requirement of the district.

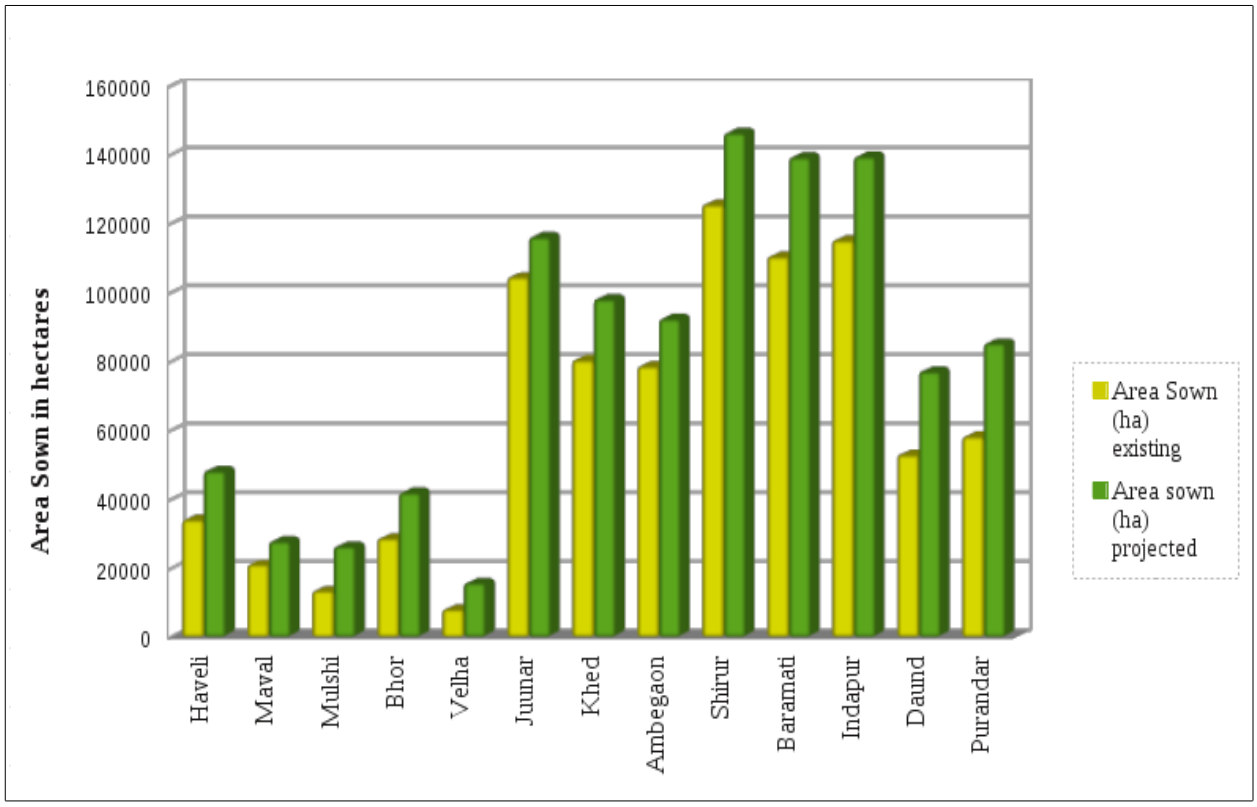


Figure 4.2 Area Sown current and projected 2020

The below table revels that the projected cropped area of the district is 1053761 ha by 2020 of which 422652 ha will be irrigated. Considering the water requirement of proposed crops, as recommended by State Agricultural Universities the annual net water requirements of all the

crops of the district have been worked out and is estimated to be 673326400 cubic meter i.e. 6.7332640 billion cubic meters (BCM) by 2020. The existing water potential of the district is 6.0056788BCM and additional potential of 0.732384 BCM needs to be created by 2020.

4.2 Crop water requirement

Name of State : Maharashtra

District: Pune

Sr. No.	Name of the Block	Area Sown (ha) existing	Area sown (ha) projected	Irrigated area (ha)	Water potential required (BCM)	Existing Water Demand (BCM)	Water Potential to be Created (BCM)
1	Haveli	34092	48236	28834	0.4009475	0.3289239	0.0720237
2	Maval	21023	27764	13767	0.1336175	0.0866934	0.0469241
3	Mulshi	13396	26291	20999	0.0924410	0.0616324	0.0308087
4	Bhor	28659	41918	12567	0.1522236	0.1266953	0.0255283
5	Velha	7952	15674	1774	0.0338638	0.0200972	0.0137666
6	Juunar	104454	116099	37771	0.7356190	0.7020108	0.0336083
7	Khed	80427	98082	32010	0.4831940	0.4534038	0.0340023
8	Ambegaon	78609	92373	25830	0.5112840	0.4516435	0.0602271
9	Shirur	125563	146340	77980	0.9363295	0.8322892	0.1040403
10	Baramati	110489	139226	34422	0.8122235	0.7600027	0.0522208
11	Indapur	115179	139441	59602	1.1980315	1.1656732	0.0323582
12	Daund	52944	77084	54438	0.8445415	0.8153972	0.0291443
13	Purandar	58197	85233	22658	0.3989478	0.2012165	0.1977313
	Total	**830984**	**1053761**	**422652**	**6.7332640**	**6.0056788**	**0.732384**

Table No: 4.2 Crop Water Requirement

The above table indicates that the projected sown area of Indapur block is 139441 ha and its estimated water requirement is highest i.e. 1.19803145 BCM, whereas it is lowest in Velha block i.e. 0.0338638 BCM with projected irrigated area of 15674 ha. The projected irrigated area of Shirur block is observed to be 146340 ha and its potential water requirement is 0.9363295 BCM. The table also revels that the blocks like Baramati (139226 ha), Indapur (139441ha), Daund (77084 ha) with slightly lesser projected irrigated area than Shirur, the water requirement of these blocks is greater than Shirur by 2020. This means the cropping pattern of Shirur block is comparatively rational than Baramati, Indapur and Daund blocks.

The data also states that the block Khed needs to be developed maximum water potential since it is the most water deficit block with .0340023 BCM by 2020. On the other side Mulshi block needs minimum water potential since its water requirement seems to be only 0.0308087 BCM.

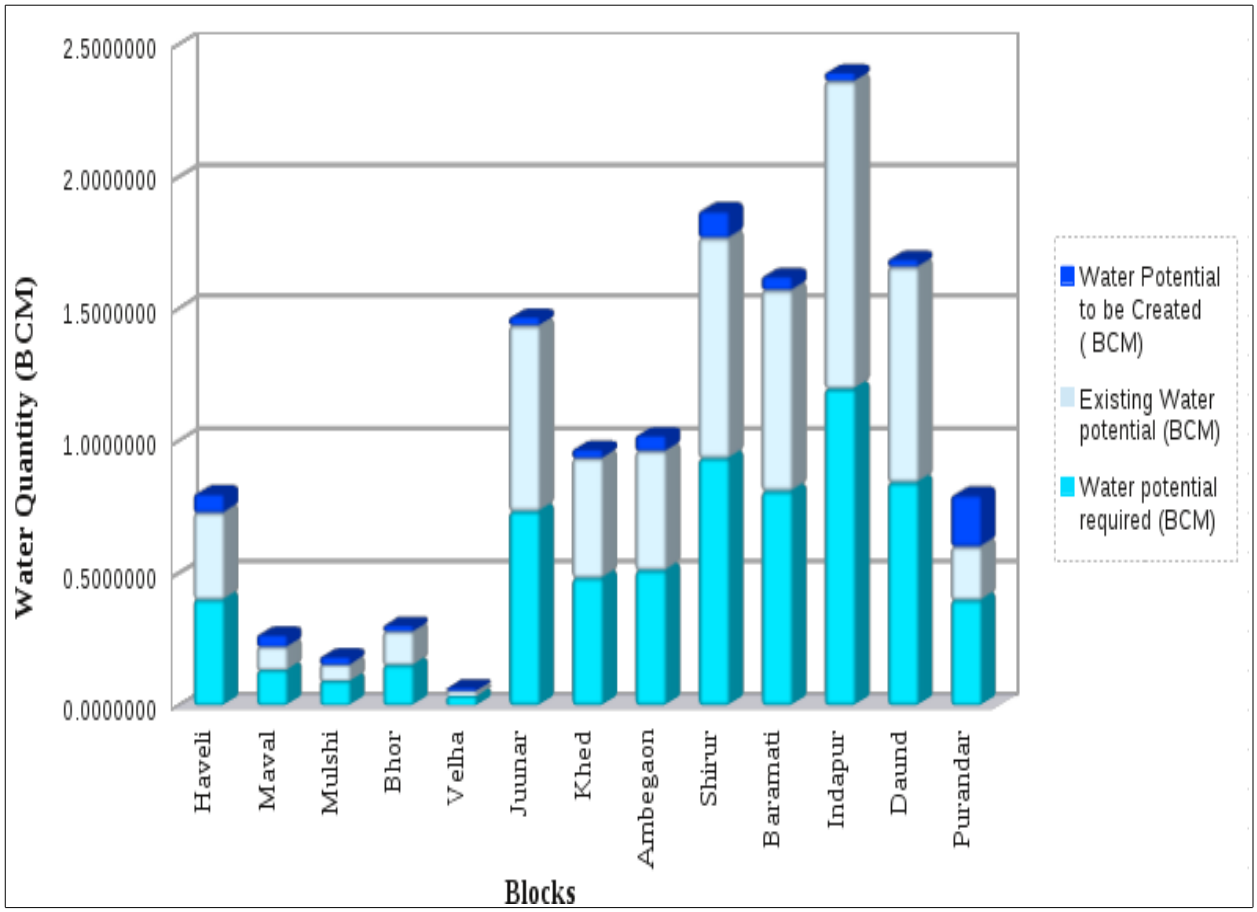


Figure 4.2.1 Crop Water Requirement

The crop water requirement shown in above table is estimated by considering water demand of crop necessarily through protective irrigations only, since these demands are normally in productive phase of crop cycle or at critical stages of its survival during dry spell of monsoon. The water demand during vegetative growth stages are normally compensated with soil moisture available through the rain.

Therefore while calculating crop water requirement the water which is available through rain has not been considered since it is available naturally. Only the additional water requirement is considered which is meant to be provided through protective irrigation.
The district irrigation plan try to compensate this crop water requirement by applying two types of interventions. Firstly by constructing number of water harvesting structures in the district, which will be a source of protective irrigation for the said crops in post project situation.

However the water available in the district has some legal obligation in harnessing it totally, since there is interstate policy which inhibits all the available water. So in that case maximum emphasis will be provided on moisture conservation works which will help to arrest the rain water and enhance the ground water level. Whatever the rain water available in different storages will be used for compensating domestic and animal water requirements whereas crop water requirement will be covered through soil and moisture conservation works.

Secondly efforts will be placed for augmenting area under micro-irrigation systems which will save 30 to 35% available water and cover more crop area for irrigation.

4.3: Livestock Water Demand:

Livestock of the Pune district is 20486182 which obviously need substantial quantity of water for its not only survival but also for contributing in economy of the farming community. The present demand of the livestock is 0.27343 BCM, whereas the projected water requirement of livestock by 2020 is 0.02968 BCM . The blockwise livestock population, and its present and projected water requirement is presented in following table.

4.3 Livestock Water Demand						Source : Dept of Animal Husbandry
Name of State : Maharashtra						
District: Pune						
Sr. No.	Name of Block	Total Number of livestock	Present Water Demand (BCM)	Water Demand in 2020 (BCM)	Existing Water potential (BCM)	Water potential to be created (BCM)
1	2	3	4	5	6	7
1	Bhor	1092685	0.00110	0.00120	0.00110	0.00010
2	Purandar	601167	0.00120	0.00130	0.00120	0.00010
3	Baramati	452949	0.00290	0.00310	0.00290	0.00020
4	Indapur	1286886	0.00340	0.00370	0.00340	0.00030
5	Mulshi	386648	0.00072	0.00079	0.00072	0.00007
6	Maval	1556971	0.00130	0.00140	0.00130	0.00010
7	Khed	2164891	0.00250	0.00270	0.00250	0.00020
8	Haveli	2149492	0.00190	0.00200	0.00190	0.00010
9	Shirur	2944653	0.00330	0.00360	0.00330	0.00030
10	Velha	125133	0.00050	0.00055	0.00050	0.00005
11	Ambegaon	2727333	0.00250	0.00260	0.00250	0.00010
12	Daund	2159761	0.00300	0.00330	0.00300	0.00030
13	Junnar	2799921	0.00290	0.00320	0.00290	0.00030
14	Pune City	37692	0.00021	0.00024	0.00021	0.00003
	Total	**20486182**	**0.02743**	**0.02968**	**0.00520**	**0.00040**

4.3 Livestock Water Demand

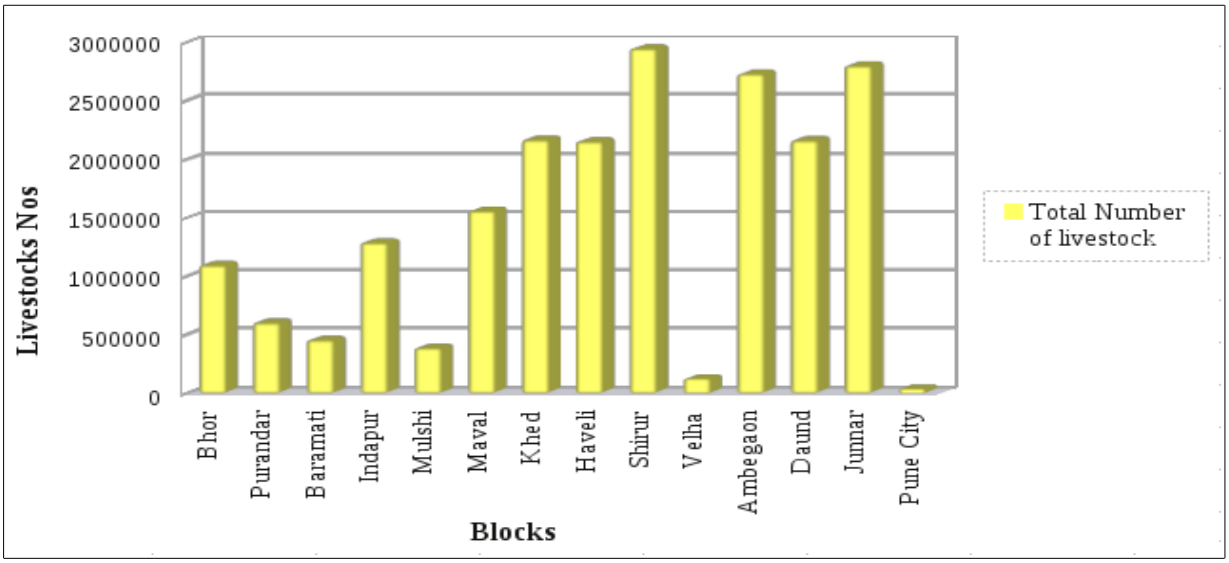


Figure 4.3.1 Total Number of Liverstocks

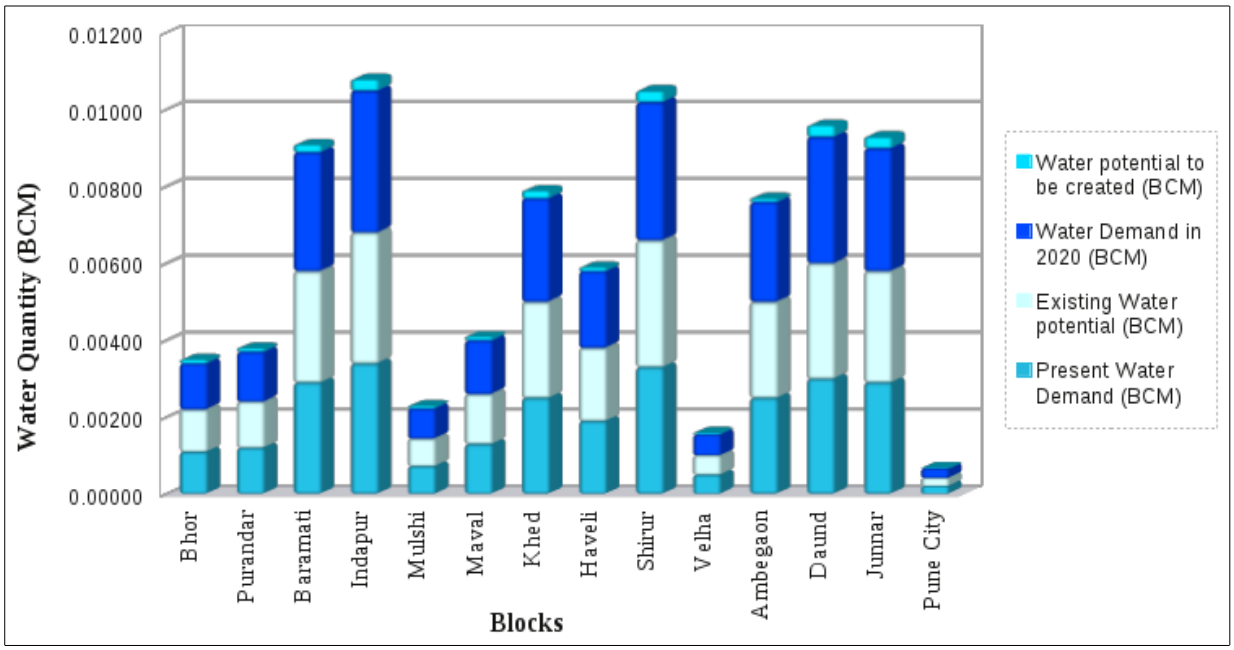


Figure 4.3.1 Livestock Water Demand

4.4: Industrial Water Demand:

With the globalization there is fast changing industries which are coming into existence varying from mechanical industries(automobiles etc), production, textiles, chemical, agricultural allied industries, manufacturing, IT etc., it is the need of the time to allocate water for the industry in optimum quantity for its sustenance. The rapid changing industrialization demand more and more water. The ever expanding count of Pune industry is 4039 with the water Demand of 0.0753 BCM and the existing water potential for industry 0.075827 BCM almost matching with existing with the minor gap of 0.0001 BCM.

The estimated water demand of the industry by 2020 is 0.1215 BCM. In that case additional water potential of 0.046173 BCM needs to be created.

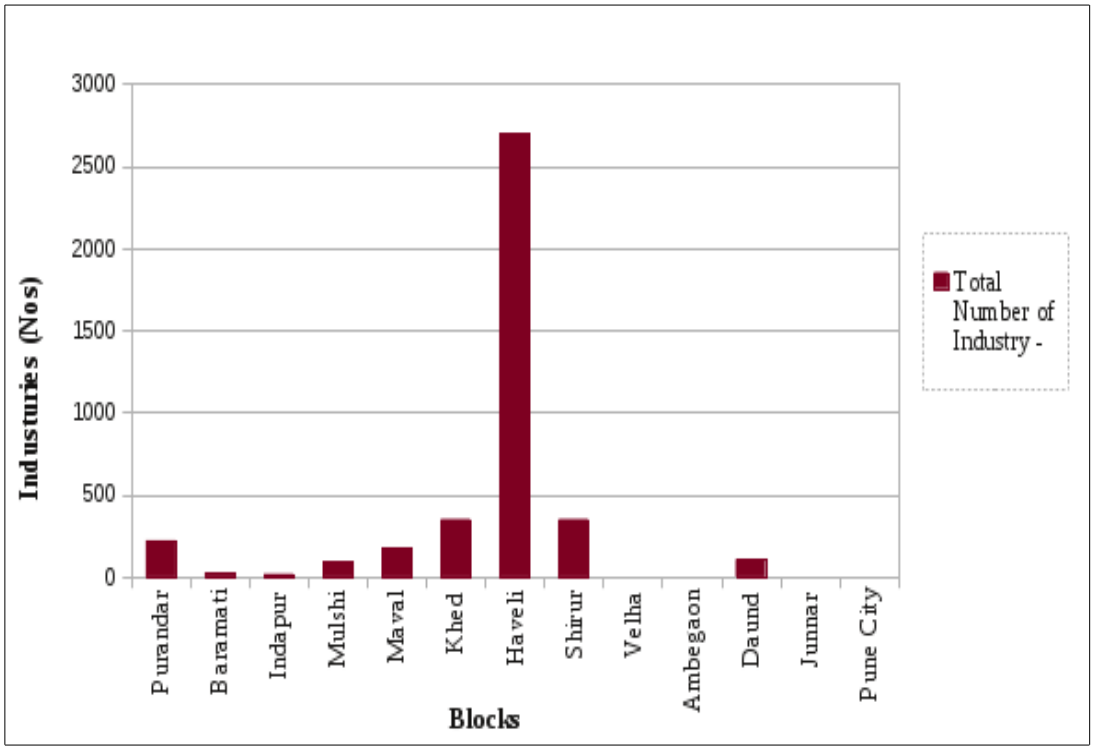


Figure 4.4 Blockswise Number of Industries in the District

The block wise number of industries existed in the districts and there present and projected water demand are calculated and is presented in the table given below

4.4 Industrial Water Demand						Source : MIDC
Name of the State : Maharashtra						
Name of District : Pune						
Sr. No.	Name of Block	Total Number of Industry	Water Demand (BCM)	Water Demand in 2020 (BCM)	Existing Water potential (BCM)	Water potential to be created (BCM)
1	2	3	4	5	6	7
1	Bhor	-	-	-	-	-
2	Purandar	220	0.00440	0.00550	0.00440	0.00110
3	Baramati	26	0.00100	0.00100	0.00100	0.00000
4	Indapur	16	0.00228	0.00228	0.00228	0.00000
5	Mulshi	94	0.00456	0.01095	0.00456	0.00639
6	Maval	178	0.00475	0.01273	0.00475	0.00799
7	Khed	350	0.00730	0.02300	0.00730	0.01570
8	Haveli	2700	0.03650	0.04950	0.03650	0.01300
9	Shirur	350	0.00850	0.01050	0.00850	0.00200
10	Velha	-	-	-	-	-
11	Ambegaon	-	-	-	-	-
12	Daund	105	0.00600	0.00600	0.00600	0.00000
13	Junnar	-	-	-	-	-
14	Pune City	-	-	-	-	-
	Total	**4039**	**0.0753**	**0.1215**	**0.075287**	**0.046173**

4.4 Industrial Water Demand

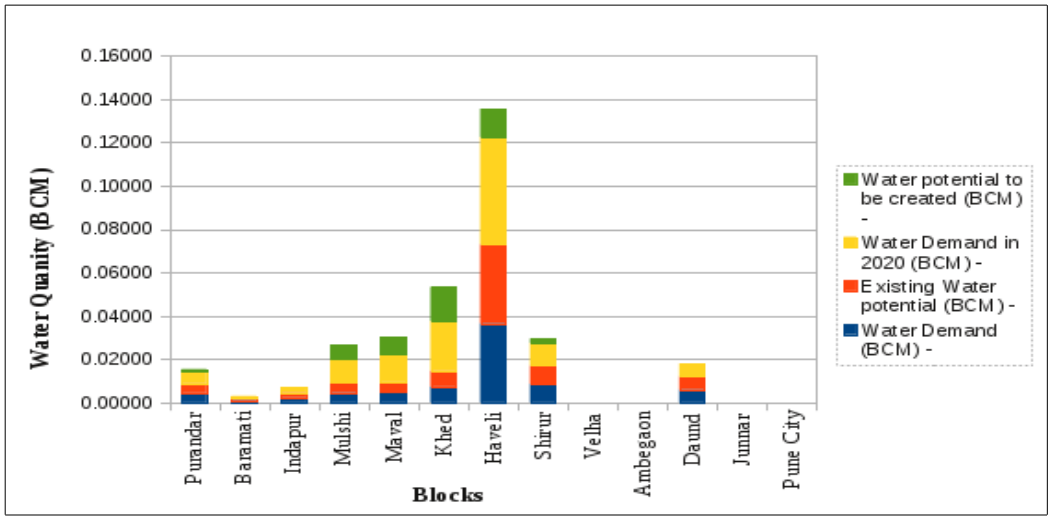


Figure 4.4.1 Blockwise Industrial Water Demand

4.6: Total Water Demand of the District for Various Sectors:

The present and projected water demand of the district including domestic, agriculture, livestock and industry sector's is 6.7227258 BCM and 7.6807600 BCM respectively. Here the current and projected water demands of the sector's mentioned above are estimated separately and presented in table below

Table. 4.6 Total Water Demand of the District for Various Section (projected)							
Name of State : Maharashtra							
District : Pune							
Sr. No.	Name of Block	Components					
	Blocks	Domestic	Crop	Livestock	Industrial	Power Generation	Total BCM
1	Bhor	0.0045965	0.15222355	0.0012000	0.00000000	0.00000000	0.15802005
2	Purandar	0.0072610	0.39894775	0.0013000	0.00550000	0.00000000	0.41300870
3	Baramati	0.0152000	0.8122235	0.0031000	0.00100000	0.00000000	0.83152350
4	Indapur	0.0093725	1.19803145	0.0037000	0.00228000	0.00000000	1.21338395
5	Mulshi	0.0031000	0.092441	0.0007900	0.01095000	0.00000000	0.10728100
6	Maval	0.0198650	0.1336175	0.0014000	0.01273000	0.00000000	0.16761250
7	Khed	0.0153775	0.483194	0.0027000	0.02300000	0.00000000	0.52427150
8	Haveli	0.6885500	0.4009475	0.0022400	0.04950000	0.00000000	1.14123750
9	Shirur	0.0093725	0.9363295	0.0036000	0.01050000	0.00000000	0.95980200
10	Velha	0.0011000	0.03386375	0.0005500	0.00000000	0.00000000	0.03551375
11	Ambegaon	0.0045000	0.511284	0.0026000	0.00000000	0.00000000	0.51838400
12	Daund	0.0092200	0.8445415	0.0033000	0.00600000	0.00000000	0.86306150
13	Junnar	0.0088410	0.735619	0.0032000	0.00000000	0.00000000	0.74766000
	Total	**0.7963560**	**6.7332640**	**0.0296800**	**0.1214600**	**0.0000000**	**7.6807600**

Table No: 4.6 (a) Total Water Demand of the District for Various sectors for 2020

The current and projected agriculture water demand is 6.0056788 BCM and 0.7275852 BCM respectively. The data shows that this is the highest demand among all other sector's mentioned above.

The domestic water demand which is currently 0.6143300 BCM and by year 2020 it would be

0.7963560 BCM. The difference between current and projected domestic levevel water requirement is estimated to be 0.182026 BCM, which is 29.63% percent more than the current demand.

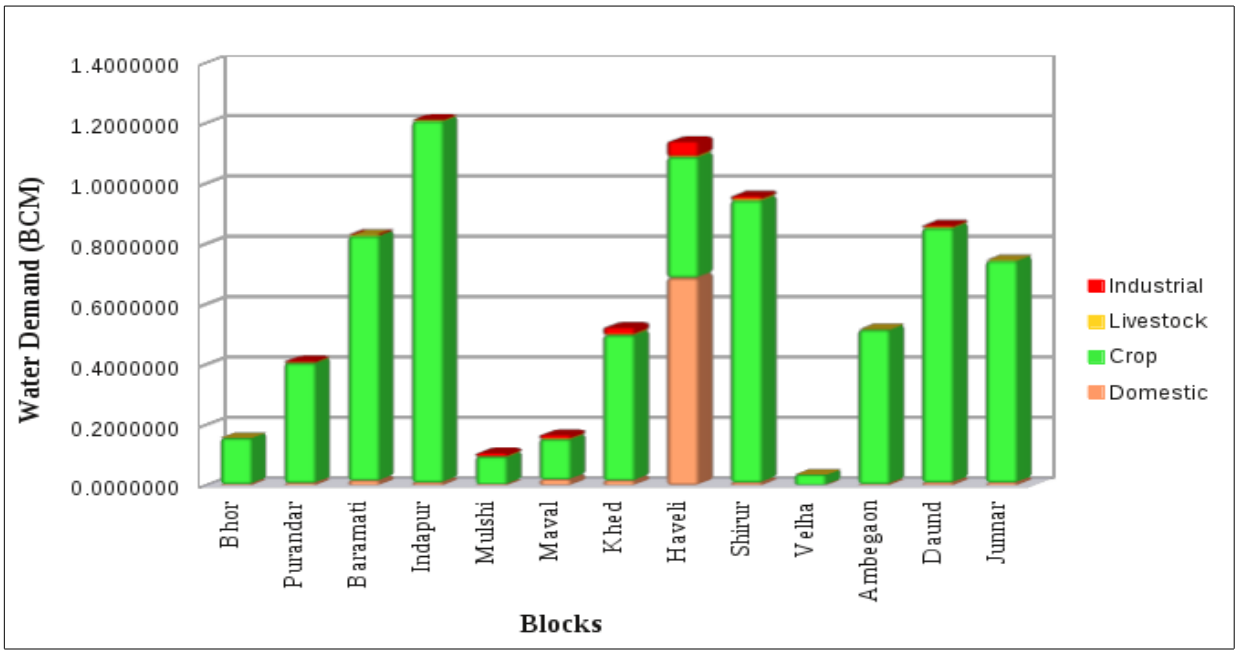


Figure 4.6 Total Water Demand from Various Sector of the District for 2020

4.7 Water Budget: Water availability, demand and gaps

The water budget means estimation of water available and required for the community residing in the district. The availability could be worked out on the basis of precipitation and geographical area of the district. The community fulfills its water demands jointly through surface and ground water and therefore water availability is estimated by considering both the water resources viz. surface water and ground water. On the other side the water requirement of the district depends on various needs of community such as drinking/domestic, animal, crop, industries and power generation etc. The demand supply scenario of water in the district needs to be thought of, in reference to pre and post project situations. Here while designing the district irrigation plan, the existing status is assumed for 2015-16 and projected means by 2020-21.

While computing the domestic level water demand the existing population i.e of 2015-16 is considered whereas the projected population has been predicted considering average annual

growth rate of district. The domestic water requirement is calculated separately for rural and urban population, assuming their water needs in respective habitations. The need of rural population is considered to be 50 LPCD whereas in case of urban population it is average 150 LPCD (100 LPCD in towns and 300 LPCD in Municipal Corporations). The crop water requirement has been worked out considering general cropping pattern of the district, water requirement of each crop (hacm), area under each crop, number of watering provided, approximate quantity of water required for each watering, method of irrigation (flood, drip or sprinkler) etc. The industrial water requirement also varies according to type of industry. In case of Pune district the spread of industries is much wider from typology of industries point of view. There are various type of industries like ISPAT, automobile, steel, production, textile, pharmaceutical, IT, service etc. So water requirement of these industries have been computed accordingly. In case of power generation presently the district does not exist any power project and the requirement is considered to be zero.

The below table reveals that total water available in the district is 3.957085 BCM of which 0.014000 BCM is in form of ground water and 3.943085 is surface water. On the other side the present and projected water demands is estimated to be 6.7227258 BCM and 7.6807600 BCM respectively. From the table it is seen that the present deficit is 2.765641 BCM and the projected water deficit is 3.723675 BCM.

Table 4.7 Water Budget								
Name of State : Maharashtra								
District : Pune								
Sr No	Name of Blocks	Existing Water avalibility (BCM)		Total (BCM)	Water Demand		Water Gap (BCM)	
		Surface Water	Ground Water		Present	Projected (2020)	Present	Projected (2020)
1	Bhor	0.077283	0.000246	0.077529	0.13173825	0.15802005	-0.054209	-0.080491
2	Purandar	0.103040	0.000567	0.103606	0.21287450	0.41300870	-0.109268	-0.309402
3	Baramati	0.267782	0.000101	0.267882	0.77775270	0.83152350	-0.509870	-0.563641
4	Indapur	0.586448	0.000137	0.586585	1.17998020	1.21338395	-0.593395	-0.626799
5	Mulshi	0.130383	0.000333	0.130715	0.06978435	0.10728100	0.060931	0.023434
6	Maval	0.126948	0.003617	0.130564	0.11113640	0.16761250	0.019428	-0.037048
7	Khed	0.194338	0.000447	0.194784	0.47609675	0.52427150	-0.281312	-0.329487
8	Haveli	0.540848	0.000479	0.541328	0.88500885	1.14123750	-0.343681	-0.599910
9	Shirur	0.483617	0.007226	0.490843	0.85257020	0.95980200	-0.361727	-0.468959
10	Velha	0.006770	0.000040	0.006810	0.02161620	0.03551375	-0.014806	-0.028704
11	Ambegaon	0.355781	0.000219	0.355999	0.45831045	0.51838400	-0.102311	-0.162385
12	Daund	0.573338	0.000513	0.573851	0.83273720	0.86306150	-0.258886	-0.289211
13	Junnar	0.496510	0.000077	0.496587	0.71311975	0.74766000	-0.216533	-0.251073
	Total	**3.943085**	**0.014000**	**3.957085**	**6.7227258**	**7.6807600**	**-2.765641**	**-3.723675**

Tabel 4.7 Water Budget

The above table shows that the block Mulshi presently has surplus water of 0.060931 BCM, whereas Indapur block is highly deficited by 0.593395 BCM. In case of projected water demands the block Mulshi would be surplus by 0.023434 BCM, whereas Indapur would be deficited by 0.0626799 BCM.

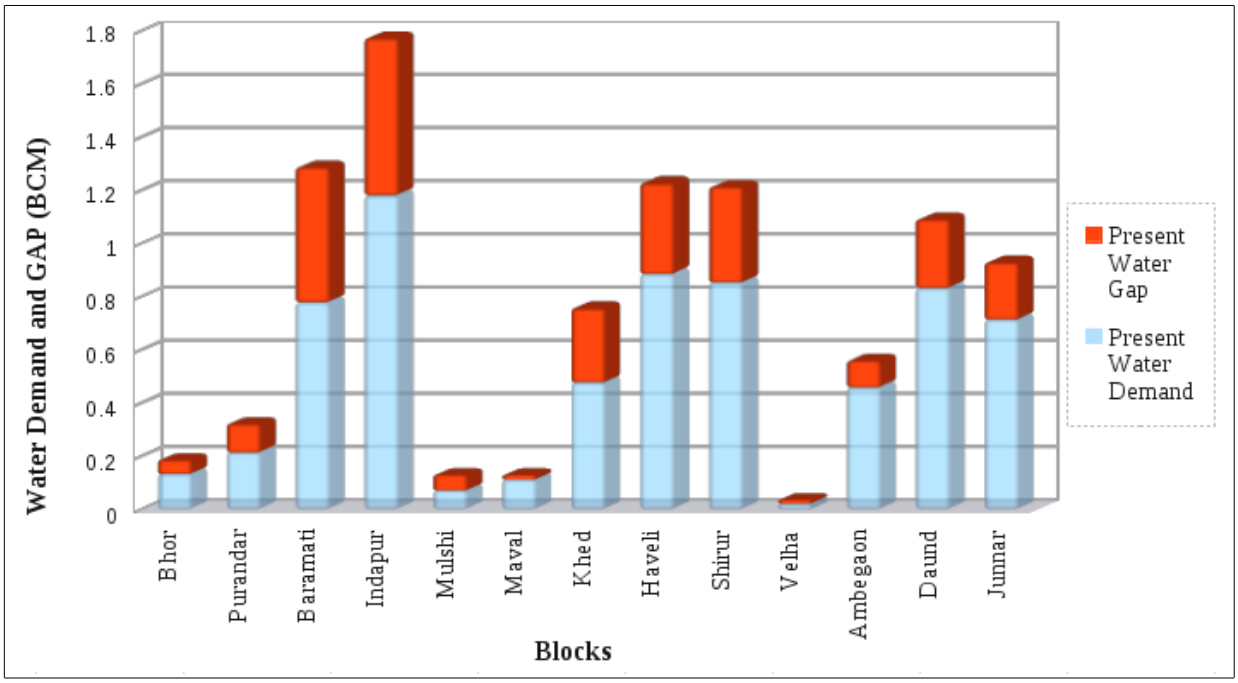


Figure 4.7 Present Water Demand

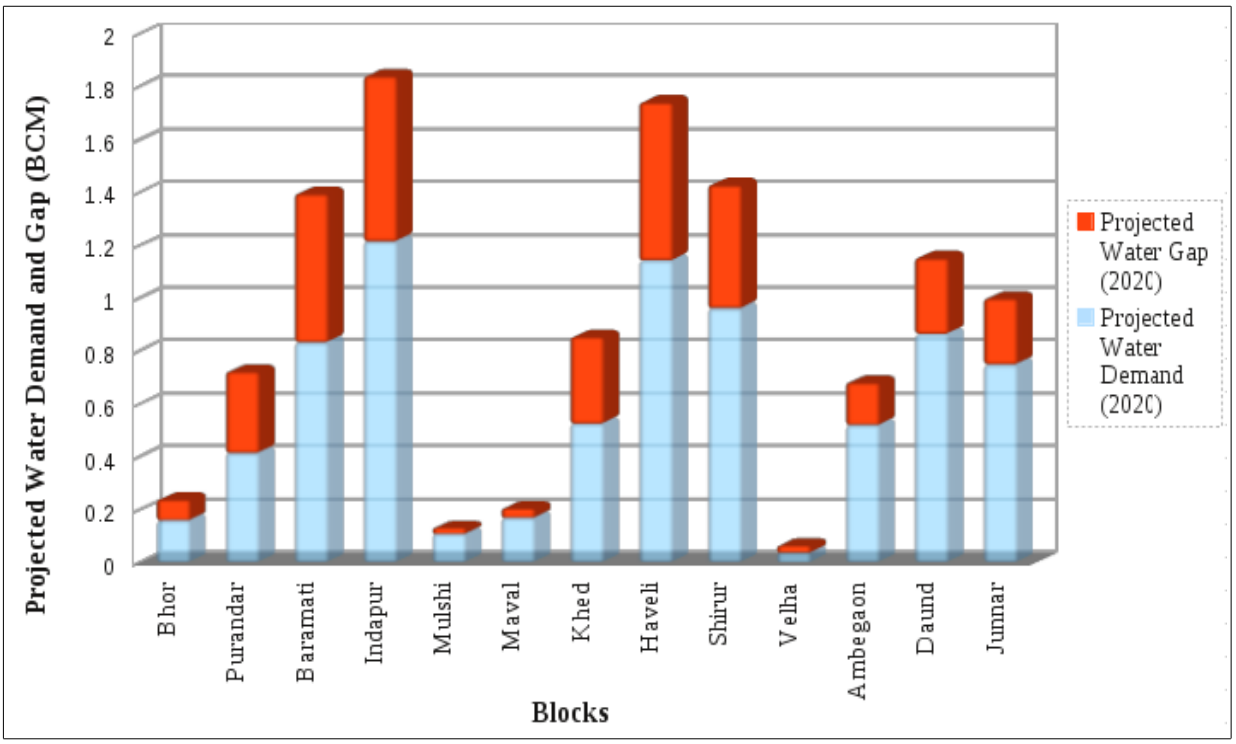


Figure 4.7 Future 2020 Water Demand

Chapter V: Strategic Action Plan for Irrigation in District under PMKSY

Strategy for Developing Action Plan:

The Pradhan Mantri Krishi Sichai Yojana (PMKSY) envisages expansion of irrigated area by accessing maximum water resource to each farm (Har Khet ko Pani). The programme anticipates development, management and economic utilization of available water resources, so as to fulfill community water needs such as domestic, animal, agriculture, industries etc. The PMKSY encourages convergence of Government investments in irrigation on all fronts starting from creation of water resources, its distribution and appropriate utilization both in domestic and production sector needs. The programme accentuates development of rainfed areas by adopting watershed approach not only for development of natural resources but also livelihoods of the families residing in the watersheds. The watershed approach also stress on enhancement of agriculture production by taking up new and modern agriculture technologies and boosting up small and micro-enterprises. The PMKSY highlights need of recycling and reuse of waste water generated by urban habitations for irrigating nearby agriculture farms.

Considering above aspects of PMKSY the district has designed the strategy for planning, designing, implementation and monitoring of the project components. The planned strategy mainly contemplates identification of water demands (both existing and projected) of district community and evolves mechanism for fetching required quantity of water by 2020. The strategy has two sides, one supply side management and other demand side management. The planned strategy will be realized by converging different on-going state or centrally sponsored Government schemes of different line departments.

Supply Side Management:

This mainly endeavors efforts for creating and developing water resources by harnessing maximum precipitation through typology of relevant interventions both for creating surface storages and ground water potential. This consists of construction of cement nalla bunds, earthen nalla bunds, check dams, farm ponds, diversion bunds, gabian bandharas etc. This will be also included works like desiltation of old water bodies, nalla training, nalla deepening etc. All these activities will be coupled up with dug wells and bore wells recharging structures, so as

to augment ground water storages wherever possible. The water harvesting inputs will be supported by soil and moisture conservation works like compartment bunding, terracing, Padkai development, continuous contour trenching etc. This will help to check the runoff in the field itself and make moisture available for crops during its crucial growth stages. These interventions will also support to enhance the ground water recharge to nearby wells and bore wells. The on-going schemes of agriculture departments like Integrated Watershed Management Programme will be availed for implementing said interventions.

The supply side management interventions will also include measures for strengthening of water distribution systems at field level. It will consist of key works generally undertaken by Department of Irrigation and Minor Irrigation viz. repairs, restoration and renovation of water bodies, construction of lined field channels and unlined field channels, support for procurement of drip or sprinkler irrigation sets, on farm development through distribution of pipes/raised beds and furrows etc. These interventions will also be united with minor irrigation lining and medium irrigation lining works.

Demand Side Management:

The demand side management interventions will primarily insist on restricting water demands of community by 2020. The key components considered for this is to optimize projected crop water requirement of the district. The presents trends, experiences, aspirations have been taken into consideration while designing of relevant strategies.

First of all, the district experiences fast urbanization and industrialization, since last couple of decades. The estimated data shows that so far 45000 ha area of district has been brought under urbanization and industrialization in last 5 years. Moreover it is predicted that approximately about 50000 ha agriculture area will be utilized by urban habitations or industries in next five years. Thus all together about 1.00 lakh ha agriculture area of the district will required to be excluded from agriculture business. The conversion of agriculture land into urban habitation is whether good or bad, may be a point of debate, but the fact is that this much area will minimize crop water requirement of the district. Yes it true that it needs be compensated by either domestic or industrial water needs of population or industries to be established on that area.

The last 4-5 year’s data shows that area under sorghum, bajara, wheat, gram, ground nut etc is experienced to be decreasing day by day. The data revels that approximately 2.0 lakhs ha area of

these crops will be declined in next five years. The area under sugarcane is also decreasing which is one of the best indications for optimizing water demand of farming business. The proposed strategy looks forward to minimize the sugarcane area by about 20000 ha in next five years. The strategy proposes that the decreased area of above crops will be used for cultivation of crops like maize, soyabean, horticulture /fruit crops etc. which needs comparatively less water.

Besides, the strategy also suggests that around 40% of proposed sugarcane area will be made mandatory for adopting drip irrigation facility, so as to save about 50% water loss in its cultivation. In addition other crops will be encouraged to grow with water saving micro-irrigation systems which will also save around 30% of its water requirements.

The strategy insists on recycling and reuse of waste water, especially generated in Municipal Corporation cities like Pune, Pimpari Chinchwad etc and further use it to irrigate the agriculture nearby area of these cities. The equal efforts will be placed to promote municipal towns like Indapur, Baramati, Bhor, Shirur, Khed, Purandar etc to take required sewerage treatment plants for recycling of waste water if these towns. The said water will also be used for irrigation purpose.

The strategy also envisages that equal efforts will be made to aware rural and urban community on water literacy. It will be also supported by organizing different capacity building events like trainings, crop demonstrations, exposure visits etc both for farmers and extension workers so as to strengthen them in taking up related extension work more effectively.

Action Plan

The action plan mainly consists of four key components pertaining to the strategy designed by the district. These components are

E) Har Khed ko Pani
F) Per drop More Crop
G) PMKSY Watershed
H) State Plan Schemes

A) Har Khet ko Pani:

The interventions proposed under this component are RRR of water bodies, construction of lined field channels, unlined field channels and micro-irrigation. In the district there seems to be very negligible scope for taking up lift irrigation schemes.

Pune district is gifted with many rivers running from west to east. There are nearly 200 K.T. Weirs on these rivers. These K.T. Weirs are 5 to 50 years old. Naturally they need some structural renovation and new metal leaf gates. Under this RRR head 55 K.T. Weirs (Kolhapur Type Bandhara) are proposed for Repairs, Renovation and Restoration. This will benefit in storing water at full capacity, stoppage of leakages through gates and structure. It is estimated that this will support to provide irrigation to 3195 ha area. This work is planned to be completed in next five years and will require Rs. 4719.38 lakhs.

Conveyance efficiency of Irrigation system varies with respect to type of canal viz. main canal, distributory, minor, field channel etc. Field channels needs lining, so as to improve efficiency. It also helps in establishing channel route within a chak. It gives water to the tail end farmers. We have proposed Nazare Medium Project. These projects are in DPAP area. Also we have proposed Kukadi and Chaskaman Major Project for field channel lining.

As a part of Command Area Development Work (CAD) unlined field channels are proposed in Kukadi and Chaskaman Irrigation Project. This will include field channels below outlet for bridging the gap between potential created and actual irrigation.

The construction of 2 lined and unlined canals proposed in the action plan will need Rs 480 and 3150 lakhs respectively. The project also proposed to support micro-irrigation facilities to crop on about 1000 ha area and hence provision of Rs 500.00lakhs has been made in the action plan.

The various schemes of Ministry of Water Resource will be availed for accomplishing the above

interventions during project period of five years.

B) Per Drop More Crop:

The plan intends to provide micro- irrigation systems to crops such as drip, sprinkler irrigation in both DPAP and Non-DPAP areas. It is proposed to provide support to 61502 and 252 drip and sprinkler irrigation sets respectively in DPAP areas whereas 13 drip irrigation sets in non-DPAP area. The cost required for developing this micro-irrigation facility would be Rs 13629.52 Lakhs.

The project also proposed on farm development with close pipe systems. The action plan decided to cover total 6 projects from the district for close pipe systems. The estimated command area under these planned 6 projects is 27235 ha and cost involved is Rs 15550 lakhs.

The action also assisted with drought proofing through check dams or water harvesting structures. Altogether 650 such structures are planned to complete during next 5 years and the cost required to achieve this work would be Rs. 16250 lakhs.

All these works are proposed to be completed by availing MoA & F.W. DAC schemes of irrigation department.

C) PMKSY Watershed:

The watershed interventions mainly purposed for arresting surface runoff, increasing soil moisture and ground water potential. These interventions are of two categories viz. newly created and renovation of old structures.

The newly interventions are comprised of Farm Pond (3130), Cement Nalla Bund (1287), Earthen Nalla Bund (2422), diversion bund (423) etc. The moisture conservation works like compartment bunding (69200 ha), continuous contour trenching (55819 ha), Deep CCT (5852 ha) loose boulder structures (27300) are also proposed in the district. Besides, deepening and desilting of 601 structures also planned to be undertaken in next five years. All these works will require cost of Rs 54882.90 lakhs.

The action plan also comprises of newly water harvesting structures such as earthen structures (5155), well recharging structures (2775), terracing (475), check dams (6435) and other ground water recharging structures (5000) which will need a fund of Rs. 5087.22 lakhs.

The action plan also proposed renovation of CNB (390), Nalla Bunds(635), desilting of nalla(1815) and repairs of KT weirs (90) in next five years. The cost required for undertaking this work will be Rs. 12402.50 Lakhs.

All the above proposed works under PMKSY-Watershed component will be completed through the grants of DoLR-MoRD under its watershed programme.

D) Convergence with MNRGA:

The district irrigation plan also emphasizes on converging the MGNREGA funds for carrying out water conservation, water harvesting and land development works. It has proposed 91 water conservation works, 247 water harvesting structures and some land development works of amounting to Rs 147.29 Lakhs. All these works are planned to be completed in next five years of project period.

E) State Planned Irrigation schemes:

1) Chaskaman Project

Chaskaman canal is designed as lined canal. But lining is still incomplete. Capacity of canal is 950 cusecs however; maximum discharge through it is 550 cusecs. Hence it takes more time for rotation. Also hampers irrigated area. Leakage through canal is nearly 40%. Hence we have proposed lining to Chaskaman canal which will save 2 TMC water. It includes establishment of Water Users Association for participatory irrigation management. This will bridge the gap between potential area declared & area actually irrigated.

2) Khadakwasla Project :

New Mutha Right Bank Canal was constructed in 1965. After 50 years, there are leakages through structures andcanal beds. Also due to paucity of funds canal system is paralyzed by silting, dilapidated structures, disturbance in canal section and garbage from city area. This has resulted in reduced efficiency and heavy water loss. We have proposed works of Extension and Improvement for this old systems main canal, distributaries and minors. Extension and Improvement programme work will be carried out in specifically damaged sections of canal. It also includes handing over command to farmers through participatory Irrigation Management and formation of Water Users Associations (WUAs).

3) Nira Left Bank Canal :

This is three seasonal canals running from Purandar, Baramati and Indapur talukas. This canal is

constructed in Pre-Independence era between 1910 and 1920. This canal needs immediate attention for Extension & Improvement. We have proposed renovation of 100 year old structures & disturbed Earthworks of main canal, distributaries and minors. This will improve efficiency of canal & will also reduce leakages. Works will be carried out at specific locations. It includes formation of Water Users Associations (WUAs) and handing over command area to WUAs for participatory irrigation management.

D) Scheme for Use of Recycled Waste Water (Mundhva Jackwell to Old Mutha Canal Project)

Mula-mutha River receives recycled waste water from urban areas upstream side i.e. Pune & Pimpri-Chinchwad Municipal Corporations.

Due to non planned water use of urban area, it is reducing water allotted for agriculture purpose. To come over this imbalance, a scheme for use of recycle waste water to agricultural land is in progress.

This scheme envisages construction of pump house, rising main and Old Mutha canal scheme is partially completed.

Total planned utilization is 6.5 TMC out of which 2.56 TMC water can be utilized with completed infrastructure. Balance 3.94 TMC water will be utilized only after renovation of Old Mutha canal. This canal was constructed in 1880. Land acquisition is already completed. Canal length is 108 km out of which work upto 82 km is completed.

Balance work from km.83 to 108 is proposed here. This is important project as it is directly linked to availability of water supply for Pune. If we use more waste water to agriculture there fresh water will be saved in dam. The balance cost of scheme is 20 Cr & it’s possible to complete within a year.

Total proposed district irrigation plan is Rs 158048.81 lakhs consisting Rs 8849.38 for handling the component Har Khet ko Pani, Rs. 45429.52 for component Per Drop More Crop, Rs 72372.62 for PMKSY Watershed, Rs. 147.29 for works through MGNREGA and Rs 31250.00 from state irrigation schemes.

5 – Strategic Action Plan for Irrigation in District Under PMKSY.

Name of the State : Maharashtra

Name of District : Pune

Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period of implementation (5/7 Years)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	8	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0	0
2		MoWR		Medium Irrigation	0	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0	0
4		MoWR	Har Khet ko Pani	Lift Irrigation	0	0	0	0
5		MoWR		Ground Water Development	0	0	0	0
6		MoWR		RRR of water bodies	55	3490.00	5 Years	4719.38
7		MoWR		**Construction of Field channels (for command area only)**				
7.1		MoWR		LinedField Channels	2	3195.00	5 Years	480.00
7.2		MoWR		Unlined Channels	2	9000.00	5 Years	3150.00
8		MoWR		Micro Irrigation	2	1000.00	5 Years	500.00
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	61502	45295.85	5 Years	13588.76
10		MOA & FW.DAC & FW		DPAP Sprinkler	252	208.79	5 Years	38.66
11		MOA & FW.DAC & FW		Non DPAP Drip	13	7.00	5 Years	2.10
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00		0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00		0.00
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	650	1218.75	5 Years	16250.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0.00		0.00
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	6	27235.00	5 Years	15550.00
17	Pune	DoLR-MoRD	PMKSY Watershed	**Newly Created WHS**				
17.1		DoLR-MoRD		Farm Ponds	3130	5321.00	5 Years	2505.00
17.2		DoLR-MoRD		cement nala bund	1287	5791.50	5 Years	16589.34
17.3		DoLR-MoRD		earthen nala bund	2422	12110.00	5 Years	11139.48
17.4		DoLR-MoRD		diversion bund	423	1692.00	5 Years	3589.00
17.5		DoLR-MoRD		compartment bunding	69200	31140.00	5 Years	5495.00
17.6		DoLR-MoRD		cct	55819	18420.27	5 Years	8797.00
17.7		DoLR-MoRD		Deepening/ desilting of stucture	601	1803.00	5 Years	2887.00
17.8		DoLR-MoRD		Deep cct	5852	1404.48	5 Years	2691.46
17.9				LBS	27300	0.00	5 Years	1189.62
17.10				Earthen Str.	5155	1288.75	5 Years	307.50
17.11				Well Recharge	2775	346.88	5 Years	555.00
17.12				Terracing	475	0.00	5 Years	1187.50
17.13				Check Dams	6435	0.00	5 Years	1287.22
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	5000	625.00	5 Years	1750.00
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00		0.00
18			PMKSY Watershed	**Renovated WHS**				
18.1		DoLR-MoRD		Farm Ponds	0	0.00		0
18.2		DoLR-MoRD		CNB	390		5 Years	1560.00
18.3		DoLR-MoRD		Nallah Bunds	635	2540.00	5 Years	952.50
18.4				Desilting of Nala	1815	7260.00	5 Years	9075.00
18.5		DoLR-MoRD		KTW repair	90	1145.00	5 Years	815.00

5 – Strategic Action Plan for Irrigation in District Under PMKSY.								
Name of the State : Maharashtra								
Name of District : Pune								
Sr. No.	**Name of District**	**Concerned Ministry / Department**	**Component**	**Activity**	**Total Number**	**Command Area / Irrigation Potential (Ha)**	**Period of implementation (5/7 Years)**	**Estimated cost (in Rs. Lakh)**
19	Pune	DoLR-MoRD	**Convergence with MGNREGA**	**Newly Created**				
19.1		DoLR-MoRD		Water Consurvation	91	154.96	5 Years	15.47
19.2		DoLR-MoRD		Water Harvesting	247	209.95	5 Years	104.91
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0		0
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0		0
19.5		DoLR-MoRD		Land development	0	170.04	5 Years	26.91
20		DoLR-MoRD		**Renovation**				
20.1		DoLR-MoRD		Renovation water bodies including	0	0		0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0		0
21				**State Planned Scheme of Irrigation**				
21		State Irrigation Department		Major Irrigation (Lining)	284	0.00	5 Years	29000.00
21.1		State Irrigation Department		Medium Irrigation (Lining)	45	0.00	5 Years	2250.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00		0.00
21.3		State Irrigation Department			0	0.00		0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00		0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00		0.00
24		Externaly aded projects			0	0.00		0.00
25		Other loan projects like NABARD			0	0.00		0.00
Total					**251955**	**183243.22**	**5 Years**	**158048.81**

Strategic Action Plan for Irrigation of Blocks Pune District (PMKSY)

Block-wise Action Plan for Irrigation – Pune (PMKSY)			
Sr. No.	**Name of Blocks**	**(District Pune)**	**Page.no.**
1	Haveli		78
2	Maval		79
3	Mulshi		80
4	Bhor		81
5	Velha		82
6	Khed		83
7	Ambegaon		84
8	Junnar		85
9	Shirur		86
10	Baramati		87
11	Indapur		88
12	Daund		89
13	Purandar		90

1. Strategic Action Plan for Irrigation Block - Haveli

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka :Haveli							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	5 Years	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation				
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	3	158	5 Years	316
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	1252	858.04	5 Years	257.412
10	MOA & FW.DAC & FW		DPAP Sprinkler			5 Years	
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	35	65.625	5 Years	875
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**	1	880	5 Years	452.11
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	250	425	5 Years	125
17.2	DoLR-MoRD		cement nala bund	150	675	5 Years	1733.51
17.3	DoLR-MoRD		earthen nala bund	271	1355	5 Years	1001.33
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	500	225	5 Years	55
17.6	DoLR-MoRD		cct	500	165	5 Years	240
17.7	DoLR-MoRD		Deepening/ desilting of stucture		0	5 Years	
17.8	DoLR-MoRD		Deep cct	1170	280.8	5 Years	538.29
			LBS	3925	0	5 Years	167.02
			Earthen Str.	250	62.5	5 Years	15
			Well Recharge	150	18.75	5 Years	30
			Terracing		0	5 Years	
			Check Dams	1170		5 Years	234.04
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	300	37.5	5 Years	105
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	25	75	5 Years	100
18.3	DoLR-MoRD		Nallah Bunds	25	100	5 Years	37.5
			Desilting /deepening of Nala	100	400	5 Years	500
18.4	DoLR-MoRD		KTW repair			5 Years	
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	33		5 Years	1000
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects						
25	Other loan projects like NABARD						
					5822.37		7793.54

2. Strategic Action Plan for Irrigation Block - Maval

5 - Strategic Action Plan for Irrigation in District Under PMKSY.

Name of the State : Maharashtra

Name of Taluka : Maval

Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period of implementation (5/7 Years)	Estimated cost (in Rs. Lakh)
1	3	4	5	6	7		8
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	1	20	5 Years	40
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	74	70.5	5 Years	21.15
10	MOA & FW.DAC & FW		DPAP Sprinkler	0	0	5 Years	0
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	50	93.75	5 Years	1250
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	70	119	5 Years	70
17.2	DoLR-MoRD		cement nala bund	170	765	5 Years	780
17.3	DoLR-MoRD		earthen nala bund	231	1155	5 Years	944.83
17.4	DoLR-MoRD		diversion bund	40	160	5 Years	343
17.5	DoLR-MoRD		compartment bunding		0	5 Years	
17.6	DoLR-MoRD		cct	5800	1914	5 Years	845
17.7	DoLR-MoRD		Deepening/ desilting of stucture	10	30	5 Years	48
17.8	DoLR-MoRD		Deep cct	878	210.72	5 Years	403.72
			LBS	3425	0	5 Years	142.02
			Earthen Str.	125	31.25	5 Years	7.5
			Well Recharge	100	12.5	5 Years	20
			Terracing		0	5 Years	
			Check Dams	1170	0	5 Years	234.04
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	100	12.5	5 Years	35
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	15	45	5 Years	60
18.3	DoLR-MoRD		Nallah Bunds	25	100	5 Years	37.5
			Desilting /deepening of Nala	40	160	5 Years	200
18.4	DoLR-MoRD		KTW repair	1	18	5 Years	25
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					4958.37	5 Years	5518.09

3. Strategic Action Plan for Irrigation Block - Mulshi

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Mulshi							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	5 Years	Estimated cost (in Rs. Lakh)
1	3	4	5	6	7	5 Years	8
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation				
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**			5 Years	
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	106	86.36	5 Years	25.908
10	MOA & FW.DAC & FW		DPAP Sprinkler			5 Years	
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	30	56.25		750
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	95	161.5	5 Years	95
17.2	DoLR-MoRD		cement nala bund	75	337.5	5 Years	920.85
17.3	DoLR-MoRD		earthen nala bund	206	1030	5 Years	796.83
17.4	DoLR-MoRD		diversion bund	35	140	5 Years	308
17.5	DoLR-MoRD		compartment bunding		0	5 Years	
17.6	DoLR-MoRD		cct	5900	1947	5 Years	859
17.7	DoLR-MoRD		Deepening/ desilting of stucture	10	30	5 Years	48
17.8	DoLR-MoRD		Deep cct	878	210.72	5 Years	403.72
			LBS	3425	0	5 Years	142.02
			Earthen Str.	250	62.5	5 Years	15
			Well Recharge	100	12.5	5 Years	20
			Terracing		0	5 Years	
			Check Dams	1170			234.04
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	300	37.5	5 Years	105
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	15	45	5 Years	60
18.3	DoLR-MoRD		Nallah Bunds	50	200	5 Years	75
			Desilting /deepening of Nala	25	100		125
18.4	DoLR-MoRD		KTW repair	2	22	5 Years	25
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains				
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects						
25	Other loan projects like NABARD						
					4519.98		5019.70

4. Strategic Action Plan for Irrigation Block - Bhor

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Bhor							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period of implementation (5/7 Years)	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	4	146	5 Years	292
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	196	124.02	5 Years	37.206
10	MOA & FW.DAC & FW		DPAP Sprinkler	16	10.3	5 Years	1.87
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	25	46.875	5 Years	625
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**	1	432	5 Years	650.00
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	180	306	5 Years	180
17.2	DoLR-MoRD		cement nala bund	75	337.5	5 Years	1199
17.3	DoLR-MoRD		earthen nala bund	90	450	5 Years	536
17.4	DoLR-MoRD		diversion bund	101	404	5 Years	836
17.5	DoLR-MoRD		compartment bunding			5 Years	
17.6	DoLR-MoRD		cct	6300	2079	5 Years	917
17.7	DoLR-MoRD		Deepening/ desilting of stucture	15	45	5 Years	72
17.8	DoLR-MoRD		Deep cct			5 Years	
			LBS	1250		5 Years	62.5
			Earthen Str.	530	132.5	5 Years	30
			Well Recharge	125	15.625	5 Years	25
			Terracing			5 Years	
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	100	12.5	5 Years	35
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		cement nalla bund	50	150	5 Years	200
18.3	DoLR-MoRD		Nallah Bunds	75	300	5 Years	112.5
			Desilting /deepening of Nala	125	500	5 Years	625
18.4	DoLR-MoRD		KTW repair	1	13	5 Years	30
18.5	DoLR-MoRD		Other Ground Water Recharge			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					5545.47	5 Years	6477.41

5. Strategic Action Plan for Irrigation Block - Velha

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Velha							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	5 Years	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation				
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	4	150	5 Years	300
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip			5 Years	
10	MOA & FW.DAC & FW		DPAP Sprinkler			5 Years	
11	MOA & FW.DAC & FW		Non DPAP Drip	13	7	5 Years	2.1
12	MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0	5 Years	0
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	25	46.875		625
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	130	221	5 Years	130
17.2	DoLR-MoRD		cement nala bund	45	202.5	5 Years	720
17.3	DoLR-MoRD		earthen nala bund	95	475	5 Years	566
17.4	DoLR-MoRD		diversion bund	92	368	5 Years	751
17.5	DoLR-MoRD		compartment bunding		0	5 Years	
17.6	DoLR-MoRD		cct	6500	2145	5 Years	947
17.7	DoLR-MoRD		Deepening/ desilting of stucture	11	33	5 Years	53
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	1500	0	5 Years	75
			Earthen Str.	700	175	5 Years	42
			Well Recharge	175	21.875	5 Years	35
			Terracing			5 Years	
			Check Dams				
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	300	37.5	5 Years	105
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	30	90	5 Years	120
18.3	DoLR-MoRD		Nallah Bunds	50	200	5 Years	75
			Desilting /deepening of Nala	100	400		500
18.4	DoLR-MoRD		KTW repair			5 Years	
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains				
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects						
25	Other loan projects like NABARD						
					4613.90		5057.43

6. Strategic Action Plan for Irrigation Block - Khed

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Khed							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	5 Years	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**			5 Years	
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	744	420.61	5 Years	126.183
10	MOA & FW.DAC & FW		DPAP Sprinkler	119	105.15	5 Years	19.164
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	200	340	5 Years	200
17.2	DoLR-MoRD		cement nala bund	70	315	5 Years	1119
17.3	DoLR-MoRD		earthen nala bund	120	600	5 Years	715
17.4	DoLR-MoRD		diversion bund	85	340	5 Years	684
17.5	DoLR-MoRD		compartment bunding	350	157.5	5 Years	38.5
17.6	DoLR-MoRD		cct	5400	1782	5 Years	786
17.7	DoLR-MoRD		Deepening/ desilting of stucture	30	90	5 Years	144
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	750	0	5 Years	37.5
			Earthen Str.	300	75	5 Years	18
			Well Recharge	250	31.25	5 Years	50
			Terracing	275	0	5 Years	687.5
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	400	50	5 Years	140
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	75	225	5 Years	300
18.3	DoLR-MoRD		Nallah Bunds	125	500	5 Years	187.5
			Desilting /deepening of Nala	225	900	5 Years	1125
18.4	DoLR-MoRD		KTW repair	1	15	5 Years	15
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	67		5 Years	3600
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					6100.16	5 Years	11503.68

7. Strategic Action Plan for Irrigation Block - Ambegaon

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Ambegaon							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7)years	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	9	740	5 Years	1414.83
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	2711	1929.4	5 Years	578.82
10	MOA & FW.DAC & FW		DPAP Sprinkler			5 Years	
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	245	416.5	5 Years	245
17.2	DoLR-MoRD		cement nala bund	94	423	5 Years	1503
17.3	DoLR-MoRD		earthen nala bund	115	575	5 Years	685
17.4	DoLR-MoRD		diversion bund	30	120	5 Years	286
17.5	DoLR-MoRD		compartment bunding	250	112.5	5 Years	27.5
17.6	DoLR-MoRD		cct	6400	2112	5 Years	932
17.7	DoLR-MoRD		Deepening/ desilting of stucture	50	150	5 Years	240
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	500	0	5 Years	25
			Earthen Str.	150	37.5	5 Years	9
			Well Recharge	125	15.625	5 Years	25
			Terracing		0	5 Years	
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	450	56.25	5 Years	157.5
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	20	60	5 Years	80
18.3	DoLR-MoRD		Nallah Bunds	25	100	5 Years	37.5
			Desilting /deepening of Nala	40	160	5 Years	200
18.4	DoLR-MoRD		KTW repair	40	514	5 Years	540
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					7675.43	5 Years	8497.48

8. Strategic Action Plan for Irrigation Block - Junnar

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Junnar							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7 Years)	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	19	870	5 Years	931.73
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**				
7.2	MoWR		**Unlined Channels**	1	2250	5 Years	787.50
8	MoWR		**Micro Irrigation**	1	250	5 Years	125.00
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	8634	5052.47	5 Years	1515.741
10	MOA & FW.DAC & FW		DPAP Sprinkler	5	3.66	5 Years	0.822
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	250	425	5 Years	250
17.2	DoLR-MoRD		cement nala bund	95	427.5	5 Years	1519
17.3	DoLR-MoRD		earthen nala bund	125	625	5 Years	745
17.4	DoLR-MoRD		diversion bund	40	160	5 Years	381
17.5	DoLR-MoRD		compartment bunding	2000	900	5 Years	150
17.6	DoLR-MoRD		cct	6300	2079	5 Years	917
17.7	DoLR-MoRD		Deepening/ desilting of stucture	90	270	5 Years	432
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	750	0	5 Years	37.5
			Earthen Str.	250	62.5	5 Years	15
			Well Recharge	175	21.875	5 Years	35
			Terracing	200	0	5 Years	500
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	500	62.5	5 Years	175
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	25	75	5 Years	100
18.3	DoLR-MoRD		Nallah Bunds	25	100	5 Years	37.5
			Desilting /deepening of Nala	60	240	5 Years	300
18.4	DoLR-MoRD		KTW repair	10	137	5 Years	50
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					14165.16	5 Years	10516.12

9. Strategic Action Plan for Irrigation Block - Shirur

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Shirur							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7 Years)	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	9	870	5 Years	352.82
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**	1	6750	5 Years	2362.50
8	MoWR		**Micro Irrigation**	1	750	5 Years	375.00
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	7517	6492.24	5 Years	1947.672
10	MOA & FW.DAC & FW		DPAP Sprinkler	63	63.4	5 Years	11.704
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**	1	152	5 Years	650.00
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	220	374	5 Years	220
17.2	DoLR-MoRD		cement nala bund	85	382.5	5 Years	1359
17.3	DoLR-MoRD		earthen nala bund	90	450	5 Years	536
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	12000	5400	5 Years	901
17.6	DoLR-MoRD		cct	3500	1155	5 Years	510
17.7	DoLR-MoRD		Deepening/ desilting of stucture	140	420	5 Years	673
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	250	0	5 Years	12.5
			Earthen Str.	500	125	5 Years	30
			Well Recharge	250	31.25	5 Years	50
			Terracing		0	5 Years	
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	600	75	5 Years	210
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	10	30	5 Years	40
18.3	DoLR-MoRD		Nallah Bunds	10	40	5 Years	15
			Desilting /deepening of Nala	75	300	5 Years	375
18.4	DoLR-MoRD		KTW repair	19	230	5 Years	40
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	67	0	5 Years	8400
21.1	State Irrigation Department		Medium Irrigation (Lining)	30	0	5 Years	1500
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					24244.04	5 Years	22082.53

10. Strategic Action Plan for Irrigation Block - Baramati

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Baramati							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7 Years)	Estimated cost (in Rs. Lakh)
1	3	4	5	6	7		8
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**			5 Years	
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**	1	1295	5 Years	192.00
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	12818	9337	5 Years	2801.1
10	MOA & FW.DAC & FW		DPAP Sprinkler	10	6.4	5 Years	1.24
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**		6398	5 Years	3287.07
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	125	212.5	5 Years	125
17.2	DoLR-MoRD		cement nala bund	91	409.5	5 Years	1126.66
17.3	DoLR-MoRD		earthen nala bund	245	1225	5 Years	1028.83
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	14300	6435	5 Years	1073
17.6	DoLR-MoRD		cct	1800	594	5 Years	262
17.7	DoLR-MoRD		Deepening/ desilting of stucture	75	225	5 Years	360
17.8	DoLR-MoRD		Deep cct	878	210.72	5 Years	403.72
			LBS	3175	0	5 Years	129.52
			Earthen Str.	250	62.5	5 Years	15
			Well Recharge	200	25	5 Years	40
			Terracing		0	5 Years	
			Check Dams	1170		5 Years	234.04
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	500	62.5	5 Years	175
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	25	75	5 Years	100
18.3	DoLR-MoRD		Nallah Bunds	100	400	5 Years	150
			Desilting /deepening of Nala	250	1000	5 Years	1250
18.4	DoLR-MoRD		KTW repair	3	40	5 Years	25
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	50		5 Years	1500
21.1	State Irrigation Department		Medium Irrigation (Lining)	8		5 Years	400
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					28166.77	5 Years	16190.51

11. Strategic Action Plan for Irrigation Block - Indapur

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Indapur							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7 Years)	Estimated cost (in Rs. Lakh)
1	3	4	5	6	7		8
1	MoWR	AIBP	Major Irrigation			5 Years	
2	MoWR		Medium Irrigation			5 Years	
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	4	375	5 Years	750
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**			5 Years	
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	19196	14936	5 Years	4480.809
10	MOA & FW.DAC & FW		DPAP Sprinkler	6	4.48	5 Years	0.898
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	65	121.875	5 Years	1625
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**			5 Years	
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	1000	1700	5 Years	500
17.2	DoLR-MoRD		cement nala bund	166	747	5 Years	2202.66
17.3	DoLR-MoRD		earthen nala bund	497	2485	5 Years	2013.83
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	7500	3375	5 Years	825
17.6	DoLR-MoRD		cct	1500	495	5 Years	720
17.7	DoLR-MoRD		Deepening/ desilting of stucture		0	5 Years	
17.8	DoLR-MoRD		Deep cct	878	210.72	5 Years	403.72
			LBS	3175	0	5 Years	129.52
			Earthen Str.	350	87.5	5 Years	21
			Well Recharge	200	25	5 Years	40
			Terracing		0	5 Years	
			Check Dams	585		5 Years	117.02
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	500	62.5	5 Years	175
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	25	75	5 Years	100
18.3	DoLR-MoRD		Nallah Bunds	25	100	5 Years	37.5
			Desilting /deepening of Nala	275	1100	5 Years	1375
18.4	DoLR-MoRD		KTW repair	4	45	5 Years	30
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.			5 Years	
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	33		5 Years	8500
21.1	State Irrigation Department		Medium Irrigation (Lining)			5 Years	
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					25986.23	5 Years	24058.29

12. Strategic Action Plan for Irrigation Block - Daund

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Dund							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period of (5/7 Years)	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			Period of implementation (5/7 Years)	
2	MoWR		Medium Irrigation				
3	MoWR		Surface Minor Irrigation			5 Years	
4	MoWR	Har Khet ko Pani	Lift Irrigation			5 Years	
5	MoWR		Ground Water Development			5 Years	
6	MoWR		**RRR of water bodies**	1	108	5 Years	216
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**			5 Years	
7.2	MoWR		**Unlined Channels**				
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	5050	3976.16	5 Years	1192.848
10	MOA & FW.DAC & FW		DPAP Sprinkler	5	2.5	5 Years	0.53
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**		3648	5 Years	1874.22
17	DoLR-MoRD	PMKSY Watershed	Newly Created WHS				
17.1	DoLR-MoRD		Farm Ponds	110	187	5 Years	110
17.2	DoLR-MoRD		cement nala bund	76	342	5 Years	887.66
17.3	DoLR-MoRD		earthen nala bund	197	985	5 Years	736.83
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	16300	7335	5 Years	1224
17.6	DoLR-MoRD		cct	1419	468.27	5 Years	207
17.7	DoLR-MoRD		Deepening/ desilting of stucture	20	60	5 Years	96
17.8	DoLR-MoRD		Deep cct	1170	280.8	5 Years	538.29
			LBS	3175	0	5 Years	129.52
			Earthen Str.	250	62.5	5 Years	15
			Well Recharge	175	21.875	5 Years	35
			Terracing		0	5 Years	
			Check Dams	1170		5 Years	234.04
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	450	56.25	5 Years	157.5
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	Renovated WHS				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	25	75		100
18.3	DoLR-MoRD		Nallah Bunds	50	200	5 Years	75
			Desilting /deepening of Nala	150	600	5 Years	750
18.4	DoLR-MoRD		KTW repair			5 Years	
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	Newly Created				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		Renovation				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.				
21			State Planned Scheme of Irrigation				
21	State Irrigation Department		Major Irrigation (Lining)	34		5 Years	6000
21.1	State Irrigation Department		Medium Irrigation (Lining)				
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					18562.01	5 Years	16090.77

13. Strategic Action Plan for Irrigation Block -Purandar

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of Taluka : Purander							
Sr. No.	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Period (5/7 Years)	Estimated cost (in Rs. Lakh)
1	MoWR	AIBP	Major Irrigation			(5/7 Years)	
2	MoWR		Medium Irrigation				
3	MoWR		Surface Minor Irrigation				
4	MoWR	Har Khet ko Pani	Lift Irrigation				
5	MoWR		Ground Water Development				
6	MoWR		**RRR of water bodies**	1	53	5 Years	106
7	MoWR		**Construction of Field channels (for command area only)**				
7.1	MoWR		**LinedField Channels**	1	1900	5 Years	288.00
7.2	MoWR		**Unlined Channels**				
8	MoWR		**Micro Irrigation**			5 Years	
9	MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	3204	2013.05	5 Years	603.915
10	MOA & FW.DAC & FW		DPAP Sprinkler	28	12.9	5 Years	2.43
11	MOA & FW.DAC & FW		Non DPAP Drip			5 Years	
12	MOA & FW.DAC & FW		Non DPAP Sprinkler			5 Years	
13	MOA & FW.DAC & FW		Topping up of MGNREGA			5 Years	
14	MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	60	112.5	5 Years	1500
15	MOA & FW.DAC & FW		Secondary Storage Structure			5 Years	
16	MOA & FW.DAC & FW		**On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)**	3	15725	5 Years	8636.60
17	DoLR-MoRD	PMKSY Watershed	**Newly Created WHS**				
17.1	DoLR-MoRD		Farm Ponds	255	433.5	5 Years	255
17.2	DoLR-MoRD		cement nala bund	95	427.5	5 Years	1519
17.3	DoLR-MoRD		earthen nala bund	140	700	5 Years	834
17.4	DoLR-MoRD		diversion bund		0	5 Years	
17.5	DoLR-MoRD		compartment bunding	16000	7200	5 Years	1201
17.6	DoLR-MoRD		cct	4500	1485	5 Years	655
17.7	DoLR-MoRD		Deepening/ desilting of stucture	150	450	5 Years	721
17.8	DoLR-MoRD		Deep cct		0	5 Years	
			LBS	2000	0	5 Years	100
			Earthen Str.	1250	312.5	5 Years	75
			Well Recharge	750	93.75	5 Years	150
			Terracing		0	5 Years	
			Check Dams			5 Years	
17.9	DoLR-MoRD		Other Ground Water Recharge Structure	500	62.5	5 Years	175
17.1	DoLR-MoRD		Fishery Ponds / cattle pond			5 Years	
18		PMKSY Watershed	**Renovated WHS**				
18.1	DoLR-MoRD		Farm Ponds			5 Years	
18.2	DoLR-MoRD		CNB	50	150		200
18.3	DoLR-MoRD		Nallah Bunds	50	200	5 Years	75
			Desilting /deepening of Nala	350	1400	5 Years	1750
18.4	DoLR-MoRD		KTW repair	9	111	5 Years	35
18.5	DoLR-MoRD		Other Ground Water Recharge Structure			5 Years	
19	DoLR-MoRD	Convergence with MGNREGA	**Newly Created**				
19.1	DoLR-MoRD		Water Consurvation	7	11.92	5 Years	1.19
19.2	DoLR-MoRD		Water Harvesting	19	16.15	5 Years	8.07
19.3	DoLR-MoRD		Creation of irrigation canals and drains			5 Years	
19.4	DoLR-MoRD		Providing infrastructure Irrigation			5 Years	
19.5	DoLR-MoRD		Land development		13.08	5 Years	2.07
20	DoLR-MoRD		**Renovation**				
20.1	DoLR-MoRD		Renovation water bodies including			5 Years	
20.2	DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.				
21			**State Planned Scheme of Irrigation**				
21	State Irrigation Department		Major Irrigation (Lining)			5 Years	
21.1	State Irrigation Department		Medium Irrigation (Lining)	7			350
21.2	State Irrigation Department		Surface minor Irrigation			5 Years	
21.3	State Irrigation Department					5 Years	
22	Irrigation Scheme of State Agriculture Department					5 Years	
23	Irrigation Scheme of other Line Departments of State Govt.					5 Years	
24	Externaly aded projects					5 Years	
25	Other loan projects like NABARD					5 Years	
					32883.35	5 Years	19243.28

Annexure – II

Production and Productivity of Major Crops

Production and Productivity of Major Crops			
Sr. No.	**Name of Blocks**	**(District Pune)**	**Page.no.**
1	Haveli		92
2	Maval		93
3	Mulshi		94
4	Bhor		95
5	Velha		96
6	Khed		97
7	Ambegaon		98
8	Junnar		99
9	Shirur		100
10	Baramati		101
11	Indapur		102
12	Daund		103
13	Purandar		104

Production and Productivity of Major Crops – Haveli

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat.
Name of the State : Maharashtra												
Name of the District : Pune												
Block : Haveli												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	2744	39870	1453	67000	205	4223	2060	97427	44093	1757	82214
	Coarse Cereals	7957	106544	1339	45337.75	221	3448	1560	60043	109992	1450	52690
	Pules	798	4828	605	38558.25	145	1160	800	55337	5988	703	46948
	Oil Seeds	991	12893	1301	37686.83	970	13599	1402	55908	26492	1352	46797
	Cotton		0	0			0		0	0	0	0
	Vegetables	1225	93100	7600	70625	2150	170925	7950	85000	264025	7775	77813
B) Rabi	Cereals	1570	25418	1619	62084	1920	35328	1840	67084	60746	1730	64584
	Coarse Cereals	28501	372508	1307	30592	2795	39130	1400	65256	411638	1354	47924
	Pulses	1223	10298	842	37727	838	7123	850	52770	17421	846	45249
	Oil Seeds	0	0	0		30	135	450	35200	135	450	17600
	Vegetables	793	515450	65000	75625	1768	135287	7652	98000	650737	36326	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	110	880	800	40000	880	800	880
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	200	5234	2617	60000	5234	2617	5234
	Sugarcane		0	0	0	9328	9794	105	120540	9794	105	9794
	Vegetables		0	0	0	270	189000	70000	90000	189000	70000	189000
	Fruit crop	261	156600	60000	98000	1298	1038400	80000	313520	1195000	70000	205760
Total	Cereals	4314	65289	1536	64542	2125	39551	1950	82256	104840	1743	73399
	Coarse Cereals	36458	479052	1323	37965	3126	43458	1253	55100	522510	1288	46532
	Pules	2021	15126	724	38143	983	8283	825	54054	23409	774	46098
	Oil Seeds	991	12893	651	18843	1200	18968	1490	50369	31861	1070	34606
	Cotton	0	0	0	0	0	0	0	0	0	0	0
	Sugarcane	0	0	0	0	9328	9794	105	120540	9794	105	120540
	Vegetables	2018	608550	36300	73125	4188	495212	28534	91000	1103762	32417	82063
	Fruit crop	261	156600	60000	98000	1298	1038400	80000	313520	1195000	70000	205760

Production and Productivity of Major Crops – Maval

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Maval												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	4309	69030	1602	67000	4460	103472	2320	97427	172502	1961	82214
	Coarse Cereals	3043	9890	325	45337.75	310	1891	610	60043	11781	468	52690
	Pules	163	1304	800	38558.25	140	1330	950	55337	2634	875	46948
	Oil Seeds	1110	12987	1170	37686.83	700	12880	1840	55908	25867	1505	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	46	15180	33000	70625	2022	707700	35000	85000	722880	34000	77813
B) Rabi	Cereals	1086	15552	1432	62084	1084	16986	1567	67084	32538	1500	64584
	Coarse Cereals	9240	75306	815	30592	1270	12611	993	65256	87917	904	47924
	Pulses	687	4891	712	37727	173	1640	948	52770	6531	830	45249
	Oil Seeds	10	60	600	28038		0		35200	60	600	31619
	Vegetables	41	15170	37000	75625	1002	360720	36000	98000	375890	36500	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0		0		40000	0	0	0
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	300	5100	1700	60000	5100	1700	5100
	Sugarcane		0	0	0	1753	140240	8000	120540	140240	8000	140240
	Vegetables		0	0	0	25	8250	33000	90000	8250	33000	8250
	Fruit crop	95	34200	36000	98000	510	204000	40000	313520	238200	38000	205760
Total	Cereals	5395	84582	1517	64542	5544	120458	1944	82256	205040	1730	73399
	Coarse Cereals	12283	85196	570	37965	1580	14502	534	55100	99698	552	46532
	Pules	850	6195	756	38143	313	2970	949	54054	9165	853	46098
	Oil Seeds	1120	13047	885	32862	1000	17980	1180	50369	31027	1033	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	1753	140240	8000	120540	140240	8000	120540
	Vegetables	87	30350	35000	73125	3049	1076670	34667	91000	1107020	34833	82063
	Fruit crop	95	34200	36000	98000	510	204000	40000	313520	238200	38000	205760

Production and Productivity of Major Crops - Mulshi

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block:Mulshi												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	13122	167962	1280	67000	4335	73262	1690	97427	241223	1485	82214
	Coarse Cereals	2947	28556	969	45337.75	68	782	1150	60043	29338	1060	52690
	Pules	243	1094	450	38558.25	62	384	620	55337	1478	535	46948
	Oil Seeds	259	2883	1113	37686.83	2000	25000	1250	55908	27883	1182	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	47	27730	59000	70625	75	45000	60000	85000	72730	59500	77813
B) Rabi	Cereals	265	4264	1609	62084	2055	37011	1801	67084	41274	1705	64584
	Coarse Cereals	1616	12928	800	30592	8644	82291	952	65256	95219	876	47924
	Pulses	157	1209	770	37727	1253	10513	839	52770	11722	805	45249
	Oil Seeds	30	168	560	28038	50	300	600	35200	468	580	31619
	Vegetables	87	40890	47000	75625	231	138600	60000	60000	179490	53500	67813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0		0		40000	0	0	0
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	120	2390	1992	60000	2390	1992	2390
	Sugarcane		0	0	0	1651	1320800	80000	120540	1320800	80000	1320800
	Vegetables		0	0	0	12	7560	63000	42000	7560	63000	7560
	Fruit crop	250	87500	35000	98000	421	168400	40000	313520	255900	37500	205760
Total	Cereals	13387	172225	1445	64542	6390	110272	1746	82256	282498	1595	73399
	Coarse Cereals	4563	41484	885	37965	8712	83073	701	55100	124557	793	46532
	Pules	400	2302	610	38143	1315	10897	730	54054	13199	670	46098
	Oil Seeds	289	3051	837	32862	2170	27690	1281	50369	30741	1059	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane		0	0	0	1651	1320800	80000	120540	1320800	80000	120540
	Vegetables	134	68620	53000	73125	318	191160	61000	62333	259780	57000	67729
	Fruit crop	250	172225	35000	98000	421	168400	40000	313520	340625	37500	205760

Production and Productivity of Major Crops - Bhor

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat.
Name of the State : Maharashtra												
Name of the District : Pune												
Block : Haveli												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	2744	39870	1453	67000	205	4223	2060	97427	44093	1757	82214
	Coarse Cereals	7957	106544	1339	45337.75	221	3448	1560	60043	109992	1450	52690
	Pules	798	4828	605	38558.25	145	1160	800	55337	5988	703	46948
	Oil Seeds	991	12893	1301	37686.83	970	13599	1402	55908	26492	1352	46797
	Cotton		0	0			0		0	0	0	0
	Vegetables	1225	93100	7600	70625	2150	170925	7950	85000	264025	7775	77813
B) Rabi	Cereals	1570	25418	1619	62084	1920	35328	1840	67084	60746	1730	64584
	Coarse Cereals	28501	372508	1307	30592	2795	39130	1400	65256	411638	1354	47924
	Pulses	1223	10298	842	37727	838	7123	850	52770	17421	846	45249
	Oil Seeds	0	0	0		30	135	450	35200	135	450	17600
	Vegetables	793	515450	65000	75625	1768	135287	7652	98000	650737	36326	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	110	880	800	40000	880	800	880
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	200	5234	2617	60000	5234	2617	5234
	Sugarcane		0	0	0	9328	9794	105	120540	9794	105	9794
	Vegetables		0	0	0	270	189000	70000	90000	189000	70000	189000
	Fruit crop	261	156600	60000	98000	1298	1038400	80000	313520	1195000	70000	205760
Total	Cereals	4314	65289	1536	64542	2125	39551	1950	82256	104840	1743	73399
	Coarse Cereals	36458	479052	1323	37965	3126	43458	1253	55100	522510	1288	46532
	Pules	2021	15126	724	38143	983	8283	825	54054	23409	774	46098
	Oil Seeds	991	12893	651	18843	1200	18968	1490	50369	31861	1070	34606
	Cotton	0	0	0	0	0	0	0	0	0	0	0
	Sugarcane	0	0	0	0	9328	9794	105	120540	9794	105	120540
	Vegetables	2018	608550	36300	73125	4188	495212	28534	91000	1103762	32417	82063
	Fruit crop	261	156600	60000	98000	1298	1038400	80000	313520	1195000	70000	205760

Production and Productivity of Major Crops - Velha

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Velha												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	7410	107223	1447	67000	224	3851	1719	97427	111073	1583	82214
	Coarse Cereals	1976	14622	740	45337.75	25	238	953	60043	14861	847	52690
	Pules	90	324	360	38558.25	75	307	409	55337	631	385	46948
	Oil Seeds	1224	10796	882	37686.83	20	226	1128	55908	11021	1005	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	96	31680	33000	70625	23	9200	40000	85000	40880	36500	77813
B) Rabi	Cereals	280	2867	1024	62084	520	6911	1329	67084	9778	1177	64584
	Coarse Cereals	5392	37744	700	30592	548	4877	890	65256	42621	795	47924
	Pulses	527	3478	660	37727	93	660	710	52770	4139	685	45249
	Oil Seeds		0		28038	0	0		35200	0	0	31619
	Vegetables	128	38400	30000	75625	5	1850	37000	98000	40250	33500	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0		0		40000	0	0	0
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	20	240	1200	60000	240	1200	240
	Sugarcane		0	0	0	41	2870	7000	120540	2870	7000	2870
	Vegetables		0	0	0		0		90000	0	0	0
	Fruit crop	572	194480	34000	98000	180	75600	42000	313520	270080	38000	205760
Total	Cereals	7690	110090	1236	64542	744	10761	1524	82256	120851	1380	73399
	Coarse Cereals	7368	52366	720	37965	573	5115	614	55100	57482	667	46532
	Pules	617	3802	510	38143	168	967	560	54054	4769	535	46098
	Oil Seeds	1224	10796	441	32862	40	466	776	50369	11261	609	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	41	2870	7000	120540	2870	7000	120540
	Vegetables	224	70080	31500	73125	28	11050	25667	91000	81130	28583	82063
	Fruit crop	572	194480	34000	98000	180	75600	42000	313520	270080	38000	205760

Production and Productivity of Major Crops - Khed

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block:Khed												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	6505	78060	1200	67000	468	6800	1453	97427	84860	1327	82214
	Coarse Cereals	15820	168958	1068	45337.75	3544	49616	1400	60043	218574	1234	52690
	Pules	1482	10418	703	38558.25	450	6741	1498	55337	17159	1101	46948
	Oil Seeds	13622	152839	1122	37686.83	1000	11660	1166	55908	164499	1144	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	4283	3083760	72000	70625	717	602280	84000	85000	3686040	78000	77813
B) Rabi	Cereals	2650	29892	1128	62084	2860	35636	1246	67084	65528	1187	64584
	Coarse Cereals	29476	155928	529	30592	3565	24206	679	65256	180134	604	47924
	Pulses	4718	27034	573	37727	3252	22992	707	52770	50026	640	45249
	Oil Seeds	50	950	1900	28038	200	2200	1100	35200	3150	1500	31619
	Vegetables	3378	2195700	65000	75625	1465	1172000	80000	98000	3367700	72500	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	110	495	450	40000	495	450	495
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	300	9903	3301	60000	9903	3301	9903
	Sugarcane		0	0	0	700	525000	75000	120540	525000	75000	525000
	Vegetables		0	0	0	135	70200	52000	90000	70200	52000	70200
	Fruit crop	2346	1290300	55000	98000	384	230400	60000	313520	1520700	57500	205760
Total	Cereals	9155	107952	1164	64542	3328	42436	1350	82256	150388	1257	73399
	Coarse Cereals	45296	324886	799	37965	7219	74317	843	55100	399203	821	46532
	Pules	6200	37453	638	38143	3702	29733	1103	54054	67185	870	46098
	Oil Seeds	13672	153789	1511	32862	1500	23763	1856	50369	177552	1683	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	700	525000	75000	120540	525000	75000	120540
	Vegetables	7661	5279460	68500	73125	2317	1844480	72000	91000	7123940	70250	82063
	Fruit crop	2346	1290300	55000	98000	384	230400	60000	313520	1520700	57500	205760

Production and Productivity of Major Crops – Ambegaon

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: ambegaon												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	3075	30535	993	67000	328	4300	1311	97427	34835	1152	82214
	Coarse Cereals	7853	103817	1322	45337.75	1493	29860	2000	60043	133677	1661	52690
	Pules	1676	16894	1008	38558.25	185	2520	1362	55337	19414	1185	46948
	Oil Seeds	4438	38921	877	37686.83	1025	24600	2400	55908	63521	1639	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	3050	2196000	72000	70625	3950	3318000	84000	85000	5514000	78000	77813
B) Rabi	Cereals	1712	26536	1550	62084	2918	51649	1770	67084	78185	1660	64584
	Coarse Cereals	18461	204917	1110	30592	3520	105600	3000	65256	310517	2055	47924
	Pulses	3600	48060	1335	37727	3190	52188	1636	52770	100248	1486	45249
	Oil Seeds	190	703	370	28038	150	615	410	35200	1318	390	31619
	Vegetables	1142	799400	70000	75625	2678	2195960	82000	98000	2995360	76000	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	110	770	700	40000	770	700	770
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	500	10730	2146	60000	10730	2146	10730
	Sugarcane		0	0	0	3403	3437030	101000	120540	3437030	101000	3437030
	Vegetables		0	0	0	180	102600	57000	90000	102600	57000	102600
	Fruit crop	1909	935410	49000	98000	850	382500	45000	313520	1317910	47000	205760
Total	Cereals	4787	57071	1272	64542	3246	55949	1541	82256	113019	1406	73399
	Coarse Cereals	26314	308734	1216	37965	5123	136230	1900	55100	444964	1558	46532
	Pules	5276	64954	1172	38143	3375	54708	1499	54054	119662	1335	46098
	Oil Seeds	4628	39624	624	32862	1675	35945	1652	50369	75569	1138	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	3403	3437030	101000	120540	3437030	101000	120540
	Vegetables	4192	2995400	71000	73125	6808	5616560	74333	91000	8611960	72667	82063
	Fruit crop	1909	935410	49000	98000	850	382500	45000	313520	1317910	47000	205760

Production and Productivity of Major Crops – Junnar

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Junnar												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	6710	56767	846	67000	315	4615	1465	97427	61381	1156	82214
	Coarse Cereals	10973	92722	845	45337.75	5069	48510	957	60043	141232	901	52690
	Pules	2859	35109	1228	38558.25	325	5359	1649	55337	40468	1439	46948
	Oil Seeds	9142	118297	1294	37686.83	401	8020	2000	55908	126317	1647	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	5668	3514160	62000	70625	1572	1289040	82000	85000	4803200	72000	77813
B) Rabi	Cereals	2550	36899	1447	62084	3230	54103	1675	67084	91001	1561	64584
	Coarse Cereals	16049	94529	589	30592	5741	49373	860	65256	143901	725	47924
	Pulses	12390	79668	643	37727	780	5905	757	52770	85572	700	45249
	Oil Seeds	200	0		28038		0		35200	0	0	31619
	Vegetables	2980	1937000	65000	75625	1900	1463000	77000	98000	3400000	71000	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	210	1407	670	40000	1407	210	1407
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	690	15635	2266	60000	15635	2266	15635
	Sugarcane		0	0	0	9461	10028660	106000	120540	10028660	106000	10028660
	Vegetables		0	0	0	120	74400	62000	90000	74400	62000	74400
	Fruit crop	2923	1373810	47000	98000	3505	1927750	55000	313520	3301560	51000	205760
Total	Cereals	9260	93665	1147	64542	3545	58717	1570	82256	152382	1358	73399
	Coarse Cereals	27022	187250	717	37965	11020	99290	829	55100	286540	773	46532
	Pules	15249	114776	936	38143	1105	11264	1203	54054	126040	1069	46098
	Oil Seeds	9342	118297	647	32862	1091	23655	1422	50369	141953	1035	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	9461	10028660	106000	120540	10028660	106000	120540
	Vegetables	8648	5451160	63500	73125	3592	2826440	73667	91000	8277600	68583	82063
	Fruit crop	2923	1373810		98000	3505	1927750	55000	313520	3301560	27500	205760

Production and Productivity of Major Crops – Shirur

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Shirur												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	80	720	900	67000		0		97427	720	900	82214
	Coarse Cereals	8976	89940	1002	45337.75	6345	170427	2686	60043	260366	1844	52690
	Pules	2340	16450	703	38558.25	3218	24135	750	55337	40585	727	46948
	Oil Seeds	514	5767	1122	37686.83	1000	15950	1595	55908	21717	1359	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	1885	1357200	72000	70625	6115	4892000	80000	85000	6249200	76000	77813
B) Rabi	Cereals	4064	78232	1925	62084	4616	96751	2096	67084	174983	2011	64584
	Coarse Cereals	38130	351940	923	30592	23799	226091	950	65256	578030	937	47924
	Pulses	2566	15807	616	37727	5264	38848	738	52770	54655	677	45249
	Oil Seeds	1980	2554	129	28038	1150	2300	200	35200	4854	165	31619
	Vegetables	1085	759500	70000	75625	5730	4354800	76000	98000	5114300	73000	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	100	800	800	40000	800	800	800
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	1100	34155	3105	60000	34155	3105	34155
	Sugarcane		0	0	0	9303	9954210	107000	120540	9954210	107000	9954210
	Vegetables		0	0	0	185	99900	54000	90000	99900	54000	99900
	Fruit crop	1032	546960	53000	98000	1830	1079700	59000	313520	1626660	56000	205760
Total	Cereals	4144	78952	1413	64542	4616	96751	1048	82256	175703	1230	73399
	Coarse Cereals	47106	441879.42	963	37965	30244	397317	1479	55100	839197	1221	46532
	Pules	4906	32257	660	38143	8482	62983	744	54054	95240	702	46098
	Oil Seeds	2494	8321	626	32862	3250	52405	1633	50369	60726	1129	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	9303	9954210	107000	120540	9954210	107000	120540
	Vegetables	2970	2116700	71000	73125	12030	9346700	70000	91000	11463400	70500	82063
	Fruit crop	1032	546960	53000	98000	1830	1079700	59000	313520	1626660	56000	205760

Production and Productivity of Major Crops – Baramati

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Baramati												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	0	0		67000	0	0		97427	0	0	82214
	Coarse Cereals	1703	21952	1289	45337.75	245	7350	3000	60043	29302	2145	52690
	Pules	1160	10985	947	38558.25	170	2032	1195	55337	13017	1071	46948
	Oil Seeds	857	17954	2095	37686.83	250	7000	2800	55908	24954	2448	46797
	Cotton	60	0		69398		0		0	0	0	34699
	Vegetables	2102	1177120	56000	70625	995	656700	66000	85000	1833820	61000	77813
B) Rabi	Cereals	9350	205233	2195	62084	1300	29939	2303	67084	235172	2249	64584
	Coarse Cereals	50523	690649	1367	30592	2755	23059	837	65256	713709	1102	47924
	Pulses	9069	71736	791	37727	321	8025	2500	52770	79761	1646	45249
	Oil Seeds	1910	5482	287	28038		0		35200	5482	144	31619
	Vegetables	3591	1903230	53000	75625	297	203445	68500	98000	2106675	60750	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	200	1400	700	40000	1400	700	1400
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	190	5366	2824	60000	5366	2824	5366
	Sugarcane		0	0	0	21630	23576700	109000	120540	23576700	109000	23576700
	Vegetables		0	0	0	105	63000	60000	90000	63000	60000	63000
	Fruit crop	2176	1218560	56000	98000	1615	1082050	67000	313520	2300610	61500	205760
Total	Cereals	9350	205232.5	2195	129084	1300	29939	1152	82256	235172	1673	105670
	Coarse Cereals	52226	712601	1328	37965	3200	31809	1512	55100	744410	1420	46532
	Pules	10229	82721	869	38143	491	10057	1848	54054	92777	1358	46098
	Oil Seeds	2767	23436	1191	32862	440	12366	1875	50369	35801	1533	41616
	Cotton	60	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	21630	23576700	109000	120540	23576700	109000	120540
	Vegetables	5693	3080350	54500	73125	1397	923145	64833	91000	4003495	59667	82063
	Fruit crop	2176	1218560	56000	98000	1615	1082050	67000	313520	2300610	61500	205760

Production and Productivity of Major Crops – Indapur

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block: Indapur												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	80	640	800	67000		0		97427	640	800	82214
	Coarse Cereals	371	5602	1510	45337.75	3990	997500	25000	60043	1003102	13255	52690
	Pules	522	1655	317	38558.25	805	7656	951	55337	9310	634	46948
	Oil Seeds	231	2864	1240	37686.83	890	12015	1350	55908	14879	1295	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	219	118260	54000	70625	281	137690	49000	85000	255950	51500	77813
B) Rabi	Cereals	505	8615	1706	62084	6445	148428	2303	67084	157044	2005	64584
	Coarse Cereals	52281	1026104	1962.67	30592	9589	331492	3457	65256	1357595	2710	47924
	Pulses	2223	10159	457	37727	1857	19146	1031	52770	29305	744	45249
	Oil Seeds	480	1930	402	28038	100	469	469	35200	2399	436	31619
	Vegetables	1370	664450	48500	75625	2900	1236850	42650	98000	1901300	45575	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	100	650	650	40000	650	650	650
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	400	8896	2224	60000	8896	2224	8896
	Sugarcane		0	0	0	29040	31363200	108000	120540	31363200	108000	31363200
	Vegetables		0	0	0	150	58500	39000	90000	58500	39000	58500
	Fruit crop	137	80830	59000	98000	3034	1579500	52060	313520	1660330	55530	205760
Total	Cereals	585	9255	1253	64542	6445	148428	1152	82256	157684	1202	73399
	Coarse Cereals	52652	1031706	1736	37965	13679	1329642	9702	55100	2361347	5719	46532
	Pules	2745	11814	387	38143	2662	26801	991	54054	38615	689	46098
	Oil Seeds	711	4794	821	32862	1390	21380	1348	50369	26174	1084	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	29040	31363200	108000	120540	31363200	108000	120540
	Vegetables	1589	782710	51250	73125	3331	1433040	43550	91000	2215750	47400	82063
	Fruit crop	137	80830	59000	98000	3034	1579500	52060	313520	1660330	55530	205760

Production and Productivity of Major Crops - Daund

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block:Daund												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	80	896	1120	67000		0		97427	896	1120	82214
	Coarse Cereals	247	4009	1623	45337.75	2857	94481	3307	60043	98490	2465	52690
	Pules	65	330	508	38558.25	1056	12609	1194	55337	12939	851	46948
	Oil Seeds	252	6202	2461	37686.83	350	10938	3125	55908	17139	2793	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	137	87680	64000	70625	900	504000	56000	85000	591680	60000	77813
B) Rabi	Cereals	1536	36372	2368	62084	5864	150060	2559	67084	186432	2464	64584
	Coarse Cereals	51656	581647	1126	30592	5542	204444	3689	65256	786091	2408	47924
	Pulses	1396	10526	754	37727	1704	17159	1007	52770	27685	881	45249
	Oil Seeds	70	100	143	28038	250	1075	430	35200	1175	287	31619
	Vegetables	142	8364	5890	75625	2416	119471	4945	98000	127835	5418	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	200	2796	1398	40000	2796	1398	2796
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	1000	21750	2175	60000	21750	2175	21750
	Sugarcane		0	0	0	23800	25466000	107000	120540	25466000	107000	25466000
	Vegetables		0	0	0	142	69580	49000	90000	69580	49000	69580
	Fruit crop	518	269360	52000	98000	1520	820800	54000	313520	1090160	53000	205760
Total	Cereals	1616	37268.48	3488	129084	5864	150060	1280	82256	187328	2384	105670
	Coarse Cereals	51903	585655	1375	37965	8599	301721	2798	55100	887377	2086	46532
	Pules	1461	10856	631	38143	2760	29768	1101	54054	40624	866	46098
	Oil Seeds	322	6302	1302	32862	1600	33763	1910	50369	40064	1606	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	23800	25466000	107000	120540	25466000	107000	120540
	Vegetables	279	96044	34945	73125	3458	693051	36648	91000	789095	35797	82063
	Fruit crop	518	269360	52000	98000	1520	820800	54000	313520	1090160	53000	205760

Production and Productivity of Major Crops - Purandar

2.2 Production and Productivity of major Crops												Source: Department of Agriculture, Agriculture Statistic of State Agristat
Name of the State : Maharashtra												
Name of the District : Pune												
Block:Purandar												
Season	Crop Category	Rainfed				Irrigated				Total		
		Area (Ha.)	Production (qtn/Yr.)	Productivity / Yield(Kgs/ha)	Cost of Cultivation (Rs./ha.)	Area (Ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)	Production (qtn/Yr.)	Productivity (Kgs/ha)	Cost of Cultivation (Rs./ha.)
A) Kharif	Cereals	1445	12817	887	67000	120	1716	1430	97427	14533	1159	82214
	Coarse Cereals	5597	171044	3056	45337.75	875	28490	3256	60043	199534	3156	52690
	Pules	2212	8406	380	38558.25	45	272	605	55337	8678	493	46948
	Oil Seeds	346	4467	1291	37686.83	140	3492	2494	55908	7958	1893	46797
	Cotton		0		69398		0		0	0	0	34699
	Vegetables	309	153048	49530	70625	1191	550361	46210	85000	703409	47870	77813
B) Rabi	Cereals	352	5590	1588	62084	5088	95044	1868	67084	100634	1728	64584
	Coarse Cereals	52612	406165	772	30592	6265	62149	992	65256	468313	882	47924
	Pulses	3115	31493	1011	37727	895	12100	1352	52770	43593	1182	45249
	Oil Seeds	1005	1457	145	28038	835	9344	1119	35200	10801	632	31619
	Vegetables	750	343500	45800	75625	3100	1348500	43500	98000	1692000	44650	86813
C) Summer	Cereals		0	0	0		0			0	0	0
	Coarse Cereals		0	0	0	100	346	346	40000	346	346	346
	Pules		0	0	0		0			0	0	0
	Oil Seeds		0	0	0	100	2002	2002	60000	2002	2002	2002
	Sugarcane		0	0	0	2274	1932900	85000	120540	1932900	85000	1932900
	Vegetables		0	0	0	150	74250	49500	90000	74250	49500	74250
	Fruit crop	183	93696	51200	98000	1990	1032810	51900	313520	1126506	51550	205760
Total	Cereals	1797	18406.91	2475	129084	5208	96760	1649	82256	115167	2062	105670
	Coarse Cereals	58209	577209	1914	37965	7240	90985	1531	55100	668194	1723	46532
	Pules	5327	39898	696	38143	940	12373	979	54054	52271	837	46098
	Oil Seeds	1351	5924	718	32862	1075	14837	1872	50369	20761	1295	41616
	Cotton	0	0	0	69398	0	0	0	0	0	0	69398
	Sugarcane	0	0	0	0	2274	1932900	85000	120540	1932900	85000	120540
	Vegetables	1059	496548	47665	73125	4441	1973111	46403	91000	2469659	47034	82063
	Fruit crop	183	93696		98000	1990	1032810	51900	313520	1126506	25950	205760

Annexure – III

Crop Water Requirement – Block-Wise

Crop Water Requirement - Block-wise			
Sr. No.	**Name of Blocks**	**(District Pune)**	**Page.no.**
0	District Pune		106
1	Haveli		107
2	Maval		108
3	Mulshi		109
4	Bhor		110
5	Velha		111
6	Khed		112
7	Ambegaon		113
8	Junnar		114
9	Shirur		115
10	Baramati		116
11	Indapur		117
12	Daund		118
13	Purandar		119

Crop Water Requirement – Pune District

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	847.00	4075.00	1187	150	0.0061125	0.00127	0.004842
2	paddy	55256.00	62293.00	10471	0	0	0.00000	0
3	Bajra	41423.00	45049.00	17015	120	0.0540588	0.04971	0.0043512
4	Maize	15809.00	19934.00	8153	200	0.039868	0.03162	0.00825
5	Other cereals	4669.00	12837.00	140	0	0	0.00000	0
6	Tur	1565.00	2844.00	2247	300	0.008532	0.00470	0.003837
7	Mung	18201.00	18768.00	1399	75	0.014076	0.01365	0.00042525
8	Udid	1528.00	1660.00	15	75	0.001245	0.00115	0.000099
9	Other pulses	10292.00	10186.00	3045	0	0	0.00000	0
10	Groundnut	15364.00	15713.00	8056	300	0.047139	0.04609	0.001047
11	Sunflower	54.00	980.00	238	75	0.000735	0.00004	0.0006945
12	Soyabean	17991.00	20198.00	1434	300	0.060594	0.05397	0.006621
13	Sesamum	774.00	1924.00	35	0	0	0.00000	0
14	Nigerseed	889.00	2285.00	0	0	0	0.00000	0
15	Other Kh. Oil seed crops	101.00	0.00	0	0	0	0.00000	0
16	Sugarcane	117090.00	99358.00	113497	2500	2.48395	2.92725	-0.44330
17	Cotton	8.00	60.00	0	700	0.00042	0.00006	0.000364
	Total kh. Season	**301861.00**	**317667.00**	**166932.00**		**2.7167303**	**3.13**	**-0.41276905**
1	R. Jowar	172721.00	228729.00	71069.00	130	0.2973477	0.22454	0.0728104
2	Wheat	32623.00	43119.00	39515.00	400	0.172476	0.13049	0.041984
3	R. Maize	11857.00	15510.00	7336.00	500	0.07755	0.05929	0.018265
4	Other cereals	306.00	1000.00	175.00	130	0.0013	0.00040	0.0009022
5	Gram	29272.00	43660.00	19178.00	250	0.10915	0.07318	0.03597
6	Other pulses	2648.00	9000.00	1321.00	250	0.0225	0.00662	0.01588
7	Safflower	53.00	5490.00	1740.00	250	0.013725	0.00013	0.0135925
8	R.Sunflower	201.00	2580.00	335.00	350	0.00903	0.00070	0.0083265
9	Linseed	2.00	80.00	10.00	150	0.00012	0.00000	0.000117
10	Other R. oil seed crops	518.00	570.00	430.00	250	0.001425	0.00130	0.00013
	Total Rabbi crops	**250201.00**	**349738.00**	**141109.00**		**0.70**	**0.4966461**	**0.21**
1	bajri	3556.00	3696.00	1157.00	**450.00**	0.016632	0.01600	0.00063
2	Summer maize	820.00	1252.00	1218.00	**500**	0.00626	0.00410	0.00216
3	Summer Groundnut	3552.00	4544.00	3445.00	700	0.031808	0.02486	0.006944
	Total summer crops	**7928.00**	**9492.00**	**5820.00**		**0.0547**	**0.044966**	**0.009734**
	Gross cropped area	830984.00	1053761.00	422652.00		6.733264	6.0056788	0.7323837
	Total Cereals	343133.00	441038.00	156104		0.653673	0.5418764	0.1526626
	Total Pulses	63506	86118	27205		0.155503	6.2818742	0.05621125
	Total Oilseeds	39499	54364	15723		0.164576	0.77441055	0.0374725
1	Fodder crops	66,530	90,250	22969	350	0.315875	0.23286	0.08302
2	Vegetable	141,779	198,439	66411	750	1.4882925	1.06334	0.42495
3	Horticultural & Plantation	62685	88175	20004	1500	1.322625	0.94028	0.38235

Crop Water Requirement – Haveli

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Haveli								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	0.00	403.00	57	150	0.0006045	0.00000	0.0006045
2	paddy	1629.00	2869.00	205	0	0	0.00000	0
3	Bajra	1350.00	2000.00	164	120	0.0024	0.00162	0.00078
4	Maize	147.00	1200.00		200	0.0024	0.00029	0.002106
5	Other cereals		80.00		0	0	0.00000	0
6	Tur	37.00	300.00	145	300	0.0009	0.00011	0.000789
7	Mung	314.00	314.00		75	0.0002355	0.00024	0
8	Udid	67.00	70.00		75	0.0000525	0.00005	0.00000225
9	Other pulses	757.00	220.00		0	0	0.00000	0
10	Groundnut	660.00	700.00	970	300	0.0021	0.00198	0.00012
11	Sunflower		0.00			0	0.00000	0
12	Soyabean	142.00	300.00		300	0.0009	0.00043	0.000474
13	Sesamum	10.00	50.00		0	0	0.00000	0
14	Nigerseed		0.00		0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	5330.00	5728.00	9328	2500	0.1432	0.13325	0.00995
17	Cotton					0	0.00000	0
	Total kh. Season	**10443.00**	**14234.00**	**10869.00**		**0.15**	**0.14**	**0.01**
1	R. Jowar	1402.00	2000.00	2350.00	130	0.0026	0.00182	0.0007774
2	Wheat	1335.00	2490.00	2200.00	400	0.00996	0.00534	0.00462
3	R. Maize	140.00	960.00	445.00	500	0.0048	0.00070	0.0041
4	Other cereals	60.00				0	0.00000	0
5	Gram	747.00	1360.00	715.00	250	0.0034	0.00187	0.0015325
6	Other pulses	140.00	700.00	122.00	250	0.00175	0.00035	0.0014
7	Safflower	1.00	0.00			0	0.00000	0
8	R.Sunflower	2.00	10.00		350	0.000035	0.00001	0.000028
9	Linseed	2.00	0.00			0	0.00000	0
10	Other R. oil seed crops	92.00				0	0.00000	0
	Total Rabbi crops	**3921.00**	**7520.00**	**5832.00**		**0.02**	**0.0100871**	**0.01**
1	Bajri	30.00	60.00	60.00	**450.00**	0.00027	0.00014	0.000135
2	Summer maize					0	0.00000	0
3	Summer Groundnut	50.00	150.00	150.00	700	0.00105	0.00035	0.0007
	Total summer crops	**80.00**	**210.00**	**210.00**		**0.00132**	**0.000485**	**0.000835**
	Gross cropped area	34092.00	48236.00	28834.00		0.4	0.33	0.07
	Total Cereals	6083.00	12212.00	5631		0.0240845	0.0102616	0.0138229
	Total Pulses	2062	2964	982		0.006338	0.33698885	0.00372375
	Total Oilseeds	959	1210	1120		0.004085	0.0150896	0.001322
1	Fodder crops	4,190	6,500	2587	500	0.0325	0.02095	0.01155
2	Vegetable	9,658	13,972	8038	750	0.10479	0.07244	0.032355
3	Horticultural & Plantation	5800	5800	1298	1500	0.087	0.08700	0

Crop Water Requirement – Maval

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Maval								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	0.00	100.00	90	150	0.00015	0.00000	0.00015
2	paddy	11700.00	8689.00	4460	0	0	0.00000	0
3	Bajra	85.00	150.00	25	120	0.00018	0.00010	0.000078
4	Maize		89.00	80	200	0.000178	0.00000	0.000178
5	Other cereals		1282.00	115	0	0	0.00000	0
6	Tur	55.00	30.00	30	300	0.00009	0.00017	-0.000075
7	Mung					0	0.00000	0
8	Udid		50.00	15	75	0.0000375	0.00000	0.0000375
9	Other pulses	307.00	286.00	95	0	0	0.00000	0
10	Groundnut	296.00	300.00	700	300	0.0009	0.00089	0.000012
11	Sunflower					0	0.00000	0
12	Soyabean	236.00	236.00		300	0.000708	0.00071	0
13	Sesamum	9.00	211.00		0	0	0.00000	0
14	Nigerseed				0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	570.00	753.00	1753	2500	0.018825	0.01425	0.00458
17	Cotton					0	0.00000	0
	Total kh. Season	**13258.00**	**12176.00**	**7363.00**		**0.02**	**0.02**	**0**
1	R. Jowar	988.00	2500.00	1230.00	130	0.00325	0.00128	0.0019656
2	Wheat	518.00	1000.00	1084.00	400	0.004	0.00207	0.001928
3	R. Maize	7.00	40.00	40.00	500	0.0002	0.00004	0.000165
4	Other cereals					0	0.00000	0
5	Gram	792.00	660.00	173.00	250	0.00165	0.00198	-0.00033
6	Other pulses	286.00	200.00	0.00	250	0.0005	0.00072	-0.000215
7	Safflower					0	0.00000	0
8	R.Sunflower					0	0.00000	0
9	Linseed					0	0.00000	0
10	Other R. oil seed crops		10.00		250	0.000025	0.00000	0.000025
	Total Rabbi crops	**2591.00**	**4410.00**	**2527.00**		**0.01**	**0.0060864**	**0.00**
1	Bajri	232.00	232.00	232.00	**450.00**	0.001044	0.00104	0
2	Summer maize	12.00	12.00	10.00	**500**	0.00006	0.00006	0
3	Summer Groundnut	180.00	180.00	180.00	700	0.00126	0.00126	0
	Total summer crops	**424.00**	**424.00**	**422.00**		**0.002364**	**0.002364**	**0**
	Gross cropped area	21023.00	27764.00	13767.00		0.13	0.09	0.05
	Total Cereals	13722.00	14274.00	7546		0.010322	0.0058574	0.0044646
	Total Pulses	1440	1226	313		0.0022775	0.0890674	-0.0005825
	Total Oilseeds	721	937	880		0.002893	0.0081394	0.000037
1	Fodder crops	60	2,700	18	500	0.0135	0.00030	0.0132
2	Vegetable	1,136	4,500	2880	750	0.03375	0.00852	0.02523
3	Horticultural & Plantation	3554	3554	557	1500	0.05331	0.05331	0

Crop Water Requirement – Mulshi

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Mulhsi								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	80.00	294.00	0	150	0.000441	0.00012	0.000321
2	paddy	7631.00	15370.00	4335	0	0	0.00000	0
3	Bajra	25.00	25.00		120	0.00003	0.00003	0
4	Maize	13.00	372.00	68	200	0.000744	0.00003	0.000718
5	Other cereals	214.00	770.00		0	0	0.00000	0
6	Tur	81.00	156.00	62	300	0.000468	0.00024	0.000225
7	Mung	59.00	59.00		75	0.00004425	0.00004	0
8	Udid	34.00	65.00		75	0.00004875	0.00003	0.00002325
9	Other pulses	402.00	70.00		0	0	0.00000	0
10	Groundnut	231.00	250.00	1959	300	0.00075	0.00069	0.000057
11	Sunflower					0	0.00000	0
12	Soyabean	39.00	50.00	16	300	0.00015	0.00012	0.000033
13	Sesamum	35.00	300.00	35	0	0	0.00000	0
14	Nigerseed				0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	1610.00	2200.00	1651	2500	0.055	0.04025	0.01475
17	Cotton					0	0.00000	0
	Total kh. Season	**10454.00**	**19981.00**	**8126.00**		**0.06**	**0.04**	**0.02**
1	R. Jowar	657.00	1500.00	8644.00	130	0.00195	0.00085	0.0010959
2	Wheat	348.00	500.00	2055.00	400	0.002	0.00139	0.000608
3	R. Maize	15.00				0	0.00000	0
4	Other cereals	51.00				0	0.00000	0
5	Gram	362.00	810.00	810.00	250	0.002025	0.00091	0.00112
6	Other pulses	197.00	600.00	443.00	250	0.0015	0.00049	0.0010075
7	Safflower					0	0.00000	0
8	R.Sunflower					0	0.00000	0
9	Linseed					0	0.00000	0
10	Other R. oil seed crops		80.00	50.00	250	0.0002	0.00000	0.0002
	Total Rabbi crops	**1630.00**	**3490.00**	**12002.00**		**0.01**	**0.0036436**	**0.00**
1	Bajri	0.00				0	0.00000	0
2	Summer maize	0.00	0.00			0	0.00000	0
3	Summer Groundnut	0.00	120.00	120.00	700	0.00084	0.00000	0.00084
	Total summer crops	**0.00**	**120.00**			**0.00084**	**0**	**0.00084**
	Gross cropped area	13396.00	26291.00	20999.00		0.09	0.06	0.03
	Total Cereals	8983.00	18951.00	15102		0.006005	0.0024221	0.0035829
	Total Pulses	1135	1760	1315		0.004086	0.06332335	0.00237575
	Total Oilseeds	305	800	2180		0.00194	0.00367035	0.00113
1	Fodder crops	60	300	22	500	0.0015	0.00030	0.0012
2	Vegetable	352	1,500	318	750	0.01125	0.00264	0.00861
3	Horticultural & Plantation	900	900	531	1500	0.0135	0.01350	0

Crop Water Requirement – Bhor

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Bhor								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	119.00	294.00	150	150	0.000441	0.00018	0.0002625
2	paddy	6799.00	7798.00	16	0	0	0.00000	0
3	Bajra	294.00	569.00	383	120	0.0006828	0.00035	0.00033
4	Maize		822.00	517	200	0.001644	0.00000	0.001644
5	Other cereals	1602.00	4139.00		0	0	0.00000	0
6	Tur	5.00	243.00	188	300	0.000729	0.00002	0.000714
7	Mung	53.00	109.00		75	0.00008175	0.00004	0.000042
8	Udid	64.00	80.00		75	0.00006	0.00005	0.000012
9	Other pulses	1046.00	290.00		0	0	0.00000	0
10	Groundnut	2432.00	2500.00	650	300	0.0075	0.00730	0.000204
11	Sunflower					0	0.00000	0
12	Soyabean	2229.00	3600.00	450	300	0.0108	0.00669	0.004113
13	Sesamum		147.00		0	0	0.00000	0
14	Nigerseed	85.00	397.00		0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	1590.00	930.00	1113	2500	0.02325	0.03975	-0.01650
17	Cotton					0	0.00000	0
	Total kh. Season	**16318.00**	**21918.00**	**3467.00**		**0.05**	**0.05**	**-0.01**
1	R. Jowar	3814.00	6000.00	2666.00	130	0.0078	0.00496	0.0028418
2	Wheat	1650.00	2570.00	1235.00	400	0.01028	0.00660	0.00368
3	R. Maize	7.00	30.00		500	0.00015	0.00004	0.000115
4	Other cereals	0.00				0	0.00000	0
5	Gram	1260.00	970.00	430.00	250	0.002425	0.00315	-0.000725
6	Other pulses	35.00	900.00	150.00	250	0.00225	0.00009	0.0021625
7	Safflower					0	0.00000	0
8	R.Sunflower					0	0.00000	0
9	Linseed		30.00	10.00	150	0.000045	0.00000	0.000045
10	Other R. oil seed crops		20.00	20.00	250	0.00005	0.00000	0.00005
	Total Rabbi crops	**6766.00**	**10520.00**	**4511.00**		**0.02**	**0.0148307**	**0.01**
1	Bajri	210.00	210.00	210.00	**450.00**	0.000945	0.00095	0
2	Summer maize					0	0.00000	0
3	Summer Groundnut	370.00	370.00	370.00	700	0.00259	0.00259	0
	Total summer crops	**580.00**	**580.00**	**580.00**		**0.003535**	**0.003535**	**0**
	Gross cropped area	28659.00	41918.00	12567.00		0.15	0.13	0.03
	Total Cereals	14865.00	22802.00	5547		0.0245328	0.0156595	0.0088733
	Total Pulses	2463	2592	768		0.00554575	0.13369805	0.0022055
	Total Oilseeds	5116	7064	1500		0.020985	0.01925205	0.004412
1	Fodder crops	450	1,550	867	500	0.00775	0.00225	0.0055
2	Vegetable	2,195	5,000	2390	750	0.0375	0.01646	0.0210375
3	Horticultural & Plantation	2350	2350	752	1500	0.03525	0.03525	0

Crop Water Requirement - Velha

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Velha								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	0.00	12.00		150	0.000018	0.00000	0.000018
2	paddy	4948.00	7394.00	224	0	0	0.00000	0
3	Bajra		5.00		120	0.000006	0.00000	0.000006
4	Maize		600.00		200	0.0012	0.00000	0.0012
5	Other cereals	754.00	2215.00	25	0	0	0.00000	0
6	Tur		70.00	25	300	0.00021	0.00000	0.00021
7	Mung		0.00			0	0.00000	0
8	Udid		25.00		75	0.00001875	0.00000	0.00001875
9	Other pulses	88.00	70.00	50	0	0	0.00000	0
10	Groundnut	79.00	90.00	20	300	0.00027	0.00024	0.000033
11	Sunflower	1.00				0	0.00000	0
12	Soyabean	208.00	250.00		300	0.00075	0.00062	0.000126
13	Sesamum	25.00	132.00		0	0	0.00000	0
14	Nigerseed	33.00	661.00		0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	120.00	180.00	41	2500	0.0045	0.00300	0.00150
17	Cotton					0	0.00000	0
	Total kh. Season	**6256.00**	**11704.00**	**385.00**		**0.01**	**0.00**	**0**
1	R. Jowar	67.00	150.00	548.00	130	0.000195	0.00009	0.0001079
2	Wheat	245.00	859.00	520.00	400	0.003436	0.00098	0.002456
3	R. Maize	22.00				0	0.00000	0
4	Other cereals	7.00	100.00		130	0.00013	0.00001	0.0001209
5	Gram	217.00	320.00	93.00	250	0.0008	0.00054	0.0002575
6	Other pulses	65.00	300.00		250	0.00075	0.00016	0.0005875
7	Safflower					0	0.00000	0
8	R.Sunflower					0	0.00000	0
9	Linseed					0	0.00000	0
10	Other R. oil seed crops					0	0.00000	0
	Total Rabbi crops	**623.00**	**1729.00**	**1161.00**		**0.01**	**0.0017812**	**0.00**
1	Bajri	0.00					0.00000	
2	Summer maize	0.00				0	0.00000	0
3	Summer Groundnut	0.00	20.00	20.00	700	0.00014	0.00000	0.00014
	Total summer crops	**0.00**	**20.00**	**20.00**		**0.00014**	**0**	**0.00014**
	Gross cropped area	7952.00	15674.00	1774.00		0.03	0.02	0.01
	Total Cereals	6036.00	11255.00	1317		0.004995	0.0010671	0.0039279
	Total Pulses	370	785	168		0.00177875	0.0210863	0.00107375
	Total Oilseeds	346	1153	40		0.00116	0.0016187	0.000299
1	Fodder crops	50	700	30	500	0.0035	0.00025	0.00325
2	Vegetable	152	650	28	750	0.004875	0.00114	0.003735
3	Horticultural & Plantation	871	871	150	1500	0.013065	0.01307	0

Crop Water Requirement – Khed

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Khed								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	3.00	199.00	84	150	0.0002985	0.00000	0.000294
2	paddy	7191.00	6653.00	468	0	0	0.00000	0
3	Bajra	1164.00	2000.00	2790	120	0.0024	0.00140	0.0010032
4	Maize	4046.00	951.00	670	200	0.001902	0.00809	-0.00619
5	Other cereals	388.00	2044.00		0	0	0.00000	0
6	Tur	190.00	502.00	450	300	0.001506	0.00057	0.000936
7	Mung	188.00	517.00		75	0.00038775	0.00014	0.00024675
8	Udid	95.00	100.00		75	0.000075	0.00007	0.00000375
9	Other pulses	908.00	580.00	0	0	0	0.00000	0
10	Groundnut	4348.00	4400.00	1000	300	0.0132	0.01304	0.000156
11	Sunflower	10.00	33.00		75	0.00002475	0.00001	0.00001725
12	Soyabean	4102.00	5000.00		300	0.015	0.01231	0.002694
13	Sesamum	136.00	334.00		0	0	0.00000	0
14	Nigerseed	169.00	309.00		0	0	0.00000	0
15	Other Kh. Oil seed crops	0.00			0	0	0.00000	0
16	Sugarcane	3220.00	700.00	700	2500	0.0175	0.08050	-0.06300
17	Cotton					0	0.00000	0
	Total kh. Season	**26158.00**	**24322.00**	**6162.00**		**0.05**	**0.12**	**-0.06**
1	R. Jowar	9632.00	13500.00	2860.00	130	0.01755	0.01252	0.0050284
2	Wheat	1801.00	2310.00	2860.00	400	0.00924	0.00720	0.002036
3	R. Maize	1803.00	710.00	705.00	500	0.00355	0.00902	-0.005465
4	Other cereals	12.00	200.00		130	0.00026	0.00002	0.0002444
5	Gram	2396.00	7370.00	3252.00	250	0.018425	0.00599	0.012435
6	Other pulses	868.00	600.00		250	0.0015	0.00217	-0.00067
7	Safflower		10.00		250	0.000025	0.00000	0.000025
8	R.Sunflower		110.00	100.00	350	0.000385	0.00000	0.000385
9	Linseed					0	0.00000	0
10	Other R. oil seed crops		130.00	100.00	250	0.000325	0.00000	0.000325
	Total Rabbi crops	**16512.00**	**24940.00**	**9877.00**		**0.05**	**0.0369162**	**0.01**
1	Bajri	936.00	936.00	110.00	**450.00**	0.004212	0.00421	0
2	Summer maize	424.00	424.00	424.00	**500**	0.00212	0.00212	0
3	Summer Groundnut	1360.00	1360.00	300.00	700	0.00952	0.00952	0
	Total summer crops	**2720.00**	**2720.00**	**724.00**		**0.01164**	**0.015852**	**0**
	Gross cropped area	80427.00	98082.00	32010.00		0.48	0.45	0.03
	Total Cereals	26452.00	28791.00	10861		0.0370605	0.0540859	-0.0032934
	Total Pulses	4645	9669	3702		0.02189375	0.47969715	0.0129515
	Total Oilseeds	10125	11686	1500		0.03847975	0.0793063	0.00360225
1	Fodder crops	7,570	7,600	2860	500	0.038	0.03785	0.00015
2	Vegetable	22,047	33,000	12003	750	0.2475	0.16535	0.0821475
3	Horticultural & Plantation	5420	5500	384	1500	0.0825	0.08130	0.0012

Crop Water Requirement – Ambegaon

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Ambegaon								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	391.00	497.00	48	150	0.000495	0.00059	0.0009085
2	paddy	5150.00	4330.00	328	0	0	0.00000	0
3	Bajra	2150.00	3000.00	990	120	0.0036	0.00258	0.00102
4	Maize	3850.00	4200.00	455	200	0.0084	0.00770	0.0007
5	Other cereals	1148.00	921.00		0	0	0.00000	0
6	Tur	90.00	235.00	185	300	0.000705	0.00027	0.000435
7	Mung	936.00	1000.00		75	0.00075	0.00070	0.000048
8	Udid	351.00	350.00		75	0.0002625	0.00026	-0.000001
9	Other pulses	158.00	950.00		0	0	0.00000	0
10	Groundnut	1172.00	1172.00	1025	300	0.003516	0.00352	0
11	Sunflower		14.00		75	0.0000105	0.00000	0.0000105
12	Soyabean	1113.00	1200.00		300	0.0036	0.00334	0.000261
13	Sesamum	15.00	373.00		0	0	0.00000	0
14	Nigerseed	285.00	415.00		0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	3430.00	3403.00	3403	2500	0.085075	0.08575	-0.00068
17	Cotton					0	0.00000	0
	Total kh. Season	**20239.00**	**21563.00**	**6386.00**		**0.11**	**0.10**	**0**
1	R. Jowar	10759.00	15000.00	3768.00	130	0.0195	0.01399	0.0055133
2	Wheat	2270.00	2430.00	2918.00	400	0.00972	0.00908	0.00064
3	R. Maize	1717.00	490.00		500	0.00245	0.00859	-0.006135
4	Other cereals	0.00	200.00		130	0.00026	0.00000	0.00026
5	Gram	3870.00	5890.00	3300.00	250	0.014725	0.00968	0.00505
6	Other pulses	753.00	900.00	190.00	250	0.00225	0.00188	0.0003675
7	Safflower		40.00		250	0.0001	0.00000	0.0001
8	R.Sunflower		110.00		350	0.000385	0.00000	0.000385
9	Linseed		20.00		150	0.00003	0.00000	0.00003
10	Other R. oil seed crops		170.00	150.00	250	0.000425	0.00000	0.000425
	Total Rabbi crops	**19369.00**	**25250.00**	**10326.00**		**0.05**	**0.0432092**	**0.01**
1	Bajri	320.00	320.00	110.00	**450.00**	0.00144	0.00144	0
2	Summer maize	50.00	50.00	50.00	**500**	0.00025	0.00025	0
3	Summer Groundnut	190.00	190.00	190.00	700	0.00133	0.00133	0
	Total summer crops	**560.00**	**560.00**	**350.00**		**0.00302**	**0.00302**	**0**
	Gross cropped area	78609.00	92373.00	25830.00		0.51	0.45	0.06
	Total Cereals	27485.00	30918.00	8557		0.050415	0.0455382	0.0076468
	Total Pulses	6158	9325	3675		0.0186925	0.47985845	0.00589975
	Total Oilseeds	2775	3704	1365		0.0093965	0.0750502	0.0012115
1	Fodder crops	8,720	7,500	1110	500	0.0375	0.04360	-0.0061
2	Vegetable	25,161	33,000	6808	750	0.2475	0.18871	0.0587925
3	Horticultural & Plantation	4560	4500	850	1500	0.0675	0.06840	-0.0009

Crop Water Requirement – Junnar

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Junnar								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	254.00	562.00	79	150	0.000843	0.00038	0.000462
2	paddy	9300.00	7705.00	315	0	0	0.00000	0
3	Bajra	800.00	1000.00	3610	120	0.0012	0.00096	0.00024
4	Maize	1290.00	1500.00	1380	200	0.003	0.00258	0.00042
5	Other cereals	560.00	1066.00		0	0	0.00000	0
6	Tur	400.00	234.00	200	300	0.000702	0.00120	-0.000498
7	Mung	320.00	587.00		75	0.00044025	0.00024	0.00020025
8	Udid	340.00	340.00		75	0.000255	0.00026	0
9	Other pulses	2130.00	2260.00	125	0	0	0.00000	0
10	Groundnut	3800.00	3900.00	17	300	0.0117	0.01140	0.0003
11	Sunflower	25.00	47.00		75	0.00003525	0.00002	0.0000165
12	Soyabean	8900.00	7000.00	401	300	0.021	0.02670	-0.0057
13	Sesamum	45.00	311.00		0	0	0.00000	0
14	Nigerseed	260.00	385.00		0	0	0.00000	0
15	Other Kh. Oil seed crops	19.00			0	0	0.00000	0
16	Sugarcane	8140.00	5940.00	9461	2500	0.1485	0.20350	-0.05500
17	Cotton					0	0.00000	0
	Total kh. Season	**36583.00**	**32837.00**	**15588.00**		**0.19**	**0.25**	**-0.06**
1	R. Jowar	4945.00	12000.00	5260.00	130	0.0156	0.00643	0.0091715
2	Wheat	6240.00	6680.00	3230.00	400	0.02672	0.02496	0.00176
3	R. Maize	1860.00	520.00	481.00	500	0.0026	0.00930	-0.0067
4	Other cereals	0.00	100.00		130	0.00013	0.00000	0.00013
5	Gram	3350.00	1170.00	780.00	250	0.002925	0.00838	-0.00545
6	Other pulses		1500.00		250	0.00375	0.00000	0.00375
7	Safflower		40.00		250	0.0001	0.00000	0.0001
8	R.Sunflower	74.00	100.00		350	0.00035	0.00026	0.000091
9	Linseed		10.00		150	0.000015	0.00000	0.000015
10	Other R. oil seed crops	10.00	50.00		250	0.000125	0.00003	0.0001
	Total Rabbi crops	**16479.00**	**22170.00**	**9751.00**		**0.05**	**0.0493475**	**0.00**
1	Bajri	1713.00	1713.00	210.00	**450.00**	0.0077085	0.00771	0
2	Summer maize	334.00	334.00	334.00	**500**	0.00167	0.00167	0
3	Summer Groundnut	675.00	675.00	675.00	700	0.004725	0.00473	0
	Total summer crops	**2722.00**	**2722.00**	**1219.00**		**0.0141035**	**0.0141035**	**0**
	Gross cropped area	104454.00	116099.00	37771.00		0.74	0.7	0.03
	Total Cereals	25583.00	31367.00	14689		0.051633	0.058713	0.0053535
	Total Pulses	6540	6091	1105		0.00807225	0.74101075	-0.00199775
	Total Oilseeds	13808	12518	1093		0.03805025	0.081368	-0.0050775
1	Fodder crops	12,220	12,500	3539	500	0.0625	0.06110	0.0014
2	Vegetable	28,870	35,870	4169	750	0.269025	0.21653	0.0525
3	Horticultural & Plantation	7580	10000	3505	1500	0.15	0.11370	0.0363

Crop Water Requirement – Shirur

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Shirur								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar					0	0.00000	0
2	paddy					0	0.00000	0
3	Bajra	13800.00	14000.00	6345	120	0.0168	0.01656	0.00024
4	Maize	243.00	400.00	200	200	0.0008	0.00049	0.000314
5	Other cereals		80.00	0	0	0	0.00000	0
6	Tur	410.00	158.00	150	300	0.000474	0.00123	-0.000756
7	Mung	15760.00	15000.00	1118	75	0.01125	0.01182	-0.00057
8	Udid	466.00	466.00		75	0.0003495	0.00035	0
9	Other pulses	842.00	2040.00	2000	0	0	0.00000	0
10	Groundnut	637.00	650.00	1000	300	0.00195	0.00191	0.000039
11	Sunflower	4.00	88.00		75	0.000066	0.00000	0.000063
12	Soyabean	81.00	100.00		300	0.0003	0.00024	0.000057
13	Sesamum	469.00	49.00		0	0	0.00000	0
14	Nigerseed	26.00	59.00		0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	18410.00	18410.00	9303	2500	0.46025	0.46025	0.00000
17	Cotton					0	0.00000	0
	Total kh. Season	**51148.00**	**51500.00**	**20116.00**		**0.49**	**0.49**	**0**
1	R. Jowar	31319.00	35000.00	23399.00	130	0.0455	0.04071	0.0047853
2	Wheat	3638.00	4580.00	4616.00	400	0.01832	0.01455	0.003768
3	R. Maize	244.00	400.00	400.00	500	0.002	0.00122	0.00078
4	Other cereals	0.00	100.00	75.00	130	0.00013	0.00000	0.00013
5	Gram	5554.00	7530.00	5000.00	250	0.018825	0.01389	0.00494
6	Other pulses		300.00	264.00	250	0.00075	0.00000	0.00075
7	Safflower	46.00	2630.00	890.00	250	0.006575	0.00012	0.00646
8	R.Sunflower		390.00	150.00	350	0.001365	0.00000	0.001365
9	Linseed					0	0.00000	0
10	Other R. oil seed crops	21.00	110.00	110.00	250	0.000275	0.00005	0.0002225
	Total Rabbi crops	**40822.00**	**51040.00**	**34904.00**		**0.09**	**0.0705392**	**0.02**
1	Bajri	100.00	100.00	100.00	**450.00**	0.00045	0.00045	0
2	Summer maize	0.00	100.00	100.00	**500**	0.0005	0.00000	0.0005
3	Summer Groundnut	350.00	700.00	700.00	700	0.0049	0.00245	0.00245
	Total summer crops	**450.00**	**900.00**	**900.00**		**0.00585**	**0.0029**	**0.00295**
	Gross cropped area	125563.00	146340.00	77980.00		0.94	0.83	0.1
	Total Cereals	49694.00	55360.00	35060		0.08392	0.0764327	0.0103873
	Total Pulses	23032	25494	8532		0.0316485	0.8663372	0.004364
	Total Oilseeds	1634	4776	2850		0.015431	0.1216337	0.0106565
1	Fodder crops	8,030	8,900	2525	500	0.0445	0.04015	0.00435
2	Vegetable	20,113	28,000	17697	750	0.21	0.15085	0.0591525
3	Horticultural & Plantation	5000	6000	1838	1500	0.09	0.07500	0.015

Crop Water Requirement – Baramati

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Baramati								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar			0		0	0.00000	0
2	paddy				0	0	0.00000	0
3	Bajra	11320.00	11500.00	245	120	0.0138	0.01358	0.000216
4	Maize	2170.00	3500.00		200	0.007	0.00434	0.00266
5	Other cereals				0	0	0.00000	0
6	Tur	130.00	177.00	170	300	0.000531	0.00039	0.000141
7	Mung	454.00	500.00		75	0.000375	0.00034	0.0000345
8	Udid	65.00	70.00		75	0.0000525	0.00005	0.00000375
9	Other pulses	262.00	1090.00		0	0	0.00000	0
10	Groundnut	162.00	200.00	135	300	0.0006	0.00049	0.000114
11	Sunflower	9.00	182.00		75	0.0001365	0.00001	0.00012975
12	Soyabean	742.00	900.00	115	300	0.0027	0.00223	0.000474
13	Sesamum				0	0	0.00000	0
14	Nigerseed				0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	16680.00	14190.00	21630	2500	0.35475	0.41700	-0.06225
17	Cotton	8.00	60.00		700	0.00042	0.00006	0.000364
	Total kh. Season	**32002.00**	**32369.00**	**22295.00**		**0.38**	**0.44**	**-0.06**
1	R. Jowar	40399.00	51000.00	2245.00	130	0.0663	0.05252	0.0137813
2	Wheat	3563.00	4650.00	1400.00	400	0.0186	0.01425	0.004348
3	R. Maize	2016.00	2060.00	110.00	500	0.0103	0.01008	0.00022
4	Other cereals	167.00				0	0.00000	0
5	Gram	3515.00	8990.00	321.00	250	0.022475	0.00879	0.0136875
6	Other pulses	4.00	400.00		250	0.001	0.00001	0.00099
7	Safflower	6.00	630.00		250	0.001575	0.00002	0.00156
8	R.Sunflower	89.00	1280.00		350	0.00448	0.00031	0.0041685
9	Linseed					0	0.00000	0
10	Other R. oil seed crops	68.00				0	0.00000	0
	Total Rabbi crops	**49827.00**	**69010.00**	**4076.00**		**0.12**	**0.0859747**	**0.04**
1	Bajri		10.00	10.00	**450.00**	0.000045	0.00000	0.000045
2	Summer maize		232.00	200.00	**500**	0.00116	0.00000	0.00116
3	Summer Groundnut		228.00	190.00	700	0.001596	0.00000	0.001596
	Total summer crops	**0.00**	**470.00**	**400.00**		**0.002801**	**0**	**0.002801**
	Gross cropped area	110489.00	139226.00	34422.00		0.81	0.76	0.05
	Total Cereals	59468.00	72942.00	4200		0.11716	0.0947747	0.0223853
	Total Pulses	4430	11227	491		0.0244335	0.8026487	0.01485675
	Total Oilseeds	1076	3420	440		0.0110875	0.1322967	0.00804225
1	Fodder crops	10,360	15,000	3939	500	0.075	0.05180	0.0232
2	Vegetable	12,100	14,177	1797	750	0.1063275	0.09075	0.0155775
3	Horticultural & Plantation	6200	8200	1915	1500	0.123	0.09300	0.03

Crop Water Requirement – Indapur

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Indapur								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar		350.00	65	150	0.000525	0.00000	0.000525
2	paddy				0	0	0.00000	0
3	Bajra	200.00	300.00	450	120	0.00036	0.00024	0.00012
4	Maize	3500.00	4300.00	3683	200	0.0086	0.00700	0.0016
5	Other cereals		80.00		0	0	0.00000	0
6	Tur	15.00	442.00	442	300	0.001326	0.00005	0.001281
7	Mung		155.00	155	75	0.00011625	0.00000	0.00011625
8	Udid		0.00			0	0.00000	0
9	Other pulses				0	0	0.00000	0
10	Groundnut	25.00	25.00	200	300	0.000075	0.00008	0
11	Sunflower		238.00	238	75	0.0001785	0.00000	0.0001785
12	Soyabean	16.00	712.00	452	300	0.002136	0.00005	0.002088
13	Sesamum				0	0	0.00000	0
14	Nigerseed				0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	31480.00	24850.00	29040	2500	0.62125	0.78700	-0.16575
17	Cotton					0	0.00000	0
	Total kh. Season	**35236.00**	**31452.00**	**34725.00**		**0.63**	**0.79**	**-0.16**
1	R. Jowar	40500.00	49944.00	6969.00	130	0.0649272	0.05265	0.0122772
2	Wheat	6000.00	7850.00	6445.00	400	0.0314	0.02400	0.0074
3	R. Maize	3400.00	2620.00	2620.00	500	0.0131	0.01700	-0.0039
4	Other cereals	4.00	100.00	100.00	130	0.00013	0.00001	0.0001248
5	Gram	3700.00	3580.00	1800.00	250	0.00895	0.00925	-0.0003
6	Other pulses	300.00	500.00	57.00	250	0.00125	0.00075	0.0005
7	Safflower		290.00	50.00	250	0.000725	0.00000	0.000725
8	R.Sunflower		290.00	50.00	350	0.001015	0.00000	0.001015
9	Linseed					0	0.00000	0
10	Other R. oil seed crops	290.00				0	0.00000	0
	Total Rabbi crops	**54194.00**	**65174.00**	**18091.00**		**0.12**	**0.1036552**	**0.02**
1	Bajri	15.00	15.00	15.00	**450.00**	0.0000675	0.00007	0
2	Summer maize		100.00	100.00	**500**	0.0005	0.00000	0.0005
3	Summer Groundnut	125.00	200.00	200.00	700	0.0014	0.00088	0.000525
	Total summer crops	**140.00**	**315.00**			**0.0019675**	**0.0009425**	**0.001025**
	Gross cropped area	115179.00	139441.00	59602.00		1.2	1.17	0.03
	Total Cereals	53600.00	65544.00	20332		0.1194122	0.1018325	0.0185222
	Total Pulses	4015	4677	2454		0.01164225	1.2139634	0.00159725
	Total Oilseeds	456	1755	1190		0.0055295	0.1353727	0.0045315
1	Fodder crops	9,160	14,500	500	500	0.0725	0.04580	0.0267
2	Vegetable	3,449	7,000	3252	750	0.0525	0.02587	0.0266325
3	Horticultural & Plantation	13000	21000	3034	1500	0.315	0.19500	0.12

Crop Water Requirement – Daund

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Daund								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar		589.00	589	150	0.0008835	0.00000	0.0008835
2	paddy				0	0	0.00000	0
3	Bajra	2311.00	2500.00	1263	120	0.003	0.00277	0.0002268
4	Maize	239.00	1000.00	1000	200	0.002	0.00048	0.001522
5	Other cereals	3.00	80.00		0	0	0.00000	0
6	Tur	5.00	190.00	180	300	0.00057	0.00002	0.000555
7	Mung	14.00	131.00	126	75	0.00009825	0.00001	0.00008775
8	Udid	2.00				0	0.00000	0
9	Other pulses	9.00	800.00	750	0	0	0.00000	0
10	Groundnut	11.00	15.00	240	300	0.000045	0.00003	0.000012
11	Sunflower		263.00	0	75	0.00019725	0.00000	0.00019725
12	Soyabean	7.00				0	0.00000	0
13	Sesamum				0	0	0.00000	0
14	Nigerseed				0	0	0.00000	0
15	Other Kh. Oil seed crops				0	0	0.00000	0
16	Sugarcane	25250.00	19800.00	23800	2500	0.495	0.63125	-0.13625
17	Cotton					0	0.00000	0
	Total kh. Season	**27851.00**	**25368.00**	**27948.00**		**0.5**	**0.63**	**-0.13**
1	R. Jowar	4630.00	10135.00	6365.00	130	0.0131755	0.00602	0.0071565
2	Wheat	2392.00	4200.00	5864.00	400	0.0168	0.00957	0.007232
3	R. Maize	256.00	2240.00	1035.00	500	0.0112	0.00128	0.00992
4	Other cereals	5.00	200.00		130	0.00026	0.00001	0.0002535
5	Gram	877.00	2800.00	1704.00	250	0.007	0.00219	0.0048075
6	Other pulses		300.00		250	0.00075	0.00000	0.00075
7	Safflower		90.00		250	0.000225	0.00000	0.000225
8	R.Sunflower		230.00		350	0.000805	0.00000	0.000805
9	Linseed					0	0.00000	0
10	Other R. oil seed crops					0	0.00000	0
	Total Rabbi crops	**8160.00**	**20195.00**	**14968.00**		**0.05**	**0.019066**	**0.03**
1	Bajri						0.00000	
2	Summer maize					0	0.00000	0
3	Summer Groundnut	252.00	251.00	250.00	700	0.001757	0.00176	-0.00001
	Total summer crops	**252.00**	**251.00**	**250.00**		**0.001757**	**0.001764**	**-0.00001**
	Gross cropped area	52944.00	77084.00	54438.00		0.84	0.82	0.03
	Total Cereals	9831.00	20744.00	16116		0.047059	0.0218822	0.0269408
	Total Pulses	907	4221	2760		0.00841825	0.8295179	0.00620025
	Total Oilseeds	270	849	490		0.00302925	0.0286379	0.00123225
1	Fodder crops	4,040	9,000	4472	500	0.045	0.02020	0.0248
2	Vegetable	6,641	11,770	3600	750	0.088275	0.04981	0.0384675
3	Horticultural & Plantation	6000	10500	3200	1500	0.1575	0.09000	0.0675

Crop Water Requirement – Purandar

4.2 Crop Water Requirement								
Name of the State : Maharashtra								
Name of the District : Pune								
Block : Purandar								
Sr. No	Crops	Area Sown (ha)		Irrigated area (ha)	Crop water Demand (mm)	Water potential required (BCM)	Existing water Demand (BCM)	Water Potential to be Created (BCM)
		Existing	Projected					
1	Kh. Jowar	0.00	775.00	25	150	0.0011625	0.00000	0.0011625
2	paddy	908.00	1485.00	120	0	0	0.00000	0
3	Bajra	7924.00	8000.00	750	120	0.0096	0.00951	9.12E-05
4	Maize	311.00	1000.00	100	200	0.002	0.00062	0.001378
5	Other cereals	0.00	80.00		0	0	0.00000	0
6	Tur	147.00	107.00	20	300	0.000321	0.00044	-0.00012
7	Mung	103.00	396.00		75	0.000297	0.00008	0.00021975
8	Udid	44.00	44.00		75	0.000033	0.00003	0
9	Other pulses	3383.00	1530.00	25	0	0	0.00000	0
10	Groundnut	1511.00	1511.00	140	300	0.004533	0.00453	0
11	Sunflower	5.00	115.00		75	0.00008625	0.00000	0.0000825
12	Soyabean	176.00	850.00		300	0.00255	0.00053	0.002022
13	Sesamum	30.00	17.00		0	0	0.00000	0
14	Nigerseed	31.00	59.00		0	0	0.00000	0
15	Other Kh. Oil seed crops	82.00	0.00		0	0	0.00000	0
16	Sugarcane	1260.00	2274.00	2274	2500	0.05685	0.03150	0.02535
17	Cotton	0.00			700	0	0.00000	0
	Total kh. Season	**15915.00**	**18243.00**	**3454.00**		**0.08**	**0.05**	**0.03**
1	R. Jowar	23609.00	30000.00	4765.00	130	0.039	0.03069	0.0083083
2	Wheat	2623.00	3000.00	5088.00	400	0.012	0.01049	0.001508
3	R. Maize	370.00	5440.00	1500.00	500	0.0272	0.00185	0.02535
4	Other cereals				130	0	0.00000	0
5	Gram	2632.00	2210.00	800.00	250	0.005525	0.00658	-0.001055
6	Other pulses	0.00	1800.00	95.00	250	0.0045	0.00000	0.0045
7	Safflower		1760.00	800.00	250	0.0044	0.00000	0.0044
8	R.Sunflower	36.00	60.00	35.00	350	0.00021	0.00013	0.000084
9	Linseed		20.00		150	0.00003	0.00000	0.00003
10	Other R. oil seed crops	37.00			250	0	0.00009	-0.0000925
	Total Rabbi crops	**29307.00**	**44290.00**	**13083.00**		**0.09**	**0.0498322**	**0.04**
1	Bajri		100.00	100.00	**450.00**	0.00045	0.00000	0.00045
2	Summer maize		0.00		**500**	0	0.00000	0
3	Summer Groundnut		100.00	100.00	700	0.0007	0.00000	0.0007
	Total summer crops		**200.00**	**200.00**		**0.00115**	**0**	**0.00115**
	Gross cropped area	58197.00	85233.00	22658.00		0.4	0.2	0.2
	Total Cereals	35745.00	49780.00	12348		0.0909625	0.0531645	0.037798
	Total Pulses	6309	6087	940		0.010676	0.2241303	0.00354475
	Total Oilseeds	1908	4492	1075		0.01250925	0.07224355	0.007226
1	Fodder crops	1,620	3,500	500	500	0.0175	0.00810	0.0094
2	Vegetable	9,905	10,000	3431	750	0.075	0.07429	0.0007125
3	Horticultural & Plantation	1450	9000	1990	1500	0.135	0.02175	0.11325

Annexure – IV

Strategic Action Plan of the District – 5 years

Strategic Action Plan of the District - (5 Years)			
Sr. No.	**Name of Blocks**	**(District Pune)**	**Page.no.**
0	Strategic Action Plan of Pune	5 Year	121
1	Strategic Action Plan of Pune	I year	122
2	Strategic Action Plan of Pune	II year	123
3	Strategic Action Plan of Pune	III year	124
4	Strategic Action Plan of Pune	IV year	125
5	Strategic Action Plan of Pune	V year	126

Strategic Action Plan of Pune District – 5 Years

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							
Name of the State : Maharashtra							
Name of District : Pune							
Sr No	Component	Estiamated Cost In Rs. lakh					
		2016-17	2017-18	2018-19	2019-20	2020-21	Total
1	AIBP	0.00	0.00	0.00	0.00	0.00	0.00
2	Har Khet ko Pani	1305.95	1769.88	2233.80	2233.80	1305.96	8849.38
3	Per Drop More crop (Micro Irrigation)	2631.27	2628.74	2664.60	2757.03	2947.88	13629.52
4	Per Drop More (Supplementary water Management activities	7608.33	5891.67	6641.67	6641.67	5016.67	31800.00
5	PMKSY Watershed	10844.95	14474.46	18104.13	18104.13	10844.95	72372.62
6	Convergence with MGNREGA	21.96	29.25	36.97	36.97	22.13	147.29
7	State Planned Scheme of Irrigation	4740.85	6168.31	7800.00	7800.00	4740.85	31250.00
	Total	27153.32	30962.30	37481.16	37573.60	24878.44	158048.81

Strategic Action Plan of Pune District – I Year (2016-2017)

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							Year 2016-17
State : Maharashtra							
District : Pune							
Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0
2		MoWR		Medium Irrigation	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0
4		MoWR	Har Khet ko	Lift Irrigation	0	0	0
5		MoWR		Ground Water Development	0	0	0
6		MoWR		RRR of water bodies	8	507.64	686.45
7		MoWR		Construction of Field channels (for command area only)			
7.1		MoWR		LinedField Channels	2	479.25	72.00
7.2		MoWR		Unlined Channels	2	1350.00	472.50
8		MoWR		Micro Irrigation	2	150.00	75.00
							1305.95
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	11917	8754.33	2626.30
10		MOA & FW.DAC & FW		DPAP Sprinkler	45	38.65	4.63
11		MOA & FW.DAC & FW		Non DPAP Drip	2	1.15	0.35
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00	0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00	0.00
							2631.27
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	97	181.88	2425.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0	0
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	2	9078.33	5183.33
							7608.33
17	Pune	DoLR-MoRD	PMKSY Wat	Newly Created WHS			
17.1		DoLR-MoRD		Farm Ponds	470	799.00	376.15
17.2		DoLR-MoRD		cement nala bund	193	868.50	2487.76
17.3		DoLR-MoRD		earthen nala bund	363	1815.00	1669.54
17.4		DoLR-MoRD		diversion bund	63	252.00	534.53
17.5		DoLR-MoRD		compartment bunding	10380	4671.00	824.25
17.6		DoLR-MoRD		cct	8373	2763.09	1319.57
17.7		DoLR-MoRD		Deepening/ desilting of stucture	90	270.00	432.33
17.8		DoLR-MoRD		Deep cct	878	210.72	403.81
17.9		DoLR-MoRD		LBS	4095	0.00	178.44
17.10		DoLR-MoRD		Earthen Str.	773	193.25	46.11
17.11		DoLR-MoRD		Well Recharge	416	52.00	83.20
17.12		DoLR-MoRD		Terracing	71	0.00	177.50
17.13		DoLR-MoRD		Check Dams	965	0.00	193.03
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	750	93.75	262.50
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00	0.00
18			PMKSY Watershed	Renovated WHS			
18.1		DoLR-MoRD		Farm Ponds	0	0.00	0.00
18.2		DoLR-MoRD		CNB	59	177.00	236.00
18.3		DoLR-MoRD		Nallah Bunds	95	380.00	142.50
18.4				Desilting of Nala	272	1088.00	1360.00
18.5		DoLR-MoRD		KTW repair	13	165.39	117.72
18.6		DoLR-MoRD		Other Ground Water Recharge Structure	0	0.00	0.00
							10844.95
19		DoLR-MoRD	Convergence with MGNREGA	Newly Created			
19.1		DoLR-MoRD		Water Consurvation	13	22.14	2.21
19.2		DoLR-MoRD		Water Harvesting	37	31.45	15.72
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0.00	0.00
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0.00	0.00
19.5		DoLR-MoRD		Land development	0	25.51	4.04
20		DoLR-MoRD		Renovation			
20.1		DoLR-MoRD		Renovation water bodies including	0	0	0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0	0
	Pune						**21.96**
21				State Planned Scheme of Irrigation			
21		State Irrigation Department		Major Irrigation (Lining)	43	0.00	4390.85
21.1		State Irrigation Department		Medium Irrigation (Lining)	7	0.00	350.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00	0.00
21.3		State Irrigation Department			0	0.00	0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00	0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00	0.00
24		Externaly aded projects			0	0.00	0.00
25		Other loan projects like NABARD			0	0.00	0.00
							4740.85
Total					**40496.00**	**34419.02**	**27153.32**

Strategic Action Plan of Pune District – II Year (2017-18)

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							Year 2017-18
State : Maharashtra							
District : Pune							
Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0
2		MoWR		Medium Irrigation	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0
4		MoWR	Har Khet ko Pani	Lift Irrigation	0	0	0
5		MoWR		Ground Water Development	0	0	0
6		MoWR		RRR of water bodies	11	698.00	943.876
7		MoWR		Construction of Field channels (for command area only)			
7.1		MoWR		LinedField Channels	2	639.00	96.00
7.2		MoWR		Unlined Channels	2	1800.00	630.00
8		MoWR		Micro Irrigation	2	200.00	100.00
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	12212	8734.20	2620.26
10		MOA & FW.DAC & FW		DPAP Sprinkler	49	40.65	8.13
11		MOA & FW.DAC & FW		Non DPAP Drip	2	1.15	0.35
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00	0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00	0.00
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	132	247.50	3300.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0.00	0.00
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	1	4539.17	2591.67
17	Pune	DoLR-MoRD	PMKSY Watershed	Newly Created WHS			
17.1		DoLR-MoRD		Farm Ponds	626	1064.20	501.00
17.2		DoLR-MoRD		cement nala bund	257	1156.50	3312.71
17.3		DoLR-MoRD		earthen nala bund	484	2420.00	2226.06
17.4		DoLR-MoRD		diversion bund	85	340.00	721.19
17.5		DoLR-MoRD		compartment bunding	13840	6228.00	1099.00
17.6		DoLR-MoRD		cct	11163	3683.79	1759.27
17.7		DoLR-MoRD		Deepening/ desilting of stucture	121	363.00	581.24
17.8		DoLR-MoRD		Deep cct	1170	280.80	538.11
17.9		DoLR-MoRD		LBS	5460	0.00	237.92
17.10		DoLR-MoRD		Earthen Str.	1031	257.75	61.5
17.11		DoLR-MoRD		Well Recharge	555	69.38	111.00
17.12		DoLR-MoRD		Terracing	95	0.00	237.50
17.13		DoLR-MoRD		Check Dams	1287	0.00	257.44
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	1000	125.00	350.00
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00	0.00
18			PMKSY Watershed	Renovated WHS			
18.1		DoLR-MoRD		Farm Ponds	0	0.00	0
18.2		DoLR-MoRD		CNB	78	234.00	312.00
18.3		DoLR-MoRD		Nallah Bunds	127	508.00	190.50
18.4				Desilting of Nala	363	1452.00	1815.00
18.5		DoLR-MoRD		KTW repair	18	229.00	163.00
18.6		DoLR-MoRD		Other Ground Water Recharge Structure	0	0.00	0.00
19	Pune	DoLR-MoRD	Convergence with MGNREGA	Newly Created			
19.1		DoLR-MoRD		Water Consurvation	18	30.65	3.06
19.2		DoLR-MoRD		Water Harvesting	49	41.65	20.81
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0.00	0.00
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0.00	0.00
19.5		DoLR-MoRD		Land development	0	34.01	5.38
20		DoLR-MoRD		Renovation			
20.1		DoLR-MoRD		Renovation water bodies including	0	0	0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0	0
21				State Planned Scheme of Irrigation			
21		State Irrigation Department		Major Irrigation (Lining)	56	0.00	5718.31
21.1		State Irrigation Department		Medium Irrigation (Lining)	9	0.00	450.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00	0.00
21.3		State Irrigation Department			0	0.00	0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00	0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00	0.00
24		Externaly aded projects			0	0.00	0.00
25		Other loan projects like NABARD			0	0.00	0.00
Total					50299.00	35417.39	30962.30

Strategic Action Plan of Pune District – III Year (2018-19)

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							Year 2018-19
State : Maharashtra							
District : Pune							
Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0
2		MoWR		Medium Irrigation	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0
4		MoWR	Har Khet ko Pani	Lift Irrigation	0	0	0
5		MoWR		Ground Water Development	0	0	0
6		MoWR		RRR of water bodies	14	888.36	1201.2967
7		MoWR		Construction of Field channels (for command area only)			
7.1		MoWR		LinedField Channels	2	798.75	120.00
7.2		MoWR		Unlined Channels	2	2250.00	787.50
8		MoWR		Micro Irrigation	2	250.00	125.00
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	11971	8851.64	2655.50
10		MOA & FW.DAC & FW		DPAP Sprinkler	55	43.39	8.68
11		MOA & FW.DAC & FW		Non DPAP Drip	3	1.40	0.42
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00	0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00	0.00
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	162	303.75	4050.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0.00	0.00
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	1	4539.17	2591.67
17	Pune	DoLR-MoRD	PMKSY Watershed	Newly Created WHS			
17.1		DoLR-MoRD		Farm Ponds	782	1329.40	625.85
17.2		DoLR-MoRD		cement nala bund	322	1449.00	4150.56
17.3		DoLR-MoRD		earthen nala bund	606	3030.00	2787.17
17.4		DoLR-MoRD		diversion bund	106	424.00	899.37
17.5		DoLR-MoRD		compartment bunding	17300	7785.00	1373.75
17.6		DoLR-MoRD		cct	13955	4605.15	2199.29
17.7		DoLR-MoRD		Deepening/ desilting of stucture	150	450.00	720.55
17.8		DoLR-MoRD		Deep cct	1463	351.12	672.87
17.9		DoLR-MoRD		LBS	6825	0.00	297.41
17.10		DoLR-MoRD		Earthen Str.	1289	322.25	76.89
17.11		DoLR-MoRD		Well Recharge	694	86.75	138.80
17.12		DoLR-MoRD		Terracing	119	0.00	297.50
17.13		DoLR-MoRD		Check Dams	1609	0.00	321.86
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	1250	156.25	437.50
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00	0.00
18			PMKSY Watershed	Renovated WHS			
18.1		DoLR-MoRD		Farm Ponds	0	0.00	0
18.2		DoLR-MoRD		CNB	97	291.00	388.00
18.3		DoLR-MoRD		Nallah Bunds	159	636.00	238.50
18.4				Desilting of Nala	454	1816	2270
18.5		DoLR-MoRD		KTW repair	23	292.61	208.28
18.6		DoLR-MoRD		Other Ground Water Recharge Structure	0	0.00	0.00
19	Pune	DoLR-MoRD	Convergence with MGNREGA	Newly Created			
19.1		DoLR-MoRD		Water Consurvation	23	39.17	3.91
19.2		DoLR-MoRD		Water Harvesting	62	52.70	26.33
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0.00	0.00
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0.00	0.00
19.5		DoLR-MoRD		Land development	0	42.51	6.73
20		DoLR-MoRD		Renovation			
20.1		DoLR-MoRD		Renovation water bodies including	0	0	0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0	0
21				State Planned Scheme of Irrigation			
21		State Irrigation Department		Major Irrigation (Lining)	71	0.00	7250.00
21.1		State Irrigation Department		Medium Irrigation (Lining)	11	0.00	550.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00	0.00
21.3		State Irrigation Department			0	0.00	0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00	0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00	0.00
24		Externaly aded projects			0	0.00	0.00
25		Other loan projects like NABARD			0	0.00	0.00
Total					59576.00	41085.37	37481.16

Strategic Action Plan of Pune District – IV Year (2019-20)

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							Year 2019-20
State : Maharashtra							
District : Pune							
Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0
2		MoWR		Medium Irrigation	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0
4		MoWR	Har Khet ko Pani	Lift Irrigation	0	0	0
5		MoWR		Ground Water Development	0	0	0
6		MoWR		RRR of water bodies	14	888.36	1201.2967
7		MoWR		Construction of Field channels (for command area only)			
7.1		MoWR		LinedField Channels	2	798.75	120.00
7.2		MoWR		Unlined Channels	2	2250.00	787.50
8		MoWR		Micro Irrigation	2	250.00	125.00
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	12404	9163.19	2748.96
10		MOA & FW.DAC & FW		DPAP Sprinkler	49	38.65	7.73
11		MOA & FW.DAC & FW		Non DPAP Drip	2	1.15	0.35
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00	0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00	0.00
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	162	303.75	4050.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0.00	0.00
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	1	4539.17	2591.67
17	Pune	DoLR-MoRD	PMKSY Watershed	Newly Created WHS			
17.1		DoLR-MoRD		Farm Ponds	782	1329.40	625.85
17.2		DoLR-MoRD		cement nala bund	322	1449.00	4150.56
17.3		DoLR-MoRD		earthen nala bund	606	3030.00	2787.17
17.4		DoLR-MoRD		diversion bund	106	424.00	899.37
17.5		DoLR-MoRD		compartment bunding	17300	7785.00	1373.75
17.6		DoLR-MoRD		cct	13955	4605.15	2199.29
17.7		DoLR-MoRD		Deepening/ desilting of stucture	150	450.00	720.55
17.8		DoLR-MoRD		Deep cct	1463	351.12	672.87
17.9		DoLR-MoRD		LBS	6825	0.00	297.41
17.10		DoLR-MoRD		Earthen Str.	1289	322.25	76.89
17.11		DoLR-MoRD		Well Recharge	694	86.75	138.80
17.12		DoLR-MoRD		Terracing	119	0.00	297.50
17.13		DoLR-MoRD		Check Dams	1609	0.00	321.86
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	1250	156.25	437.50
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00	0.00
18		DoLR-MoRD	PMKSY Watershed	Renovated WHS			
18.1		DoLR-MoRD		Farm Ponds	0	0.00	0
18.2		DoLR-MoRD		CNB	97	291.00	388.00
18.3		DoLR-MoRD		Nallah Bunds	159	636.00	238.50
18.4		DoLR-MoRD		Desilting of Nala	454	1816.00	2270.00
18.5		DoLR-MoRD		KTW repair	23	292.6111	208.2777
18.6		DoLR-MoRD		Other Ground Water Recharge Structure	0	0.00	0.00
19	Pune		Convergence with MGNREGA	Newly Created			
19.1		DoLR-MoRD		Water Consurvation	23	39.1657142857	3.91
19.2		DoLR-MoRD		Water Harvesting	62	52.7	26.3336842105
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0	0
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0	0
19.5		DoLR-MoRD		Land development	0	42.51	6.7275
20		DoLR-MoRD		Renovation			
20.1		DoLR-MoRD		Renovation water bodies including	0	0	0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0	0
21				State Planned Scheme of Irrigation			
21		State Irrigation Department		Major Irrigation (Lining)	71	0.00	7250.00
21.1		State Irrigation Department		Medium Irrigation (Lining)	11	0.00	550.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00	0.00
21.3		State Irrigation Department			0	0.00	0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00	0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00	0.00
24		Externaly aded projects			0	0.00	0.00
25		Other loan projects like NABARD			0	0.00	0.00
Total					60002.00	41391.93	37573.60

Strategic Action Plan of Pune District – V Year (2019-20)

5 - Strategic Action Plan for Irrigation in District Under PMKSY.							Year 2020-21
State : Maharashtra							
District : Pune							
Sr. No.	Name of District	Concerned Ministry / Department	Component	Activity	Total Number	Command Area / Irrigation Potential (Ha)	Estimated cost (in Rs. Lakh)
1	2	3	4	5	6	7	9
1	Pune	MoWR	AIBP	Major Irrigation	0	0	0
2		MoWR		Medium Irrigation	0	0	0
3		MoWR		Surface Minor Irrigation	0	0	0
4		MoWR	Har Khet ko Pani	Lift Irrigation	0	0	0
5		MoWR		Ground Water Development	0	0	0
6		MoWR		RRR of water bodies	8	507.64	686.46
7		MoWR		Construction of Field channels (for command area only)			
7.1		MoWR		LinedField Channels	2	479.25	72.00
7.2		MoWR		Unlined Channels	2	1350.00	472.50
8		MoWR		Micro Irrigation	2	150.00	75.00
9		MOA & FW.DAC & FW	Per Drop More crop (Micro Irrigation)	DPAP Drip	12998	9792.49	2937.75
10		MOA & FW.DAC & FW		DPAP Sprinkler	54	47.45	9.49
11		MOA & FW.DAC & FW		Non DPAP Drip	4	2.15	0.65
12		MOA & FW.DAC & FW		Non DPAP Sprinkler	0	0.00	0.00
13		MOA & FW.DAC & FW		Topping up of MGNREGA	0	0.00	0.00
14		MOA & FW.DAC & FW	Per Drop More (Supplementary water Management activities	Drought Proofing Through check Dams / Water Harvesting Stucture	97	181.88	2425.00
15		MOA & FW.DAC & FW		Secondary Storage Structure	0	0.00	0.00
16		MOA & FW.DAC & FW		On Farm Development (Distrubution Pipe / raised bed and furrow system etc.)	1	4539.17	2591.67
17	Pune	DoLR-MoRD	PMKSY Watershed	Newly Created WHS			
17.1		DoLR-MoRD		Farm Ponds	470	799.00	376.15
17.2		DoLR-MoRD		cement nala bund	193	868.50	2487.76
17.3		DoLR-MoRD		earthen nala bund	363	1815.00	1669.54
17.4		DoLR-MoRD		diversion bund	63	252.00	534.53
17.5		DoLR-MoRD		compartment bunding	10380	4671	824.25
17.6		DoLR-MoRD		cct	8373	2763.09	1319.57
17.7		DoLR-MoRD		Deepening/ desilting of stucture	90	270.00	432.33
17.8		DoLR-MoRD		Deep cct	878	210.72	403.81
17.9				LBS	4095	0.00	178.44
17.10				Earthen Str.	773	193.25	46.11
17.11				Well Recharge	416	52.00	83.20
17.12				Terracing	71	0.00	177.50
17.13				Check Dams	965	0.00	193.03
17.14		DoLR-MoRD		Other Ground Water Recharge Structure	750	93.75	262.50
17.15		DoLR-MoRD		Fishery Ponds / cattle pond	0	0.00	0.00
18			PMKSY Watershed	Renovated WHS			
18.1		DoLR-MoRD		Farm Ponds	0	0.00	0
18.2		DoLR-MoRD		CNB	59	177.00	236.00
18.3		DoLR-MoRD		Nallah Bunds	95	380.00	142.50
18.4				Desilting of Nala	272	1088.00	1360.00
18.5		DoLR-MoRD		KTW repair	13	165.39	117.72
18.6		DoLR-MoRD		Other Ground Water Recharge Structure	0	0.00	0.00
19	Pune	DoLR-MoRD	Convergence with MGNREGA	Newly Created			
19.1		DoLR-MoRD		Water Consurvation	14	23.84	2.38
19.2		DoLR-MoRD		Water Harvesting	37	31.45	15.7152631579
19.3		DoLR-MoRD		Creation of irrigation canals and drains	0	0	0
19.4		DoLR-MoRD		Providing infrastructure Irrigation	0	0	0
19.5		DoLR-MoRD		Land development	0	25.506	4.0365
20		DoLR-MoRD		Renovation			
20.1		DoLR-MoRD		Renovation water bodies including	0	0	0
20.2		DoLR-MoRD		Renovation & Maintenance of irrigation canals & Drains.	0	0	0
21				State Planned Scheme of Irrigation			
21		State Irrigation Department		Major Irrigation (Lining)	43	0.00	4390.85
21.1		State Irrigation Department		Medium Irrigation (Lining)	7	0.00	350.00
21.2		State Irrigation Department		Surface minor Irrigation	0	0.00	0.00
21.3		State Irrigation Department			0	0.00	0.00
22		Irrigation Scheme of State Agriculture Department			0	0.00	0.00
23		Irrigation Scheme of other Line Departments of State Govt.			0	0.00	0.00
24		Externaly aded projects			0	0.00	0.00
25		Other loan projects like NABARD			0	0.00	0.00
Total					41582.00	30929.51	24878.44

Annexure V

Areawise Cropwise Irrigation Status -Seasons

Areawise Cropwise Irrigation Status			
Sr. No.	**Name of Blocks**	**(District Pune)**	**Page.no.**
1	Season - Kharif		128
2	Season - Rabi		129
3	Season - Summer		130
4	Total of All Season		131

Areawise Cropwise Irrigation Status : Season – Kharif

2.1 Area wise Crop wise Irrigation Status																					
District : Pune																					
State : Maharashtra																					
Season : Kharif																					Area : Ha
Taluka	Cereals			Coarse Cereals			Pulses			Oilseeds			Cotton			Vegetables			Total		
Name	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total
Haveli	205	2744	2949	221	7957	8178	145	798	943	970	991	1961				2150	1225	3375	1541	12190	13731
Maval	4460	4309	8769	310	3043	3353	140	163	303	700	1110	1810				2022	46	2068	5610	7125	12735
Mulshi	4335	13122	17457	68	2947	3015	62	243	305	2000	259	2259				75	47	122	6465	13107	19572
Bhor	16	8262	8278	1050	4294	5344	188	852	1040	1000	3616	4616				114	320	434	2254	17024	19278
Velha	224	7410	7634	25	1976	2001	75	90	165	20	1224	1244				23	96	119	344	10700	11044
Juunar	315	6710	7025	5069	10973	16042	325	2859	3184	401	9142	9543				1572	5668	7240	6110	29684	35794
Khed	468	6505	6973	3544	15820	19364	450	1482	1932	1000	13622	14622				717	4283	5000	5462	37429	42891
Ambegaon	328	3075	3403	1493	7853	9346	185	1676	1861	1025	4438	5463				3950	3050	7000	3031	17585	20616
Shirur		80	80	6345	8976	15321	3218	2340	5558	1000	514	1514				6115	1885	8000	10563	13410	23973
Baramati	0	0	0	245	1703	1948	170	1160	1330	250	857	1107		60	60	995	2102	3097	665	3780	4445
Indapur	0	80	80	3990	371	4361	805	522	1327	890	231	1121				281	219	500	5685	1204	6889
Daund	0	80	80	2857	247	3104	1056	65	1121	350	252	602				900	137	1037	4263	644	4907
Purandar	120	1445	1565	875	5597	6472	45	2212	2257	140	346	486				1191	309	1500	1180	12821	14001
Total	**10471**	**53822**	**64293**	**26092**	**71757**	**97849**	**6864**	**14462**	**21326**	**9746**	**36602**	**46348**	**0**	**60**	**60**	**20105**	**19387**	**39492**	**53173**	**176703**	**229876**

Areawise Cropwise Irrigation Status : Season – Rabi

2.1 Area wise Crop wise Irrigation Status																		
District : Pune																		
State : Maharashtra																		
Season : Rabi																		Area : Ha
Taluka	Cereals			Coarse Cereals			Pulses			Oilseeds			Vegetables			Total		
Name	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total
Haveli	1920	1570	3490	2795	28501	31296	837	1223	2060	30	0	30	1768	793	2561	5582	10438	16020
Maval	1084	1086	2170	1270	9240	10510	173	687	860		10	10	1002	41	1043	2527	2523	5050
Mulshi	2055	265	2320	8644	1616	10260	1253	157	1410	50	30	80	231	87	318	12002	2068	14070
Bhor	1235	1335	2570	2746	11614	14360	580	1290	1870	50	0	50	421	1116	1537	4611	8239	12850
Velha	520	280	800	548	5392	5940	93	527	620		0	0	5	128	133	1161	2199	3360
Juunar	3230	2550	5780	5741	16049	21790	780	12390	13170		200	200	1900	2980	4880	9751	36609	46360
Khed	2860	2650	5510	3565	29476	33041	3252	4718	7970	200	50	250	1465	3378	4843	9877	26273	36150
Ambegaon	2918	1712	4630	3520	18461	21981	3190	3600	6790	150	190	340	2678	1142	3820	9778	13492	23270
Shirur	4616	4064	8680	23799	38130	61929	5264	2566	7830	1150	1980	3130	5730	1085	6815	34829	68431	103260
Baramati	1300	9350	10650	2755	50523	53278	321	9069	9390		1910	1910	297	3591	3888	4376	80184	84560
Indapur	6445	505	6950	9589	52281	61870	1857	2223	4080	100	480	580	2900	1370	4270	17991	85009	103000
Daund	5864	1536	7400	5542	51656	57198	1704	1396	3100	250	70	320	2416	142	2558	13360	69700	83060
Purandar	5088	352	5440	6265	52612	58877	895	3115	4010	835	1005	1840	3100	750	3850	13083	36527	49610
Total	39135	27255	66390	76779	365551	442330	20199	42961	63160	2815	5925	8740	23913	16603	40516	138928	441692	580620

Areawise Cropwise Irrigation Status : Season – Summer

2.1 Area wise Crop wise Irrigation Status																								
District : Pune																								
State : Maharashtra																								
Season : Summer																								Area : Ha
Taluka	Cereals			Coarse Cereals			Pulses			Oilseeds			Sugarcane			Vegetables			Fruit crop			Total		
Name	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total
Haveli				110		110				200		200	9328		9328	270		270	1298	261	1559	580	0	580
Maval										300		300	1753		1753	25		25	510	95	605	325	0	325
Mulshi										120		120	1651		1651	12		12	421	250	671	132	0	132
Bhor										300		300	1113		1113	24		24	342	442	784	324	0	324
Velha										20		20	41		41				180	572	752	20	0	20
Juunar				210		210				690		690	9461		9461	120		120	3505	2923	6428	1020	0	1020
Khed				110		110				300		300	700		700	135		135	384	2346	2730	545	0	545
Ambegaon				110		110				500		500	3403		3403	180		180	850	1909	2759	790	0	790
Shirur				100		100				1100		1100	9303		9303	185		185	1830	1032	2862	1385	0	1385
Baramati				200		200				190		190	21630		21630	105		105	1615	2176	3791	495	0	495
Indapur				100		100				400		400	29040		29040	150		150	3034	137	3171	650	0	650
Daund				200		200				1000		1000	23800		23800	142		142	1520	518	2038	1342	0	1342
Purandar				100		100				100		100	2274		2274	150		150	1990	183	2173	350	0	350
Total				1240	0	1240	0	0	0	5220	0	5220	113497	0	113497	1498	0	1498	17479	12844	30323	7958	0	7958

Areawise Cropwise Irrigation Status : Season – Total of all Season

2.1 Area wise Crop wise Irrigation Status

District : Pune

State : Maharashtra

Season : Total of All Seasons

Area : Ha

Taluka	Cereals			Coarse Cereals			Pulses			Oilseeds			Cotton			Sugarcane			Vegetables			Fruit crop			Total		
Name	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total	Irrigated	Rainfed	Total
Haveli	2125	4314	6439	3126	36458	39584	982	2021	3003	1200	991	2191	0	0	0	9328	0	9328	4188	2018	6206	1298	261	1559	22247	46063	68310
Maval	5544	5395	10939	1580	12283	13863	313	850	1163	1000	1120	2120	0	0	0	1753	0	1753	3049	87	3136	510	95	605	13749	19830	33579
Mulshi	6390	13387	19777	8712	4563	13275	1315	400	1715	2170	289	2459	0	0	0	1651	0	1651	318	134	452	421	250	671	20977	19023	40000
Bhor	1251	9597	10848	3796	15908	19704	768	2142	2910	1350	3616	4966	0	0	0	1113	0	1113	559	1436	1995	342	442	784	9179	33141	42320
Velha	744	7690	8434	573	7368	7941	168	617	785	40	1224	1264	0	0	0	41	0	41	28	224	252	180	572	752	1774	17695	19469
Juunar	3545	9260	12805	11020	27022	38042	1105	15249	16354	1091	9342	10433	0	0	0	9461	0	9461	3592	8648	12240	3505	2923	6428	33319	72444	105763
Khed	3328	9155	12483	7219	45296	52515	3702	6200	9902	1500	13672	15172	0	0	0	700	0	700	2317	7661	9978	384	2346	2730	19150	84330	103480
Ambegaon	3246	4787	8033	5123	26314	31437	3375	5276	8651	1675	4628	6303	0	0	0	3403	0	3403	6808	4192	11000	850	1909	2759	24480	47106	71586
Shirur	4616	4144	8760	30244	47106	77350	8482	4906	13388	3250	2494	5744	0	0	0	9303	0	9303	12030	2970	15000	1830	1032	2862	69755	62652	132407
Baramati	1300	9350	10650	3200	52226	55426	491	10229	10720	440	2767	3207	0	60	60	21630	0	21630	1397	5693	7090	1615	2176	3791	30073	82501	112574
Indapur	6445	585	7030	13679	52652	66331	2662	2745	5407	1390	711	2101	0	0	0	29040	0	29040	3331	1589	4920	3034	137	3171	59581	58419	118000
Daund	5864	1616	7480	8599	51903	60502	2760	1461	4221	1600	322	1922	0	0	0	23800	0	23800	3458	279	3737	1520	518	2038	47601	56099	103700
Purandar	5208	1797	7005	7240	58209	65449	940	5327	6267	1075	1351	2426	0	0	0	2274	0	2274	4441	1059	5500	1990	183	2173	23168	67926	91094
Total	49606	81077	130683	104111	437308	541419	27063	57423	84486	17781	42527	60308	0	60	60	113497	0	113497	45516	35990	81506	17479	12844	30323	375053	667229	1042282